Phase-field Modelling of Microstructure Evolution During Solidification Incorporating CALPHAD Description of Gibbs Energies

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THESIS CERTIFICATE

This is to certify that the thesis titled **Phase-field Modelling of Microstructure Evolution During Solidification Incorporating CALPHAD Description of Gibbs Energies** by **P. Gerald Tennyson**, to the Indian Institute of Technology, Madras, for the award of the degree of **Doctor of Philosophy**, is a bona fide record of the research work done by him under our supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Phase-field method is a versatile method for simulating microstructural evolution during phase transformations that uses a non-conserved order parameter (ϕ) to identify a phase. Microstrucural evolution is captured by the minimization of a Ginzburg-Landau type free energy functional with respect to the order parameter ϕ, along with other conserved fields such as concentration (c), in the case of alloys. The isothermal growth of a dendrite into a supersaturated melt was studied using a binary alloy phase-field model. Two approaches were adopted to study dendritic solidification in alloy melts. The first approach used a single concentration variable to treat both the phases (solid and liquid). This approach employed model free energy functions to describe the phases. The other approach employed separate concentration variables for each of the phases. This approach also involved coupling of the CALPHAD formulation of Gibbs energy functions to describe the phases with the phase-field method. Quantitative predictions were made using the thin-interface limit approach and the antitrapping current formulation.

Two systems of industrial relevance were chosen for the study *viz.*, Al-Cu and Al-Mg, along the Al-rich corner. The solid and the liquid phases were modelled using random substitutional solution. Phase-field model parameters were related to the interfacial energy which is a physical quantity. Parametric studies were done, by varying grid spacing and interface width, to ascertain the stability range of the growing dendrite and validated using LGK analysis of dendritic growth.

The phase-field method, which involves the solution of a set of non-linear PDEs at every point in the spatial domain, is computationally very intensive. This work also serves to highlight the implementation of the phase-field equations on heterogeneous platforms (many-core clusters and multi-core Graphical Processing Units). Different tools used to visualize the problem are also explained.

The studies were then extended to capture the morphological transition from dendrite to globular as an effect of thermal cycling during semi-solid processing. Thermal cycling was done within the freezing range of the alloy systems considered, by varying the thermal amplitude and

cycle times. The morphological evolution under such imposed thermal cycles was simulated both in 2-D and 3-D. Suitable scaling using the thermosolutal diffusion distances was used to overcome computational difficulties in quantitative comparison at system scale. It was found that in the absence of convection, solutal re-melting plays a significant role in effecting the transition from dendritic to globular. Results are interpreted in the light of existing theories of microstructure refinement and globularisation.

KEYWORDS: Solidification, Microstructure evolution, Phase-field modelling, CALPHAD, MPI, GPU, OpenCL, Semi-solid processing, Globularisation, Thin-interface limit, Antitrapping current

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NOTATION

Thermodynamic quantities

c	molar concentration, mol cm^{-3}
c_{L}	molar concentration of liquid, mol cm^{-3}
c_{S}	molar concentration of solid, mol cm^{-3}
F	total free energy functional
$f(\phi, c, T)$	homogeneous free energy density, J cm^{-3}
f_i^{L}	free energy of component i in liquid, where i=A,B, J cm^{-3}
f_i^{S}	free energy of component i in solid, where i=A,B, J cm^{-3}
${}^{\phi}G_m$	molar Gibbs energy of a phase ϕ, J mol^{-1}
${}^{\circ}G_{\mathrm{i}}$	molar Gibbs energy of element i in it stable state, J mol^{-1}
${}^{\mathrm{id}}G_{\mathrm{m}}$	ideal molar Gibbs energy, J mol^{-1}
${}^{\mathrm{E}}G_{\mathrm{m}}$	excess molar Gibbs energy, J mol^{-1}
$L_{\mathrm{A,B}}$	interaction parameter, J mol^{-1}
k	partition coefficient
m_{L}	liquidus slope
T_0	initial temperature, K
T_m^{A}	melting temperature of A, K
T_m^{B}	melting temperature of B, K
x_{B}	mole fraction of B
x_{B}^0	initial composition of the melt
$x_{\mathrm{B}}^{\mathrm{L}}$	composition of liquid
$x_{\mathrm{B}}^{\mathrm{S}}$	composition of solid
$x_{\mathrm{B}}^{\mathrm{L},e}$	equilibrium composition of liquid
$x_{\mathrm{B}}^{\mathrm{S},e}$	equilibrium composition of solid
β	interface kinetics coefficient, Ks cm^{-1}
σ	surface tension, J cm^{-2}

Variables related to dendrite characterization

P	Peclet number
R_t	dendrite tip radius, cm
V_t	dendrite tip velocity, cm s^{-1}
ΔT_t	thermal undercooling, K
ΔT_c	solutal undercooling, K
ΔT	total undercooling, K
ϵ_c	strength of capillary anisotropy
Δc	magnitude of solute rejection at the interface
Γ	Gibbs-Thomson coefficient, K cm

κ	curvature, cm^{-1}
σ^*	stability parameter

Reference scales

d_0	capillary length, cm
D_T	thermal diffusivity, $\mathrm{cm}^2\mathrm{s}^{-1}$
D_L	diffusivity in liquid, $\mathrm{cm}^2\mathrm{s}^{-1}$
D_S	diffusivity in solid, $\mathrm{cm}^2\mathrm{s}^{-1}$
R	universal gas constant, 8.314 $\mathrm{Jmol}^{-1}\mathrm{K}^{-1}$
T_r	reference temperature, K
v_m	molar volume, $\mathrm{cm}^3\mathrm{mol}^{-1}$

Numerical simulation parameters

Nx,Ny,Nz	number of grid units along x,y,z directions, respectively
$\Delta\mathrm{t}$	time resolution
$\Delta\mathrm{x}$	spatial resolution

Phase-field parameters

ϕ	order parameter (ϕ=0: liquid, ϕ=1: solid)
δ	interface width, cm
ε_ϕ	gradient energy coefficient, $\sqrt{\mathrm{Jm}^{-1}}$
θ	interface tilt angle
$g_w(\phi)$	double well potential
W	height of the double well potential $g_w(\phi)$, Jm^{-3}
$p(\phi)$	interpolating function for phase-field
$s(\phi)$	interpolating function for concentration
$h(\phi)$	interpolating function for diffusion
M_ϕ	phase-field mobility
j_{at}	antitrapping current

CHAPTER 1

INTRODUCTION

Semi-solid processing (SSP) involves treating materials between the solidus and liquidus. The most important characteristic of SSP is that the solidification microstructure changes markedly, from dendritic under conventional processing conditions to non-dendritic or globular. Of the many factors that influence the dendritic breakdown, thermal and solutal gradients play a pivotal role in the fragmentation process. Researches on semi-solid processing indicate that the thermal gradients in the melt during SSP create an excessive solute pileup at the roots of the dendrites which results in the remelting of the arms at the roots. This along with convection in the melt leads to the formation of globular crystals as the thermal cycling is done between the solidus and the liquidus. In order to simulate the thermal gradient experienced by a dendrite as it gets stirred between the solidus and the liquidus, an appropriate computational model is necessary.

Phase-field models have gained wide significance in the recent past, as they avoid the explicit tracking of the interface and are extensively used to study microstructure evolution in different situations. In this study, an isothermal phase field model for a binary alloy, developed by Wheeler *et al.* (1992) and an anti-trapping model formulation for alloys proposed by Kim (2007) were used to simulate the dendritic growth. The work also involves coupling of CALPHAD formulation of Gibbs energy functions for the solid and liquid phases with the phase-field model. The motion of a dendrite during SSP is captured by imposing thermal cycles on the computational domain through discrete, single and multiple cycles. Significant changes in the growth morphologies are brought about by changing the amplitude and the duration of the thermal fluctuations.

Thermodynamic calculations based on realistic Gibbs energy functions are quite versatile and efficient for providing global information such as composition of the phases, their fractions, etc. of an equilibrium microstructure. They convey, however, no information on the local arrangement of the phases in the microstructure, *viz.*, the size, distribution and morphology of the individual phases. These features are dependent on the growth kinetics. Coupling thermodynamic calculations to the microstructure evolution models will eventually help in the understanding

and predicting microstructure evolution in technical alloys. Salient aspects of the present work are as follows:

- Implementation of a phase-field model for binary alloy solidification : Phase-field model by Wheeler *et al.* (1992) and Kim *et al.* (1999) and implementation of multicomponent anti-trapping model by Kim (2007) adapted for binaries.
- Incorporating CALPHAD description of Gibbs energies for the solid and liquid phases for systems Al-Cu and Al-Mg.
- Validation of the model
- Simulation of dendritic to globular transition during semisolid processing

Chapter 2 provides an overview of the phase-field method. A brief account of the phase-field models pertaining to single phase solidification is made. The chapter also discuses the phase-field models that have incorporated CALPHAD description of Gibbs energies and or coupled with thermodynamic databases. Some general theories on dendritic growth are also presented.

Chapter 3 is on "Model formulation", which mainly discusses the phase-field models used and the anti-trapping current formulation adapted to binaries. This is followed by a chapter on thermodynamic models used and the relevant coupling with phase-field models (Chapter 4). It delves more into model validation with the Gibbs energies of the respective binaries chosen, *viz.*, Al-Mg and Al-Cu.

Chapter 5 deals with numerical and computer implementation of the approach. Parallel implementation on multiple CPUs and GPU platforms are detailed.

Chapter 6 discusses model validation with the Gibbs energies of the respective binary systems, *viz.*, Al-Mg and Al-Cu. It is followed by application of the phase-field approach to simulate dendritic to globular transition during semisolid processing with respect to 2-D. Special emphasis is placed on solute redistribution and morphology transitions as a result of thermal cycling.

Chapter 7 focusses on dendritic growth morphologies in 3-D. Thermal cycling incorporating Gibbs energy functions performed for a single primary dendrite is discussed. Growth morphologies were also studied using 3-D simulations on large domains.

The thesis concludes with overall remarks drawn from the present study and future perspectives.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview

The phase-field model is a phenomenological model which utilises, to analyse temporal evolution of microstructures, a set of phase-field variables that are continuous in space and time. Its distinct advantage lies in its simplistic approach of avoiding interface tracking by describing the geometry in terms of one or more phase fields. The basics of phase-field method and its applications have been well reviewed and documented (Wheeler *et al.* (1995); Ode *et al.* (2001); Boettinger *et al.* (2002); Chen (2002); Gránásy *et al.* (2004); Singer-Loginova and Singer (2008); Moelans *et al.* (2008); Emmerich (2008); Steinbach (2009); Nestler and Choudhury (2011); Plapp (2011)). In this chapter, a brief description of the different phase-field models are put forth with emphasis on the coupling of realistic Gibbs energy functions obtained through CALPHAD approach with phase-field models in the context of solidification. In the literature, two classes of phase-field models for alloy solidification are encountered, depending upon how the diffuse interface is defined. The first approach defines the interface to be composed of the solid and liquid with the same composition. The second approach defines the interface as a mixture of solid and liquid retaining their macroscopic bulk properties by defining a separate concentration field for each phase. The general concept of the phase-field method is to describe a system by a field variable (order parameter) or a set of them in the case of multiphase systems. These field variables take constant values in the respective bulk phases and vary smoothly between the phases through interfaces of characteristic thickness δ. The governing equations for the phase-field and the coupling to appropriate thermodynamic state variables (temperature, composition) can be obtained by taking a variational derivative of the free energy functional, which is generally of the Ginzburg-Landau type.

Early approaches on solidification modelling date back to 19th century through the works of Stefan, who studied the formation of ice in the Arctic regions (Stefan (1889)). This forms the classical mathematical model of solidification and is more commonly referred to as the sharp

interface formulation for time dependent moving boundary problem. The sharp interface formulation expresses the diffusion process in the bulk (solid and liquid) and the evolution of latent heat at the interface. The Gibbs-Thomson relation further accounts for the deviation of melting point due to interface curvature. Simulating the solidification process using a sharp interface approach involves explicit tracking of the interface location and application of boundary condition at the interface.

The diffuse interface approach circumvents this problem of interface tracking by using both continuum and microscopic descriptions of the interface. This is achieved by making the interface to possess a finite width and proposing an order parameter called a phase-field $\phi(x,t)$. A Ginzburg-Landau type free energy functional or an entropy functional is used, from which nonlinear diffusion equations are derived in terms of ϕ and the appropriate thermodynamic variables (such as temperature, composition). In the asymptotic limit, as the interface width approaches zero, the free boundary problems are recovered. The advantage of phase-field method lies in avoiding the interface tracking and thereby providing a numerical convenience for computing complicated microstructures. The treatment of interfaces as diffuse is not new and its origins can be traced back to the early works of van der Waals (1893), who treated density to be continuous across two phases, but his work remained largely unnoticed until it was translated by Rowlinson (1979). The origins of the phase-field equations are based on the works of Cahn and Hilliard (1958) on their work on spinodal decomposition. Here, composition is treated as the order parameter and a Ginzburg-Landau free energy functional was used to derive the diffusion equation. Allen and Cahn (1979) studied the antiphase boundary motion in the context of order-disorder transition using a non-conserved order parameter. While the Cahn-Hilliard equation provides evolution of a conserved parameter (composition), the Allen-Cahn equation describes the evolution of a non-conserved parameter (chemical order). The following sections discuss on the literature prevalent in the area of phase-field models for solidification with emphasis on single phase solidification.

2.2 Phase-field models for solidification

2.2.1 Solidification of a pure substance

The concept of a continuous non-conserved order parameter to distinguish two coexisting phases was first proposed by Fix (1983) and Langer (1986) who adopted "Model C" of Halperin *et al.* (1974). This metholodolgy evolved around the same time as Caginalp and Fife (1986), Collins and Levine (1985) and Umantsev and Roĭtburd (1988), who were individually working on their own diffuse interface models. Penrose and Fife (1990) provided a framework to derive the phase-field equations in a thermodynamically consistent manner from an entropy functional. It was only after Kobayashi (1991) showed the first pictures of dendrites following the methodology of Langer (1986), the path for the modern phase-field started emerging.

The starting point for a phase-field model would be to define an order parameter which varies in time and space. This continuous function takes on definitive values in the bulk phases say ϕ = 1 in solid and ϕ = 0 in liquid. A free energy functional is constructed over a volume V and has the form

$$\mathbf{F} = \int_V \left[f(\phi, T) + \frac{\varepsilon_\phi^2}{2} (\nabla \phi)^2 \right] dV \tag{2.1}$$

where $f(\phi, T)$ is the homogeneous bulk free energy and the gradient energy term represents the increase in free energy due to the formation of an interface. The bulk free energy is constructed by coupling a double well function with the driving potential for the transition. A simple representation of a bulk free energy is,

$$f(\phi, T) = W g(\phi) + L \frac{T - T_M}{T_M} p(\phi) \tag{2.2}$$

where $g(\phi) = \phi^2 (1 - \phi)^2$ is a double-well function which exists only at the interface and $p(\phi) = \phi$ is an interpolating function which takes care of which phase is stable at a grid location. Penrose and Fife (1990, 1993) pointed out that when postulating temperature and phase-field equations together it is necessary to ensure positive local entropy production. This led to the usage of an entropy functional for non-isothermal cases. Wang *et al.* (1993) provided a computationally tractable phase-field model that was derived in a thermodynamically consistent manner using an entropy functional to guarantee positive entropy production.

Generally, phase-field equations defined from the phenomenological approximations are not suitable for obtaining quantitative results. In such cases, quantitative results matching the sharp interface approach are realised in the vanishing limit of interface width. Phase-field simulations independent of numerical parameters could be reproduced only at higher undercoolings when the interface widths are small (Wang and Sekerka (1996*a*,*b*)). The asymptotic analysis by Wheeler *et al.* (1993) to recover the classical free boundary problem at the interface is now referred to as the sharp-interface limit. More importantly, quantitative results are possible when the interface width considered is less than the capillary length (which is usually of the order of 1 nm). This implies that the computational domain needs to be much larger in order to simulate real microstructures. In order to simulate experimentally relevant length scales, the interface width has to be artificially increased. In doing so, quantitative results can be reproduced only if both kinetic and equilibrium properties remain unaffected by the interface width. Karma and Rappel (1996*b*, 1998) proposed the thin interface limit for phase-field models, thereby allowing for quantitative analysis of phase-field methods. The thin interface limit assumes that the interface width is small when compared to the diffusion length and large when compared to the capillary length. This paved the way for quantitative analysis of dendritic growth at low undercoolings by allowing larger interface width. The model also proposed a formulation to determine the phase-field mobility at vanishing kinetics and quantitative simulations for both, two-dimensional as well as three-dimensional dendritic growth (Karma and Rappel (1996*a*); Karma *et al.* (2000)). The formulation is more tedius when considering alloy phase-field models.

2.2.2 Solidification of alloys

Two different approaches in phase-field methodology exist in the treatment of alloys which, Plapp (2011), defines as the "coarse-graining" and "two-phase" approach. The "coarse-graining" approach was proposed initially by Wheeler *et al.* (1992) when they recast the model equations of Kobayashi (1991) to incorporate solute atoms and combining diffusion of the solutes. Their model (WBM model) used a free energy functional. Solidification proceeds due to supersaturation in the melt and the interface is treated as a narrow region in space where all the quantities vary rapidly. Extending on the WBM model and deriving the phase-field and solute diffusion equations from an entropy functional, Warren and Boettinger (1995) were able to predict and

show physically relevant dendritic patterns, including the development, coarsening, and coalescence of secondary and tertiary dendrite arms. More details about WBM model are explicated in Section 3.2. In the "two-phase" approach the interface is treated as a mixture of two phases, each of which is able to retain its bulk properties even in the interface and separate concentration fields are defined for each phase. The model equations are derived from volume averaging method by Ni and Beckermann (1991), where the concentration field is obtained by taking average of the concentrations in each phase, which is the weighted by the phase-field value. This approach was adopted in parallel by Tiaden *et al.* (1998) who improvised on the multiphase field model by Steinbach *et al.* (1996) by defining volume averaged concentration fields. Kim *et al.* (1999) also introduced two separate concentration fields for either of the phases and derived the formulation for the thin interface limit, following the works of Karma and Rappel (1998).

The "coarse graining" approach has a severe limit on the interface thickness as the properties are continuous throughout the interface which is circumvented by decoupling the bulk and the interfacial properties. The addition of two new concentration fields will take away a degree of freedom and hence increases the computational complexity requiring to solve for these values at every point in time and space. Almgren (1999) demonstrated that in the case of unequal thermal conductivities, anomalous effects such as interface stretching, interface diffusion and chemical potential jump at the interface set in. This was taken up by Karma (2001) who proposed the idea of anti-trapping current, a phenomenological current that pushes the solute atoms away from the interface and which was later expounded by Echebarriaa *et al.* (2004). Following the works of Karma (2001), Kim (2007) extended the anti-trapping formulation for multicomponent alloy solidification.

George and Warren (2002) performed parallel 3-D dendritic growth simulations on a Ni-Cu system. Coupling temperature with alloy solidification was treated by Loginova *et al.* (2001), before which temperature was incorporated without having to directly solve the heat equation through the following assumptions: isothermal temperature (solutal growth neglecting latent heat release), frozen temperature gradient (directional solidification - Karma *et al.* (2000); Boettinger and Warren (1999)) and time-dependent but spatially varying uniform temperature (recalescence - Boettinger and Warren (1996)). Loginova *et al.* (2001) treated both solute and thermal diffusion, by using an implicit time stepping schemes, in which at high undercool-

ings thermal undercooling replaces supersaturation as the driving force for growth. Coupling of temperature in the study of free dendritic growth requires that the diffusion boundary layer does not hit the boundaries by using large domain sizes as done by Lan *et al.* (2003). Ramirez *et al.* (2004) extended the anti-trapping current formulation to the non-isothermal modes. They were able to obtain good agreement with 1-D steady state analytical solutions. One important concern in coupling thermal and solutal diffusion is that the diffusivities operate at totally different length scales. Thermal diffusivity is ˜$10^2 - 10^4$ orders of magnitude greater than solutal diffusivity. This constrains the coupling of the diffusivities within a Lewis number (D_T/D) not more than 100. Numerical instabilities are encountered at higher values of Lewis number.

Introducing convection into the melt is a non-trivial task; not only we encounter different time and length scales, but also require sophisticated numerical techniques and adaptive grids to solve the governing equations. Tönhardt and Amberg (1998, 2000) treated both solid and liquid phases as Newtonian fluids with the viscosity of solid much greater than that of liquid. A shear flow in the direction of the growing nucleus significantly altered the 4-fold symmetry, with the directions of growth more favoured in the upstream direction. Beckermann *et al.* (1999) incorporated melt convection phenomenologically in a phase-field model for pure as well as binary alloy solidification. The advective term is absent in the phase-field equation, whereas the Navier-Stokes equation is modified to include a distributed dissipative interfacial drag term to model the non-slip condition at the solid-liquid interface. Anderson *et al.* (2001) derived a phase-field model of a pure substance with convection and analysed the sharp interface limit.

The phase-field models traditionally used phenomenological functions to describe the free energies. The increased availability of Gibbs energy functions (in the form of thermodynamic databases) for engineering materials and powerful software to perform useful calculations using these functions, through the CALPHAD approach, has gained much prominence in recent years. The availability of realistic Gibbs energy functions has allowed the phase-field community to start exploring the possibility of replacing the traditional phenomenological functions with actual descriptions of the systems. The following sections provide a brief overview of the CALPHAD approach and the coupling of various phase-field methods with thermodynamic databases.

2.3 CALPHAD

The CALPHAD (**CAL**culation of **PHA**se **D**iagrams) method, as a branch of thermodynamics, was widely recognised as a powerful computational tool by the pioneering works of Kaufman and Bernstein (1970) and Hillert (1968) and others in the 70s. The foundations of calculation of phase equilibria were first laid out by J. Willard Gibbs. J.J. Van Laar was the first to calculate a series of simple hypothetical binary phase diagrams, using the regular solution model to represent the thermodynamic properties of the phases (Saunders and Miodownik (1998); Kattner (1997); Kaufman and Ågren (2013)). The CALPHAD method is based on the well known thermodynamic principle that a system with a given composition, temperature and pressure will attain the state of equilibrium when it is at the lowest Gibbs energy. The equilibrium state of a system can be calculated through an energy minimization procedure when the Gibbs energy as a function of pressure, temperature and composition is known for individual phases (Ågren (1996)). The CALPHAD method is neither empirical nor fundamental, rather, it provides a way or method to couple experimental and theoretical results in a thermodynamic perspective to render it applicable in much wider context (Kaufman and Ågren (2013)). The treatment itself being not only completely general and extensible, but also theoretically insightful (Kattner (1997)). Phase equilibrium calculations not only provide the information on the number of phases present and their composition, but also the numerics on enthalpy contents, temperature, concentration dependence on phase boundaries in the context of both macroscopic and microscopic coupling so much so that, in recent years, it has been aptly renamed as "Computational Thermodynamics". This reflects the fact that phase diagram forms a subset of the gamut of information that can be accessed by these calculations.

The Gibbs energy G(p,T,x..,ζ...) when available as a function of pressure, temperature, composition and internal variables, a large number of thermodynamic quantities and phase diagram information can be obtained. Metastable equilibrium calculations can also be performed to obtain information on the driving forces for nucleation, which on coupling with kinetics may be used to predict the evolution of various phases in a system.

The required Gibbs energy expressions of the various phases are determined by fitting the parameters in a thermodynamic model using experimental information on thermodynamic properties and phase equilibria, thereby providing a robust and a consistent way for constructing

temperature and composition dependent Gibbs energy expressions for multicomponent systems. There are different thermodynamic models available to describe the temperature and composition dependence of Gibbs energy of different phases. For example, a random substitutional solution model wherein an atom is free to mix over all lattice sites or a sublattice model where an atom is restricted in its mixing behaviour to distinct sublattices. The elemental Gibbs energies are combined together to form Gibbs energies of binaries through appropriate models. The same can be extended to ternaries and beyond, wherein the deviations from the model are treated using higher order interaction parameters. Due to this standardized approach almost spanning for more than four decades, a lot of effort by different researchers have been combined to form thermodynamic databases for complex multicomponent alloys.

Computer programs which were initially written by individuals have now become a collective effort in packages like Thermo-Calc (Sundman *et al.* (1985); Sundman (1991)), PANDAT (Chang *et al.* (2001); Chen *et al.* (2002, 2003); Cao *et al.* (2009)), FactSage (Bale *et al.* (2002)), MTDATA (Davies *et al.* (2002)), which are matured packages in terms of optimization of the parameters in the Gibbs energy expressions used in the calculation of phase diagrams. More such tools and packages have been collated and referenced in a special issue of CALPHAD titled "Tools for Computational Thermodynamics" (Hallstedt and Liu (2009); van de Walle (2009); Shobu (2009); Tomiska (2009); Cheynet *et al.* (2009); Petaev (2009); Bhattacharjee and Paolini (2009))

Phase equilibrium calculation is initiated by the minimization of the Gibbs energy of the phases that take part in this equilibrium given by,

$$G = \sum_{i=1}^{N} n_i G_i^{\phi} = \text{minimum} \tag{2.3}$$

where n_i is the number of moles and G_i^{ϕ} is the Gibbs energy of phase i. Gibbs energy minimization and calculation of phase equilibria is not a trivial procedure. A mathematical expression for G, rooted in statistical mechanics, contains a large number of unknown parameters that need to be determined. These parameters are then adjusted until the choice of function G for that particular model is capable of a sound representation of the information chosen. The procedure can be summarized as follows: (i) selecting the information to be used in the optimization, (ii) selecting the thermodynamic models that best represents each of the phases of a system,

(iii) optimizing the model parameters to achieve the best representation of the selected information and (iv) compilation of the parameters into a database, the whole procedure usually being termed as CALPHAD assessment. The information that is chosen to be represented can be experimental measurements and or estimates from theoretical calculations. The collated information, through the above mentioned cycle, is processed iteratively. In some cases, a more advanced thermodynamic model may be required to represent a phase or in some others, the obtained information maybe less reliable in which cases it is necessary to give a less weightage. The thermodynamic models used, vary for different phases in a system and to delve into those models is much exhaustive in the context of the present work. However, it is insightful to refer to Kattner (1997); Chang *et al.* (2004); Kattner and Campbell (2009); Liu (2009), on their reviews on the existing thermodynamic models and the context of their usage.

2.4 Coupling phase-field models with thermodynamic databases

The coupling of phase-field models with realistic Gibbs energy functions, though still in the infancy, researchers have tried to harness the huge potential it has, while simulating microstructure evolution of important engineering alloys (Kitashima (2008)). This has lead to quantitative deductions being made in the contexst of real alloy systems. There are commercial packages available, such as MICRESS® (ACCESS e.V. (1995–)), which have coupled phase-field code with Thermo-Calc, a versatile software for thermodynamic calculations in multicomponent systems. MICRESS®, for simulating microstructure evolution, through the TQ-Interface of Thermo-Calc can access the required thermodynamic quantities such as chemical potential, phase composition, etc. The other way is to directly use the Gibbs energy functions of the relevant phases in the phase-field program. A brief review of research that have coupled phase-field models with thermodynamic databases only in the context of solidification are presented here. Variations in the approach depends on the phase-field model used and the thermodynamic models used for the alloy system considered.

Grafe *et al.* (2000*a*) were the first to extend a multiphase field model by Steinbach *et al.* (1996) and Tiaden *et al.* (1998) to incorporate multicomponents. Their modified multiphase multi-

component diffusion equation is given by,

$$c_t(x,t) = -\nabla \left(\sum_{i=1}^{n} \phi_i \sum_{m=1}^{p-1} \mathbf{D} \nabla c_i^m \right) \tag{2.4}$$

where the reduced diffusivity matrix (**D**) was obtained through coupling with DICTRA, a tool for multicomponent diffusion simulation, which in turn uses Thermo-Calc. Required data can be obtained by interfacing the phase-field code through an API called TQ-API where specific routines can be called for accessing the thermodynamic data. Simulations were done to study the slow directional solidification of Ni-base superalloys using this methodology (Grafe *et al.* (2000*b*)).

Multicomponent alloy solidification with interstitial solute diffusion was first proposed by Cha *et al.* (2001) by introducing the number of moles rather than mole fraction as the concentration variable.

$$\begin{aligned} \sum_{i \in S} c_i &= 1/\Delta \\ \sum_{i \in I} c_i + c_{IE} &= \text{constant} \end{aligned}$$

where, S and I are substitutional and interstitial elements, respectively, c_{IE} is the number of moles of empty interstitial sites. The partial molar volume of the both substitutional and interstitial elements is assumed constant and given by Δ. The model was unable to be scaled-up to large computational domains as the use of a single composition variable across the interface causes stringent restrictions on the interface width. This was subsequently addressed by Cha *et al.* (2005), following the equal chemical potential approach of Kim *et al.* (1999). All the thermodynamic quantities were accessed through the TQ-API.

Kobayashi *et al.* (2003) extended the binary alloy model of Kim *et al.* (1999) to ternary alloys using an extended regular solution approximation of Gibbs energies for solid and liquid. The model was tested with Al-Si-Mg ternary for the vanishing interface kinetics condition at the thin interface limit. The model was also able to reproduce well the equilibrium interfacial compositions, the Gibbs-Thomson effect and the interface anisotropy. The Gibbs energy expressions for the solid and liquid were incorporated directly into the phase-field code and 2-D dendritic growth was simulated for lower undercoolings.

Qin and Wallach (2003) proposed a non-isothermal phase-field model to treat multicomponent alloy solidification. They derived their model in a thermodynamically consistent way using a volume averaging method to describe the compositions in the solid and liquid phases. They also were able to couple their model with MTDATA to study Al-Si alloys by extracting the thermodynamic information at each grid point in the computational domain.

Qin *et al.* (2005) further extended their study to include multiple phase fields by following the approach of Steinbach *et al.* (1996). This is achieved by introducing the constraint, $\sum_{\alpha}^{N} \phi_{\alpha}(x,t) = 1$. An Al-Si-Cu-Fe multicomponent system was simulated by considering α-Al, Si, $Cu(Al)_2$ θ and δ-AlFeSi phases. The results obtained were in perfect agreement with experiments.

Eiken *et al.* (2006) reformulated the multiphase field model of Steinbach *et al.* (1996) and Tiaden *et al.* (1998) in a thermodynamically consistent manner to include several components. The Gibbs energies were taken from assessed thermodynamic databases. In order to obtain the thermodynamic data at every grid point, the thermodynamic calculations were done at periodic intervals. This interval was determined by a specified maximum temperature deviation or a significant change in the number of phases and the data extrapolated between the quasi-equilibrium conditions. Böttger *et al.* (2006) studied the equiaxed solidification of Mg-Al alloy systems with hexagonal crystal symmetry in both 2-D and 3-D by extending the multiphase field model developed by Eiken *et al.* (2006). Thermodynamic quantities such as Gibbs energies of the phases and chemical potentials were accessed at every grid point directly using TQ-Interface of Thermo-Calc. This method was adopted to study grain refinement on AZ31 (Mg-Al-Zn) and Al-Cu-Si piston alloys. Nucleation of random seeds in the matrix required accessing thermodynamic information at each nucleation site, obtained through thermodynamic coupling.

Zhang *et al.* (2006) worked on the lines of Kim *et al.* (1999) by extending the phase-field equations to incorporate several components and the multicomponent diffusion equation was derived from solute diffusion equation of Cha *et al.* (2001). They studied only the effect of substitutional elements and the thermodynamic and diffusion data was obtained by coupling their phase-field code with Thermo-Calc using the TQ-Interface. Zhang *et al.* (2010) extended the muticomponent phase-field model which included anti-trapping current in the solute diffusion equation following Kim (2007). Dendritic solidification was studied with and without the

effect of solute interactions in a Al-Cu-Si ternary alloy. Assuming only the diagonal terms of the diffusivity matrix, no solute interaction condition was obtained.

Steinbach *et al.* (2007) gives a concise overview of thermodynamic coupling and the steps required to circumvent the issues arising due to the coupling. The multiphase multicomponent model by Eiken *et al.* (2006) was used to highlight issues such as miscibility gaps, quasi equilibrium construction of solute partitioning at the interphases and calculating the offset driving force for interface motion. Their coupling ensured the reliability of extension of Gibbs energy functions to metastable regions and also the chemical order/disorder transformation that occurs when stable γ' gets ordered from the γ phase in Ni-base superalloys. Wen-Yuan *et al.* (2008) extended the approach of Kim *et al.* (1999) to study non-isothermal dendritic growth in Al-Cu system in the Al-rich region. Thermodynamic coupling was effected by incorporating the Gibbs energy expressions. Simulations were validated against Ivantsov solution for various undercoolings. Dendritic solidification in Ni-base superalloys, both in 2-D and 3-D, were studied by Warnken *et al.* (2009) using the model proposed by Eiken *et al.* (2006). Fries *et al.* (2009) analysed the grain refinement in Mg alloys and also studied the effect of controlled perturbation during the peritectic solidification of a 13 components steel, using the model of Eiken *et al.* (2006) which has been incorporated in the microstructure calculation software MICRESS. The databases used were the TCFF5 and MOB2 from Thermo-Calc AB. Kundin and Siquieri (2011) extended on the model of Steinbach *et al.* (1996) by incorporating into their multiphase field model an anti-trapping flux. The bulk free energy functions for the individual phases are taken from a CALPHAD formulation and investigated the peritectic solidification of Al-Ni alloys.

2.5 Dendrite growth theories

Characterized by their ubiquitous solidification morphology, dendrites remain an important aspect in most material processing technologies, appearing commonly in as-solidified alloys, in all castings and in fusion weldments. The metastability of supercooled melts allows the initiation of dendritic growth by either spontaneous nucleation at a favourable site within the melt or at the container wall. Dendrites form as a result of instability of the interface as elucidated in the works of (Pelcé (1988); Tiller (1991); Kurz and Fisher (1998); Davis (2001); Dantzig and Rappaz (2009); Glicksman (2010)). In presence of high anisotropy, structure develops a

dendritic morphology. After an initial transition, the free growing dendrites reach a steady state during which the dendrite tip advances with a steady tip velocity and radius. The shape of the growing crystal is a result of the various physical forces, such as diffusive, capillary and kinetic. Kinetic and transport theories were developed to predict the solidification characteristics of dendrites as a consequence of their widespread commercial importance and hence understanding their physics is of paramount importance. Following is an outline of the various theoretical treatments for predicting dendritic behaviour using fundamental principles.

2.5.1 Growth of a needle crystal

The discussion here focuses on growth beyond an initial instability, where the preferred growth directions dominate, eventually forming fully developed dendrites. Theories outlined here are concerned with the dynamics of the dendrite tip as measured by the undercooling, tip radius and growth rate. Ivantsov (1947) was the first to provide a mathematical analysis for the temperature distribution in the melt surrounding a dendrite. He analysed the growth in the limit where both capillary and kinetic effects at the interface were neglected, which was further generalized by Horvay and Cahn (1961). The principal assumptions were: (i) the solid is a parabola in 2-D or a paraboloid of revolution in 3-D, growing in a shape preserving manner with a radius R^* and velocity v^* (ii) temperature field around a dendrite is isothermal (iii) solid-liquid interface temperature is the equilibrium temperature neglecting the effects of curvature, leading to the following expressions,

$$\Delta_{2D} = \sqrt{\pi P}\exp(P)\mathrm{erfc}(\sqrt{P}) \tag{2.5}$$

$$\Delta_{3D} = P\exp(P)\mathrm{E}_1(P) \tag{2.6}$$

Here the dimensionless melt undercooling Δ is given as a function of the Peclet number, which is related to the tip velocity and radius as $P = \frac{VR}{2\alpha}$, α being the thermal diffusivity. This analytical solution has an inherent problem *i.e.* it does not yield unique values for R^* and velocity v^*. There exists, as a result, an entire family of solutions for a given undercooling ΔT since $R^*v^* = constant$. A dendrite, whereas in reality, always selects a particular tip speed and radius for a particular undercooling. An important parameter which did not make its way into the solution is the surface energy leading to the Gibbs-Thomson condition not being satisfied

at the tip. The treatment with surface energy causes a shape correction at the tip making it to deviate from a paraboloid. Temkin (1960) applied the Gibbs-Thomson condition at the tip, resulting in a reduced temperature at the tip, but in all cases keeping the basic paraboloid shape. This analysis at the reduced undercooling at the tip ($T^* - T_\infty$) leads to a minimum tip radius $R^*_{min} = \frac{2\Gamma_{sl}}{\Delta T}$ in the limit where the velocity goes to zero at $T^* = T_\infty$. Temkin's corrections were thought of as an "extremum condition" until proved otherwise by the large offset to experiments. Nash and Glicksman (1974) developed an integral solution using Green's function.

2.5.2 Morphological stability criterion

The modern theories of dendritic growth solved the problem of tip selection radius and velocity by the concept of marginal stability. This is derived from the stability analysis of a solid-liquid interface growing at a steady state into an undercooled melt. The overall methodology of deriving the stability criterion is basically a perturbation analysis of an interface with conditions known apriori. This mathematical technique introduces the effect of time dependency into the shape of the interface which, in normal cases is assumed to move at a constant velocity. The analytical theories on dendritic growth suggest that the growth is influenced by the interactions of thermal/ solute diffusion, capillarity and strength of crystalline anisotropy (Beltran-Sanchez and Stefanescu (2003)). This perturbation analysis, referred to as the marginal stability, solves the problem of uniquely defining the tip radius and velocity at steady state by introducing a stability parameter σ^*. The marginal stability analysis, though has fundamental flaws, predictive results can be obtained which, conform close to experimental evidence. A more rigorous treatment of predicting steady state dendrite growth is the microscopic solvability theory (Ben-Jacob *et al.* (1984); Kessler *et al.* (1988)). This theory, in addition to the marginal stability analysis, establishes σ^* as a function of capillary anisotropy. The stability parameter is given by,

$$\sigma^* = \frac{2d_0 D_L}{VR^2} \tag{2.7}$$

where R and V are the tip radius and velocity, respectively. The capillary length d_0 is given by,

$$d_0 = \frac{\Gamma}{m_L \Delta c} \tag{2.8}$$

Γ is the Gibbs-Thomson co-efficient, m_L the liquidus slope and Δc, the magnitude of solute rejection at the interface. This parameter is not constant even under isotropic conditions. This is because of the liquid composition, which varies during solidification due to the non-equilibrium kinetic effects. An interesting feature of phase-field models is that this stability parameter is contained within these models.

2.6 Semi-solid processing

The overall field of semisolid processing broadly falls into two categories namely "Rheocasting" and "Thixocasting". The former starts with a liquid metal and involves agitation during partial solidification followed by forming. The latter begins with a solid metal of suitable structure and involves heating to the desired solid fraction and forming. The shearing of the dendrite morphology during agitation of the melt invariably leads to the fragmentation of the dendrites followed by subsequent coarsening into more globular form. The mechanism of semi-solid processing can be understood by splitting it into two regimes *viz.*, the submicron level, where the fragmentation of the dendrites maybe thought of solely due to solutal and on the macro level where the rheological behaviour of the melt and the convective forces with thermal gradients takes prominence. The most important characteristic of SSP is that the solidification microstructure changes markedly, from dendritic growth under conventional processing conditions to non-dendritic or globular growth.

In this study, we try to understand the dendrite fragmentation and evolution of a fine globular microstructure that takes place during a semi-solid processing technique such as electromagnetic stirring in a rheocasting setup through phase-field modelling. The physical phenomena that take place during the process of dendrite fragmentation are many. There are several experimental studies, *insitu* visualization and simulation studies have come into the open literature in the recent past. One of the significant effects that details dendritic fragmentation and coarsening is an effect of surface energy reduction, involving remelting and resolidification (Rettenmayr (2009)). The morphological changes that are brought about during such processes would be that the secondary arms can melt back radially, axially, melt off at its roots or coalesce (Kahlweit (1968); Chien and Kattamis (1970)). The melting of regions of high curvature should be accompanied by the coarsening of low curvature regions. In most cases, coarsening is seen when

one or more secondary arms are outgrown or their growth inhibited by their neighbours. The coarsening effect due to solidification is immediately affected by surface energy driven melting. This causes the entire dendrite to melt back radially and axially. Coarsening increases the diffusion distances during solidification in the liquid and solid phases, which mostly leads to a inhomogeneous microstructure due to the restriction of back diffusion in the solid. This coupled melting and resolidification effect generally ends up with a homogeneous microstructure. In multicomponent alloys, the number of solute elements present might also have a significant effect on the overall growth morphology due to different diffusivities of the individual elements present (Roosz *et al.* (1986); Dutta and Rettenmayr (2000)). In general, during semi-solid processing, both remelting and mechanical breakdown by stirring are considered to play a major role in dendritic fragmentation and coarsening.

Early studies done on fragmentation have indicated that the remelting of the dendritic arms at its roots are caused by thermal gradients and solute enrichment at the roots, confirmed by experiments on NH_4Cl, a transparent material (Christenson *et al.* (1989*a*,*b*)). Jackson *et al.* (1966) explained that remelting can be caused by partial remelting of the dendrites because of the interaction of the two diffusion fluxes, namely thermal and solutal fluxes. As solute gets enriched near the roots, the solidus temperature is increased leading to necking at the branches. Kattamis *et al.* (1967) explained that remelting can possibly occur when there is a solutal influx towards regions of high curvature *i.e.* the neck from regions of large radius/ low curvature that are solute-rich whereby the spherodisation of individual particles after remelting will happen due to atomic flux from areas of high curvature to areas of low curvature (Fan (2002), Atkinson and Liu (2008)). It is also proved that the effect of shear forces cannot be neglected when considering forced convection in the mushy zone (Smith *et al.* (1991); Vives (1990); Paradies *et al.* (1997)).

Studies on electomagnetically levitated Ni-Cu droplets performed at different undercoolings provide insights, that while constitutional melting is responsible at lower undercoolings, mechanical shearing was suggested to play a major role at higher undercoolings (Li *et al.* (1998); Herlach *et al.* (2001)).

Hellawell *et al.* (1997) and Liu *et al.* (2002) showed that deceleration of the solidifying front can also cause stagnation of the solute causing the dendrite arms to fragment and detach. Pilling and Hellawell (1996) proved that without forced convection fragmentation is unlikely to occur, but

might cause bending of the arms. Studies on Al-Cu with high resolution synchrotron by Ruvalcaba *et al.* (2007) have shown that an incoming flow of solute-rich liquid into the mushy zone, solute rejection due to local growth and subsequent pile-up at the roots can cause significant fragmentation. The buoyant transport of the solvent to the front can also cause fragmentation (Mathiesen *et al.* (2006)).

This along with convection in the melt leads to the formation of globular crystals by the thermal cycling between the solidus and the liquidus (Tao *et al.* (2006)). These fragments get carried away by the flow and get piled up, further enhancing the fragmentation process. One more significant aspect to be considered in melting and solidification of alloy is the large composition differences brought about by thermal gradients. This causes an excess solute pileup at the solid-liquid interface resulting in a solute driven melting or solutal remelting (Rettenmayr *et al.* (1997), Dutta and Rettenmayr (2000), Rettenmayr *et al.* (2001), Gamsjäger *et al.* (2007)). Significant changes in the growth morphologies are brought about by changing the amplitude and the duration of the thermal fluctuations.

2.7 Summary

Published literature on simulation of dendritic growth using phase-field method and experimental studies on semi-solid processing is extensive. However, there are every few studies on the application of phase-field method to semi-solid processing. Such simulations would throw light into the physical phenomena that take place during the morphological transition. Computational studies that aid in identification of morphological artefacts and numerical parameter windows that avoid them are also limited. Quantitative characterization of morphological transition also needs to be explored to add value to the microstructure simulations using the phase-field approach.

CHAPTER 3

MODEL FORMULATION

3.1 Sharp interface formulation

The classical sharp interface model for isothermal solidification of an alloy is described by solute diffusion in each of the bulk phases and two corresponding boundary conditions at the solid-liquid interface *viz.*, flux conservation and Gibbs-Thomson condition.

$$\frac{\partial c_{\mathrm{S}}}{\partial t} = D_{\mathrm{S}}\nabla^2 c_{\mathrm{S}} \tag{3.1}$$

$$\frac{\partial c_{\mathrm{L}}}{\partial t} = D_{\mathrm{L}}\nabla^2 c_{\mathrm{L}} \tag{3.2}$$

$$\frac{\partial f^{\mathrm{S}}}{\partial c_{\mathrm{S}}}(c_{\mathrm{S}}^i) = \frac{\partial f^{\mathrm{L}}}{\partial c_{\mathrm{L}}}(c_{\mathrm{L}}^i) = \frac{\partial f^e}{\partial c} - \frac{\Delta H_f}{T_m(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)}\beta V - \frac{2\sigma\kappa}{(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)} \tag{3.3}$$

$$(c_{\mathrm{L}}^i - c_{\mathrm{S}}^i)V = D_{\mathrm{S}}\frac{\partial c_{\mathrm{S}}}{\partial n} - D_{\mathrm{L}}\frac{\partial c_{\mathrm{L}}}{\partial n} \tag{3.4}$$

where c_{S} and c_{L} are the compositions of solid and liquid, D_{S} and D_{L}, the diffusivities of solute in solid and liquid, c_{S}^i, c_{L}^i, the compositions at the solid and liquid sides of the interface, respectively, c_{S}^e,c_{L}^e, the equilibrium compositions of solid and liquid at a given temperature. f^{S}, f^{L} are the free energies in solid and liquid, respectively and $\frac{\partial f^e}{\partial c}$ is the equilibrium chemical potential.

3.2 Models used

Alloy solidification presents a rather complex scenario than that of pure components since phase transformation kinetics involves both heat and mass transfer. However, since these two processes are occurring at two different time scales, in most cases, it is sufficient enough to consider the slower of the two, namely, mass transfer. Heat transfer, taking place on a faster scale, (nearly two orders of magnitude larger than mass transfer), can usually be treated as frozen or isothermal as temperature evolves so rapidly compared to solute redistribution, that

the process can be assumed to take place in a quasi-steady state. In most cases, the ratio of thermal (D_T) and solutal (D) diffusivities (Lewis number=D_T/D) ranges between 10^4 to 10^2. The phase-field models outlined below are isothermal alloy solidification models. For isothermal cases. the total free energy which must decrease during any process is given by,

$$F = \int_V \left[f(\phi, c, T) + \frac{\varepsilon_c^2}{2}(\nabla c)^2 + \frac{\varepsilon_\phi^2}{2}(\nabla \phi)^2 \right] dV \tag{3.5}$$

where $f(\phi, c, T)$ is the homogeneous free energy density which is a function of the order parameter ϕ, composition c and temperature T. ε_ϕ and ε_c are the gradient energy coefficients for phase-field and composition, respectively.

In the case of equilibrium, the variational derivative of F, should satisfy the condition,

$$\frac{\delta F}{\delta \phi} = \frac{\partial f}{\partial \phi} - \varepsilon_\phi^2 \nabla^2 \phi = 0 \tag{3.6}$$

$$\frac{\delta F}{\delta c} = \frac{\partial f}{\partial c} - \varepsilon_c^2 \nabla^2 c = \text{constant} \tag{3.7}$$

For time dependent cases, the simplest equations that guarantee a decrease in the total free energy with time are

$$\frac{\partial \phi}{\partial t} = -M_\phi \left(\frac{\partial f}{\partial \phi} - \varepsilon_\phi^2 \nabla^2 \phi \right) \tag{3.8}$$

$$\frac{\partial c}{\partial t} = \nabla \cdot \left(M_c c(1-c) \nabla \left(\frac{\partial f}{\partial c} - \varepsilon_c^2 \nabla^2 c \right) \right) \tag{3.9}$$

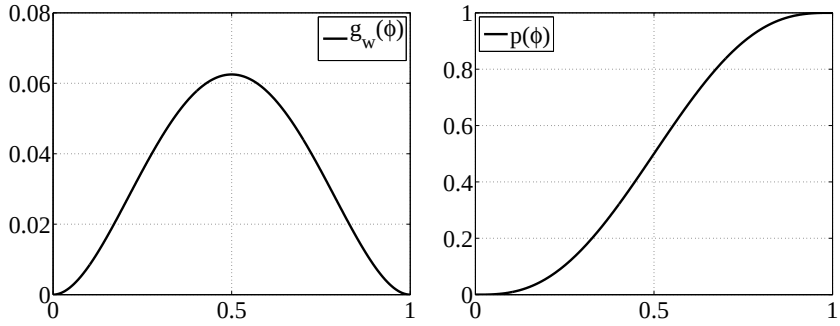
In this formulation, the two equations are coupled together through the energy function $f(\phi, c, T)$. A simple approach for a binary alloy single phase solidification would involve constructing this function from the free energies of the pure elements for the phases present, namely, $f_{\mathrm{A}}^{\mathrm{L}}(T)$, $f_{\mathrm{A}}^{\mathrm{S}}(T)$, $f_{\mathrm{B}}^{\mathrm{S}}(T)$, $f_{\mathrm{B}}^{\mathrm{L}}(T)$ and extending it to the alloy case.

$$f_{\mathrm{A}}(\phi, T) = (1 - p(\phi)) f_{\mathrm{A}}^{\mathrm{L}}(T) + p(\phi) f_{\mathrm{A}}^{\mathrm{S}}(T) + W_{\mathrm{A}} g_w(\phi) \tag{3.10}$$

$$f_{\mathrm{A}}(\phi, T) = f_{\mathrm{A}}^{\mathrm{L}}(T) + p(\phi)(f_{\mathrm{A}}^{\mathrm{S}}(T) - f_{\mathrm{A}}^{\mathrm{L}}(T)) + W_{\mathrm{A}} g_w(\phi) \tag{3.11}$$

$$f_{\mathrm{B}}(\phi, T) = f_{\mathrm{B}}^{\mathrm{L}}(T) + p(\phi)(f_{\mathrm{B}}^{\mathrm{S}}(T) - f_{\mathrm{B}}^{\mathrm{L}}(T)) + W_{\mathrm{B}} g_w(\phi) \tag{3.12}$$

The free energy density uses two functions: a double well function $g_w(\phi) = \phi^2(1-\phi)^2$ and



(a) $g_w(\phi) = \phi^2(1-\phi)^2$ (b) $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$

Figure 3.1: Functional forms of $g_w(\phi)$ and $p(\phi)$

an interpolating function $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ which can be considered as modulating the free energy between two phases being modelled. The functions are so chosen that when ϕ = 0 and 1, $p(\phi)$ and $g_w(\phi)$ goes to 0 ensuring that $\dfrac{\partial f}{\partial \phi} = 0$. The functions themselves are valid only in the interface whose forms are shown in Fig.3.1. Throughout this work, ϕ=0 represents liquid and ϕ=1 represents solid.

The term $(f_{\mathrm{A}}^{\mathrm{S}}(T) - f_{\mathrm{A}}^{\mathrm{L}}(T))$ is the driving force ΔG for the transformation and for small ΔT it can be approximated as $\Delta f = \dfrac{L_{\mathrm{A}}\Delta T}{T_{m_{\mathrm{A}}}}$, which upon substitution, taking $f_{\mathrm{A}}^{\mathrm{L}}(T) = 0$ as the reference state yields,

$$f_{\mathrm{A}}(\phi, T) = p(\phi)\left(\frac{L_{\mathrm{A}}(T - T_{m_{\mathrm{A}}})}{T_{m_{\mathrm{A}}}}\right) + W_{\mathrm{A}} g_w(\phi) \tag{3.13}$$

The model can be extended to an alloy by the addition of the contribution of the solute atoms (B) to the free energy density. The simplest approach to enable the extension from pure elements is to define a continuous composition field *c* over the interface and the free energy density is extrapolated as a mixture in composition of the energy densities of the pure elements.

$$f(\phi, c, T) = (1-c) f_{\mathrm{A}}(\phi, T) + c f_{\mathrm{B}}(\phi, T) + RT(c \ln c + (1-c)\ln(1-c)) \tag{3.14}$$

where c is the concentration of the solute. A second method to form $f(\phi, c, T)$ is to obtain the free energy functions of the respective phases, namely, $f^{\mathrm{S}}(c, T)$ and $f^{\mathrm{L}}(c, T)$, through a CALPHAD description which would be more convenient since the free energy descriptions

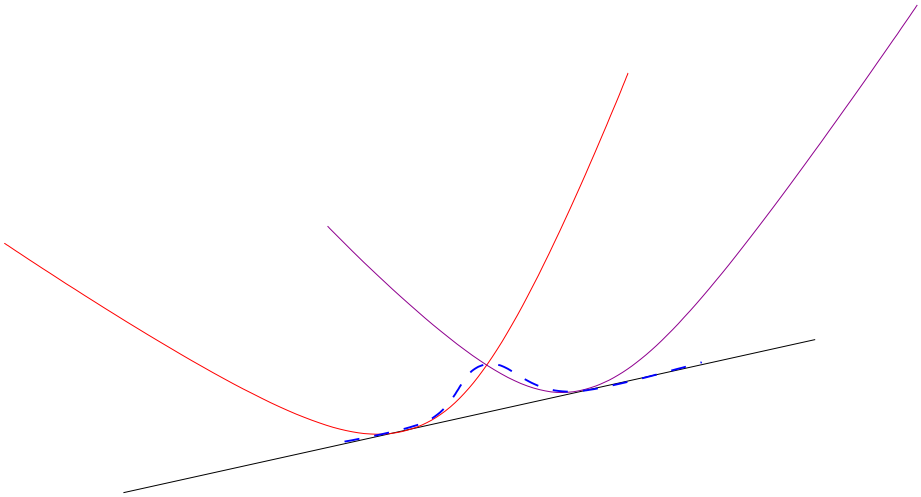


Figure 3.2: Excess interfacial energy contribution arising due to the contribution of chemical free energy as depicted by the dashed blue line. The solid curves represent the homogeneous free energy densities. The straight black line is the common tangent. The dashed blue line is taken across the interface. From Kim *et al.* (1999).

themselves are functions of T.

$$f(\phi, c, T) = W g_w(\phi) + (1 - p(\phi)) f^{\mathrm{L}}(c, T) + p(\phi) f^{\mathrm{S}}(c, T) \tag{3.15}$$

The above formulation though simplistic in its approach poses serious issues in numerical calculations. The free energy density $(1 - p(\phi)) f^{\mathrm{L}}(c, T) + p(\phi) f^{\mathrm{S}}(c, T)$ in the interfacial region lies on the dash blue line shown in Fig.3.2. This extra potential arises in addition to the double well potential $W g_w(\phi)$, which maybe considered negligible with $W g_w(\phi)$ in case of a sharp interface or an alloy with a small Δc. An increase in the interface width or a higher Δc will make the excess potential significant and hence cannot be ignored.

Kim *et al.* (1999) overcame this problem by decoupling the bulk terms from the gradients. An equal chemical potential condition was employed across the interface by introducing separate concentration fields in each phase. This addition of two new concentration fields pins the free energy density $(1 - p(\phi)) f^{\mathrm{L}}(c_{\mathrm{L}}, T) + p(\phi) f^{\mathrm{S}}(c_{\mathrm{S}}, T)$ to lie on the common tangent.

$$f(\phi, c, T) = W g_w(\phi) + (1 - p(\phi)) f^{\mathrm{L}}(c_{\mathrm{L}}, T) + p(\phi) f^{\mathrm{S}}(c_{\mathrm{S}}, T) \tag{3.16}$$

This formulation treats the interface as a mixture of two phases whose compositions reflect the compositions in the bulk. This representation of free energy, introduces two new concentration

fields which have to be solved iteratively at each point in the interface.

$$c = h(\phi)c_{\mathrm{S}} + (1 - h(\phi))c_{\mathrm{L}} \tag{3.17}$$

$$\frac{\partial f^{\mathrm{S}}}{\partial c_{\mathrm{S}}}(c_{\mathrm{S}}(\mathrm{x}, t)) = \frac{\partial f^{\mathrm{L}}}{\partial c_{\mathrm{L}}}(c_{\mathrm{L}}(\mathrm{x}, t)) \tag{3.18}$$

This leads to redefining the driving force in the phase-field equation w.r.t to c_{S} and c_{L} as

$$f^{\mathrm{L}}(c_{\mathrm{L}}) - f^{\mathrm{S}}(c_{\mathrm{S}}) - (c_{\mathrm{L}} - c_{\mathrm{S}}) f^{\mathrm{L}}_{c_{\mathrm{L}}}(c_{\mathrm{L}})$$

This is consistent with the thermodynamic driving force deduced from a parallel tangent construction.

$$\frac{1}{M_\phi}\frac{\partial \phi}{\partial t} = \varepsilon^2 \nabla^2 \phi - W \frac{dg_w(\phi)}{d\phi} - \frac{dp(\phi)}{d\phi}\left(f^{\mathrm{S}} - f^{\mathrm{L}} + \mu(c_{\mathrm{S}} - c_{\mathrm{L}})\right) \tag{3.19}$$

Based on the works of Tiaden *et al.* (1998), the concentration, $c(\mathrm{x}, t)$, of a component in a phase of a binary alloy is defined by a simpler partition relation,

$$c(\mathrm{x}, t) = \phi_{\mathrm{S}} c_{\mathrm{S}} + \phi_{\mathrm{L}} c_{\mathrm{L}} \tag{3.20}$$

The solute diffusion equation is derived by writing the conservation equation in terms of phase-field weighted values for the average concentration as,

$$\frac{\partial c}{\partial t} = \nabla\left[h(\phi) D_{\mathrm{S}} \nabla c_{\mathrm{S}}\right] + \nabla\left[(1 - h(\phi)) D_{\mathrm{L}} \nabla c_{\mathrm{L}}\right] \tag{3.21}$$

3.3 Quantitative phase-field simulations for alloy development

Although, simulating the evolution of complex morphologies is quite straightforward with this method, making quantitative predictions on experimentally relevant length and time scales has remained a major challenge. In such cases, the width of the diffuse interface δ and the characteristic time scale τ, being microscopic, will not be able to resolve a typical experimental length/ time scales even with efficient algorithms. In the light of the above, the only way to proceed with the simulations would be increase the interface width and the time scale of the simulations. But the bloating of the diffuse width, though sounding computationally feasible,

has its own inherent disadvantages.

3.3.1 Interface thickness dependent non-equilibrium effects

In the light of the above models described *viz.*, Wheeler *et al.* (1992) (WBM) and Kim *et al.* (1999) (KKS), it was shown that the thin interface limit of a phase-field model will provide sufficient freedom to choose the interface width δ arbitrarily large compared to the capillarity to simulate microstructures at larger length scales. In general, microstructural evolution is generally accompanied by nonequilibrium effects at the interface, which would be negligibly small in the context of microscopic thickness. However, these effects are artificially magnified when a mesoscopic thickness is considered, which may dominate the capillary effects. This, in turn, alters the large scale pattern evolution. To develop a quantitative phase-field model for modelling solidification at low velocity, it is necessary to have a judicious choice of parameters which will avoid these non-physical effects at the interface.

Karma and Rappel (1998), incorporated in their model the thin interface limit, which assumes that the interface width is small compared to the scale of the microstructure, thereby recovering the Gibbs-Thomson condition at the interface. However, in the case of alloys, a major source of difficulty arises in the form of solute diffusion, which is more slower in solid than in liquid. Using a mesoscopic interface width with an asymmetrical diffusion model will lead to artificially enhanced nonequilibrium effects at the interface, which were first identified by Almgren (1999) and Karma (2001) for solidification of pure metals. In the case of solidification, a mesoscopically thick moving interface tend to posses these effects: (i) solute diffusion/ surface diffusion along the arc length of the interface, (ii) interface stretching due to the moving curved interface thereby leading to a modification of the mass conservation equation at the interface and (iii) a discontinuity of the chemical potential across the interface leading to solute trapping (Echebarriaa *et al.* (2004)). The magnitude of these effects scales up with the interface width. At lower undercoolings, these effects become significantly large as δ used is generally orders of magnitude larger than the actual width.

The first two of these effects can be nullified by a nonvariational approach which can be more computationally efficient than a strictly variational formulation (Karma and Rappel (1998)). In general, a nonvariational approach involves using different interpolation functions for the

evolution equations. $p(\phi)$ for the phase-field, $s(\phi)$ for the concentration and $h(\phi)$ for the diffusion equation, thereby decoupling the phase-field equation from the diffusion equation. The third effect which is the anomalous jump in the chemical potential (solute trapping) during low speed solidification. This is avoided by using a anti-trapping current, an additional solute current introduced, which pushes the solute out of the solidifying material. The following section discusses the derivation of the anti-trapping current using the model put forth by Kim (2007), but reduced to a binary.

3.3.2 Anti-trapping current formulation

A one-sided diffusion model with the anti-trapping current formulation for multicomponent alloys by Kim (2007) was chosen for the study. The present work largely focusses on binary alloy systems. Although, Kim (2007) had derived the anti-trapping formulation in the context of multicomponents, it was deemed necessary to elucidate the formulation for the case of a binary alloy. The following section focusses on the derivation of the anti-trapping current in the case of a two component mixture.

In alloy solidification, the solid diffusivity is considerably lower than in the liquid. As a result, an increase in the interface thickness, generally increases the solute trapping effect, the magnitude of which is proportional to the interface thickness. In order to restore local equilibrium at the interface, an additional solute current is introduced that pushes out the trapped solute, given by,

$$\boldsymbol{j_{at}} = A\delta(c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e})\hat{n}\frac{\partial \phi}{\partial t} \tag{3.22}$$

where δ is related to the interface thickness, $\hat{n} = \nabla\phi\ /|\nabla\phi|$, unit normal pointing from solid to liquid and $A > 0$ is a constant in ϕ which has to be determined from matched asymptotic analysis. The current, as defined in (3.22) is directly proportional to the velocity of the interface ($\frac{\partial \phi}{\partial t}$) and the composition difference between the solid and the liquid $(c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e})$. The mass conservation equation is thus given by,

$$\frac{\partial c}{\partial t} = \nabla\cdot\left[(1-h(\phi))D_{\mathrm{L}}\nabla c_{\mathrm{L}}\right] + \nabla\cdot\left(\alpha\frac{\partial \phi}{\partial t}\frac{\nabla\phi}{|\nabla\phi|}\right) \tag{3.23}$$

where $\alpha = A\delta$.

The 1-D steady state form of the diffusion equation is given by,

$$-V\frac{dc}{d\mathrm{x}} = \frac{d}{d\mathrm{x}}\left[(1-h(\phi_0))D_{\mathrm{L}}\frac{dc_{\mathrm{L}}}{d\mathrm{x}}\right] + V\frac{d}{d\mathrm{x}}\left[\alpha\frac{d\phi}{d\mathrm{x}}\right] \tag{3.24}$$

Integrating (3.24) will provide,

$$-V(c-c^*) = (1-h(\phi_0))D_{\mathrm{L}}\frac{dc_{\mathrm{L}}}{d\mathrm{x}} + V\alpha\frac{d\phi}{d\mathrm{x}} \tag{3.25}$$

where $c^* = c(-\xi)$ since $h(\phi_0) \to 1$ as $\phi \to 1$ and $\dfrac{\partial\phi}{\partial x} = 0$ at $x \to -\xi$, where 2ξ is the width of the interface. Rewriting (3.24),

$$\begin{aligned} -V(c-c^*) - V\alpha\frac{d\phi}{d\mathrm{x}} &= (1-h(\phi_0))D_{\mathrm{L}}\frac{dc_{\mathrm{L}}}{d\mathrm{x}} && (3.26)\\ \frac{dc_{\mathrm{L}}}{d\mathrm{x}} &= \frac{-V}{(1-h(\phi_0))D_{\mathrm{L}}}\left[(c-c^*) + \alpha\frac{d\phi}{d\mathrm{x}}\right] && (3.27)\\ \mathrm{Taking}B &= (c-c^*) + \alpha\frac{d\phi}{d\mathrm{x}} && (3.28) \end{aligned}$$

The concentration gradient is thus modified as

$$\frac{dc_{\mathrm{L}}}{d\mathrm{x}} = \frac{-V}{(1-h(\phi_0))}\frac{B}{D_{\mathrm{L}}} \tag{3.29}$$

At the thin interface limit of very low Peclet number, $P \equiv \delta V/\tilde{D} \ll 1$, the gradient can be approximated as within the first order of P as

$$\frac{dc_{\mathrm{L}}}{d\mathrm{x}} \approxeq \frac{-V}{(1-h(\phi_0))}\frac{B^0}{D_{\mathrm{L}}} \tag{3.30}$$

where B^0 and ϕ_0 denote the equilibrium state with $V = 0$.

The equilibrium phase-field profile ϕ_0, is the solution to the phase-field equation, Eq.(3.19) at steady state $\dfrac{\partial\phi}{\partial t} = 0$

$$\varepsilon^2\frac{d^2\phi_0}{d\mathrm{x}^2} = Wg(\phi_0) \tag{3.31}$$

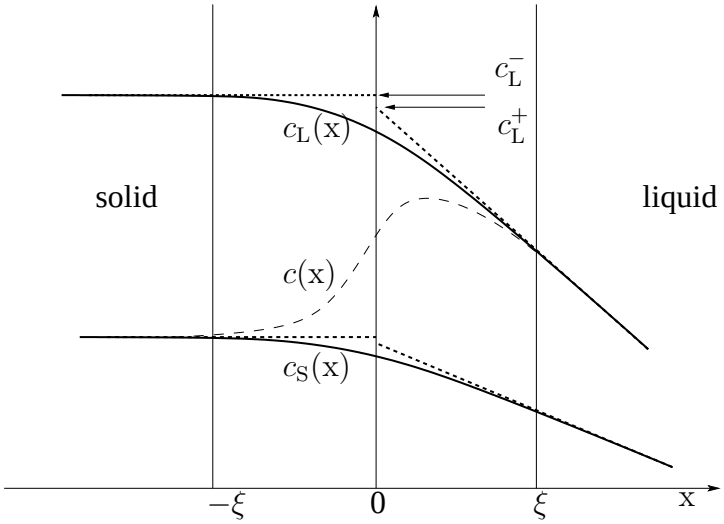


Figure 3.3: Composition profiles across the interface ($-\xi < 0 < \xi$) of the solute in the solid $c_{\mathrm{S}}(\mathrm{x})$, liquid $c_{\mathrm{L}}(\mathrm{x})$ and the mixture composition $c(\mathrm{x})$. $\mathrm{x} = 0$ is the phase-field profile at ϕ = 0.5. The extrapolations of the linear parts in $c_{\mathrm{S}}(\mathrm{x})$ and $c_{\mathrm{L}}(\mathrm{x})$ into the interfacial region are shown as broken lines

Multiplying $\dfrac{d\phi_0}{d\mathrm{x}}$ on both sides and integrating yields,

$$\frac{d\phi_0}{d\mathrm{x}} = \frac{-\sqrt{2W}}{\varepsilon}\sqrt{g(\phi_0)} = \frac{-\sqrt{2W}}{\varepsilon}\phi_0(1-\phi_0) \tag{3.32}$$

Also, substituting (3.17) in B^0 reads as,

$$\begin{aligned} B^0 &= c^e - c_{\mathrm{S}}^e + \alpha\frac{d\phi}{d\mathrm{x}} && (3.33)\\ &= s(\phi_0)c_{\mathrm{S}}^e + (1-s(\phi_0))c_{\mathrm{L}}^e - c_{\mathrm{S}}^e - \alpha\frac{\sqrt{2W}}{\varepsilon}\phi_0(1-\phi_0) && (3.34)\\ &= (1-s(\phi_0))(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e) - \alpha\frac{\sqrt{2W}}{\varepsilon}\phi_0(1-\phi_0) && (3.35) \end{aligned}$$

where c_{L}^e and c_{S}^e are the respective equilibrium compositions of liquid and solid. Substituting for B^0 in (3.27) gives,

$$\frac{dc_{\mathrm{L}}}{d\mathrm{x}} = -\frac{V}{(1-h(\phi_0))D_{\mathrm{L}}}\left[(1-s(\phi_0))(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e) - \alpha\frac{\sqrt{2W}}{\varepsilon}\phi_0(1-\phi_0)\right] \tag{3.36}$$

Integrating (3.36) yields the composition profile as shown in Fig.3.3.

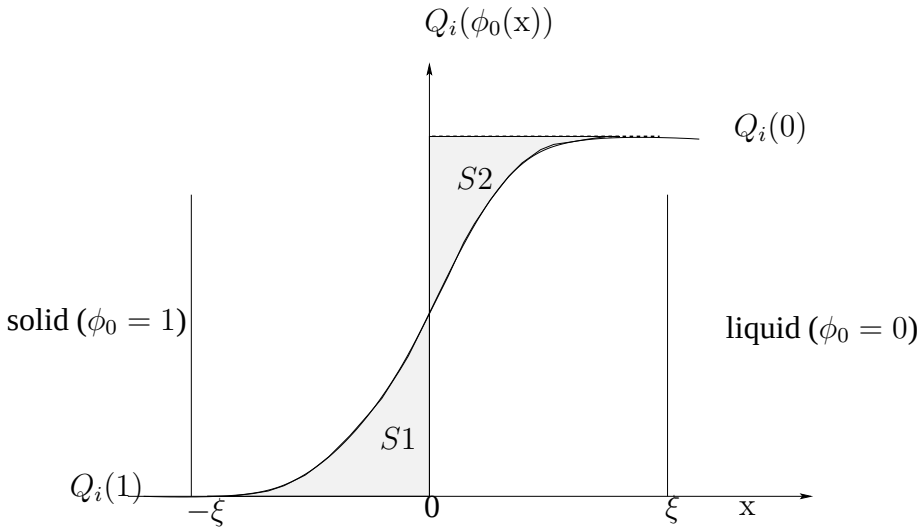


Figure 3.4: Profile of the integrand $Q_i(\phi_0(\mathrm{x}))$ in (3.37) across the interface and the shaded regions S1 and S2 at x<0 and x>0, respectively

$$c_{\mathrm{L}}(\mathrm{x}) = c^{L-} - V\int_{-\xi}^{\mathrm{x}} Q(\phi_0)d\mathrm{x} \tag{3.37}$$

where

$$Q(\phi_0) = \frac{1}{1-h(\phi_0)}\frac{B^0}{D_{\mathrm{L}}} \tag{3.38}$$

at $\phi_0 = 0$

$$Q(0) = \frac{c^e_{\mathrm{L}} - c^e_{\mathrm{S}}}{D_{\mathrm{L}}} \tag{3.39}$$

since all $p(\phi_0)$,$h(\phi_0)$ and $s(\phi_0)$ vanish at $\phi_0 = 0$. The profile of $Q(\phi_0)$ across the interface is shown in Fig.3.4. At the thin interface limit, $-\xi < \mathrm{x} < \xi \ll D/V$, the integration in (3.37) can be approximated as,

$$\int_{-\xi}^{\mathrm{x}} Q(\phi_0)d\mathrm{x} = S1 + \mathrm{x}Q(0) - S2 \tag{3.40}$$

where $S1$ and $S2$ are the shaded regions depicted in Fig.3.4 for x<0 and x>0 where x=0 is the sharp interface limit. The linear part of the composition profile $c_{\mathrm{L}}(\mathrm{x})$ is given by,

$$c_{\mathrm{L}}{}^{\mathrm{lin}}(\mathrm{x}) = c^{L-} - V[S1 + \mathrm{x}Q(0) - S2] \tag{3.41}$$

Extrapolating the straight line into the interfacial region, intersects at $\mathrm{x} = 0$. This is denoted by $c^{\mathrm{L}+}$, which brings to $c^{\mathrm{L}+} \equiv c_{\mathrm{L}}{}^{\mathrm{lin}}(0) = c^{L-} - V(S1 - S2)$. This puts the condition for the vanishing chemical potential jump *i.e.*, $c^{L+} = c^{L-}$ is satisfied when S1=S2. This in terms of

$Q(\phi_0)$ can be written as,

$$\int_{-\infty}^{0} Q(\phi_0)d\mathrm{x} = \int_{0}^{\infty} (Q(\phi_0) - Q(0))d\mathrm{x} \tag{3.42}$$

Considering the symmetry of ϕ_0 in $\dfrac{d\phi_0}{d\mathrm{x}}$ a simple condition to satisfy (3.42) can be given by,

$$Q(\phi_0) = (1 - \phi_0)Q(0) \tag{3.43}$$

Making substitutions from (3.38) and (3.39) into (3.43), we get,

$$\frac{1}{1 - h(\phi_0)} \frac{B^0}{D_\mathrm{L}} = \frac{c_\mathrm{L}^e - c_\mathrm{S}^e}{D_\mathrm{L}} \tag{3.44}$$

$$\frac{1}{1 - h(\phi_0)} \left[(1 - s(\phi_0))(c_\mathrm{L}^e - c_\mathrm{S}^e) - \alpha \frac{\sqrt{2W}}{\varepsilon} \phi_0(1 - \phi_0) \right] = (1 - \phi_0)(c_\mathrm{L}^e - c_\mathrm{S}^e) \tag{3.45}$$

$$\left[(1 - s(\phi_0))(c_\mathrm{L}^e - c_\mathrm{S}^e) - \alpha \frac{\sqrt{2W}}{\varepsilon} \phi_0(1 - \phi_0) \right] = (1 - h(\phi_0))(1 - \phi_0)(c_\mathrm{L}^e - c_\mathrm{S}^e) \tag{3.46}$$

$$(c_\mathrm{L}^e - c_\mathrm{S}^e) \left[(1 - s(\phi_0)) - (1 - h(\phi_0))(1 - \phi_0) \right] = \alpha \frac{\sqrt{2W}}{\varepsilon} \phi_0(1 - \phi_0) \tag{3.47}$$

which gives,

$$\alpha = \frac{\varepsilon}{\sqrt{2W}} (c_\mathrm{L}^e - c_\mathrm{S}^e) \frac{\left[(1 - s(\phi_0)) - (1 - h(\phi_0))(1 - \phi_0) \right]}{\phi_0(1 - \phi_0)} \tag{3.48}$$

The anti-trapping function α, which suppresses the anomalous solute trapping effect arising due to the chemical potential jump at the interface, is thus the result of the correct coupling between the interpolating functions $h(\phi)$ and $s(\phi_0)$.

3.4 Phase-field mobility

In order to incorporate the necessary kinetics in the phase-field, it is necessary to derive the phase-field mobility at the thin interface limit. To start with, it is required to determine the model parameters ε, W which are given by the 1-D phase-field profile at equilibrium. A more

detailed derivation of ε and W is given in Kim *et al.* (1999).

$$2\xi = \frac{\varepsilon}{\sqrt{2W}} \int_{\phi_S}^{\phi_L} \frac{d\phi_0}{\phi_0(1-\phi_0)} = \frac{\varepsilon}{\sqrt{2W}} \ln \frac{\phi_L(1-\phi_S)}{\phi_S(1-\phi_L)} \tag{3.49}$$

$$\sigma = \varepsilon^2 \int_{-\infty}^{\infty} \left(\frac{d\phi_0}{dx}\right)^2 dx = \frac{\varepsilon\sqrt{W}}{3\sqrt{2}} \tag{3.50}$$

The phase-field mobility M_ϕ is obtained by the ratio of the the driving force and the interface velocity β. At the thin interface limit, this is obtained from the composition profile $c_{\mathrm{L}}(\mathrm{x})$ at the condition of the vanishing chemical potential jump defined by $c^{L-}(\mathrm{x}) = c^{L+}(\mathrm{x})$. Inserting the composition and chemical potential profiles into the driving force terms of the phase-field gives the new driving force for the effective sharp interface with the straight composition profiles. Substituting the value of (3.48) in (3.35), B^0 is given by,

$$B^0 = (c_L^e - c_S^e)(1 - h(\phi_0))(1 - \phi_0) \tag{3.51}$$

The composition gradient as per (3.27) after the substitution of B^0 from (3.51) gives,

$$\frac{dc_{\mathrm{L}}}{d\mathrm{x}} = -V(1-\phi_0)\frac{(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)}{D_{\mathrm{L}}} \tag{3.52}$$

The thermodynamic driving force for the transformation from the phase-field equation is given by

$$\Delta f = f^{\mathrm{L}} - f^{\mathrm{S}} - \mu(c_{\mathrm{L}} - c_{\mathrm{S}}) \tag{3.53}$$

where f^{S}, f^{L}, c_{S}, c_{L}, μ though functions of position, an assumption of thin interface requires only the gradients of the above functions. At this limit, where the interface Peclet number $P = \xi V/\tilde{D} \ll 1$, the thermodynamic state of the interface should be as close to the local equilibrium state. Expanding all the variables around their equilibrium values within the first order of P,

$$
\begin{aligned}
c_{\mathrm{S}}(\mathrm{x}) &= c_{\mathrm{S}}^{e} + \delta c_{\mathrm{S}}(\mathrm{x}) \\
c_{\mathrm{L}}(\mathrm{x}) &= c_{\mathrm{L}}^{e} + \delta c_{\mathrm{L}}(\mathrm{x}) \\
\mu(\mathrm{x}) &= \mu^{e} + \delta\mu(\mathrm{x}) \\
f^{\mathrm{L}} &= f^{\mathrm{L},e} + \left.\frac{\partial f^{\mathrm{L}}}{\partial c_{\mathrm{L}}}\right|_{e} \delta c_{\mathrm{L}} = f^{\mathrm{L},e} + \mu^{e}\delta c_{\mathrm{L}} \\
f^{\mathrm{S}} &= f^{\mathrm{S},e} + \left.\frac{\partial f^{\mathrm{S}}}{\partial c_{\mathrm{S}}}\right|_{e} \delta c_{\mathrm{S}} = f^{\mathrm{S},e} + \mu^{e}\delta c_{\mathrm{S}}
\end{aligned}
\tag{3.54}
$$

The thermodynamic force within the first order of P is given by,

$$
\begin{aligned}
\Delta f =& f^{\mathrm{L}} - f^{\mathrm{S}} - \mu(c_{\mathrm{L}} - c_{\mathrm{S}}) \\
=& f^{\mathrm{L},e} + \mu^{e}\delta c_{\mathrm{L}} - (f^{\mathrm{S},e} + \mu^{e}\delta c_{\mathrm{S}}) - (\mu^{e} + \delta\mu(\mathrm{x})) \\
& ((c_{\mathrm{L}}^{e} + \delta c_{\mathrm{L}}(\mathrm{x})) - (c_{\mathrm{S}}^{e} + \delta c_{\mathrm{S}}(\mathrm{x}))) \\
=& (f^{\mathrm{L},e} - f^{\mathrm{S},e}) - (\mu^{e} + \delta\mu(\mathrm{x}))(c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e})
\end{aligned}
$$

Rearranging the terms,

$$
\Delta f = (f^{\mathrm{L},e} - f^{\mathrm{S},e}) - \mu(\mathrm{x})(c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e}) \tag{3.55}
$$

With this, the phase-field equation for the 1-D solidification at instantaneous steady state gives,

$$
-\frac{V}{M_{\phi}}\frac{d\phi}{d\mathrm{x}} = \varepsilon^{2}\frac{d^{2}\phi_{0}}{d\mathrm{x}^{2}} - W\frac{dg_{w}(\phi_{0})}{d\phi_{0}} + (f^{\mathrm{L},e} - f^{\mathrm{S},e} - \mu(\mathrm{x})(c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e}))\frac{dp(\phi_{0})}{d\phi_{0}} \tag{3.56}
$$

Multiplying with $d\phi_0/d\mathrm{x}$ on both sides and integrating makes the first two terms of (3.56) to vanish at steady state.

$$
\begin{aligned}
-\frac{V}{M_{\phi}}\int_{-\infty}^{\infty}\left(\frac{d\phi_{0}}{d\mathrm{x}}\right)^{2}d\mathrm{x} =& f^{\mathrm{L},e}\int_{-\infty}^{\infty}\frac{dp(\phi_{0})}{d\mathrm{x}}d\mathrm{x} - f^{\mathrm{S},e}\int_{-\infty}^{\infty}\frac{dp(\phi_{0})}{d\mathrm{x}}d\mathrm{x} \\
& - (c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e})\int_{-\infty}^{\infty}\mu(\mathrm{x})\frac{dp(\phi_{0})}{d\phi_{0}}d\mathrm{x} \\
=& - f^{\mathrm{L},e} + f^{\mathrm{S},e} - (c_{\mathrm{L}}^{e} - c_{\mathrm{S}}^{e})\int_{-\infty}^{\infty}\mu(\mathrm{x})\frac{dp(\phi_{0})}{d\phi_{0}}d\mathrm{x}
\end{aligned}
\tag{3.57}
$$

as $p(\phi_0)$ = 1 as $\mathrm{x} \to -\infty$ and $p(\phi_0)$ = 0 as $\mathrm{x} \to \infty$. The third term on the RHS can be solved

by integrating by parts.

$$\int_{-\infty}^{\infty}\mu(\mathrm{x})\frac{p(\phi_0)}{d\mathrm{x}}d\mathrm{x} = \mu(\mathrm{x})p(\phi_0)|_{\mathrm{x}=-\infty}^{\mathrm{x}=\infty} - \int_{-\infty}^{\infty}\frac{d\mu(\mathrm{x})}{d\mathrm{x}}p(\phi)d\mathrm{x} \tag{3.58}$$

$$= -\mu^0 - f^{\mathrm{L},e}\int_1^0 \frac{dc_\mathrm{L}}{d\mathrm{x}}p(\phi_0)\frac{d\mathrm{x}}{d\phi_0}d\phi_0 \tag{3.59}$$

where $\mu(-\infty)=\mu^0$ denotes the chemical potential corresponding to the constant composition $c^{\mathrm{L}-}$ as shown in Fig.3.3 and

$$\frac{d\mu(\mathrm{x})}{d\mathrm{x}} = f^{\mathrm{L}}\frac{dc_\mathrm{L}}{d\mathrm{x}} \approx f^{\mathrm{L},e}\frac{dc_\mathrm{L}}{d\mathrm{x}} \tag{3.60}$$

Substituting (3.52) for $\frac{dc_\mathrm{L}}{d\mathrm{x}}$ and (3.32) for $\frac{d\phi_0}{d\mathrm{x}}$ in (3.59) yields,

$$\begin{aligned}\int_{-\infty}^{\infty}\mu(\mathrm{x})\frac{p(\phi_0)}{d\mathrm{x}}d\mathrm{x} =& -\mu^0 - f^{\mathrm{L},e}\int_1^0\left(-V(1-\phi_0)\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)}{D_\mathrm{L}}\right)\left(-\frac{\varepsilon}{\sqrt{2W}(\phi_0(1-\phi_0))}\right)p(\phi_0)d\phi_0\\ =& -\mu^0 - VG^{\mathrm{L},e}\frac{\varepsilon}{\sqrt{2W}}\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)}{D_\mathrm{L}}\int_1^0\frac{p(\phi_0)}{\phi_0}d\phi_0\end{aligned} \tag{3.61}$$

Substituting (3.61) in the steady state phase-field equation (3.57) gives,

$$-\frac{V}{M_\phi}\frac{\sqrt{W}}{3\sqrt{2}\varepsilon} = -f^{\mathrm{L},e}+f^{\mathrm{S},e}-(c_\mathrm{L}^e-c_\mathrm{S}^e)\left(-\mu^0 - Vf^{\mathrm{L},e}\frac{\varepsilon}{\sqrt{2W}}\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)}{D_\mathrm{L}}\int_1^0\frac{p(\phi_0)}{\phi_0}d\phi_0\right) \tag{3.62}$$

The driving force for solidification after rearranging the terms get modified to,

$$\begin{aligned}f^{\mathrm{L},e} - f^{\mathrm{S},e} - (c_\mathrm{L}^e-c_\mathrm{S}^e)\mu^0 =& \frac{V}{M_\phi}\frac{\sqrt{W}}{3\sqrt{2}\varepsilon} + (c_\mathrm{L}^e-c_\mathrm{S}^e)\left(Vf^{\mathrm{L},e}\frac{\varepsilon}{\sqrt{2W}}\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)}{D_\mathrm{L}}\int_1^0\frac{p(\phi_0)}{\phi_0}d\phi_0\right)\\ =& V\left(\frac{1}{M_\phi}\frac{\sqrt{W}}{3\sqrt{2}\varepsilon} - f^{\mathrm{L},e}\frac{\varepsilon}{\sqrt{2W}}\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)^2}{D_\mathrm{L}}\int_0^1\frac{p(\phi_0)}{\phi_0}d\phi_0\right)\\ =& V\beta\end{aligned} \tag{3.63}$$

where β it the interface kinetics coefficient.

$$\beta = \frac{1}{M_\phi}\frac{\sqrt{W}}{3\sqrt{2}\varepsilon} - f^{\mathrm{L},e}\frac{\varepsilon}{\sqrt{2W}}\frac{(c_\mathrm{L}^e-c_\mathrm{S}^e)^2}{D_\mathrm{L}}\int_0^1\frac{p(\phi_0)}{\phi_0}d\phi_0 \tag{3.64}$$

The second term is the first order correction to the phase-field mobility at the thin interface limit. In the condition or very low undercooling, it is needed to find the mobility at the condition of vanishing kinetics whose physical meaning in the thin-interface alloy phase-field method

maybe interpreted as follows: A finite phase-field mobility decreases the solute composition in the solid. At the same time, the finite interface thickness bringing the solute trapping effect increases the solute composition in the solid. At the condition where the two opposing effects are exactly cancelled out, the solid composition recovers the equilibrium composition, which is the condition for the vanishing kinetics coefficient in the thin interface PFM.

For infinite interface mobility i.e when $\beta = 0$, the phase-field mobility is thus recovered as

$$M_\phi = \frac{WD_{\mathrm{L}}}{3\varepsilon^2 f^{\mathrm{L},e}(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)^2 \int_0^1 \frac{p(\phi_0)}{\phi_0} d\phi_0} \tag{3.65}$$

In order to have vanishing interface kinetics in all directions, it is required to treat mobility to change with the capillary anisotropy. Using $\varepsilon(n) = \varepsilon_0 \varepsilon(\eta)$ modifies (3.66)

$$M_\phi = \frac{WD_{\mathrm{L}}}{3\varepsilon(n)^2 f^{\mathrm{L},e}(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)^2 \int_0^1 \frac{p(\phi_0)}{\phi_0} d\phi_0} \tag{3.66}$$

3.5 Anisotropy

Dendritic growth in our simulations are all performed without interface kinetics. The Gibbs-Thomson condition in 2-D without interface kinetics is given by,

$$c_{\mathrm{L}}^i = c_{\mathrm{L}} - \frac{\sigma_{\mathrm{SL}}\kappa}{\frac{\partial^2 f_{\mathrm{L}}}{\partial c_{\mathrm{L}}^2}(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)} \tag{3.67}$$

$$= c_{\mathrm{L}} - d_0(\mathbf{n})\kappa \tag{3.68}$$

The capillary length is directly proportional to the surface tension and the angular dependence of which is given by,

$$d_0(\mathbf{n}) = \frac{1}{\frac{\partial^2 f_{\mathrm{L}}}{\partial c_{\mathrm{L}}^2}(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)}(\sigma + \sigma_{\theta\theta}) \tag{3.69}$$

where θ is the angle between the normal to the interface and x-axis. Anisotropic capillary effects are reproduced by allowing the interface thickness to possess the same anisotropy as the interfacial energy and is introduced into the phase-field equation thorough the gradient energy

term ε.

A standard four-fold anisotropy is adopted and defined by,

$$\begin{aligned} \varepsilon(\mathbf{n}) &= \varepsilon_0(1+\epsilon_4(n_{\mathrm{x}}^4+n_{\mathrm{y}}^4)) && (3.70)\\ \mathbf{n} &= \frac{\nabla\phi}{|\nabla\phi|} && (3.71) \end{aligned}$$

The anisotropic strength is characterized by the parameter ϵ_4, which can be measured experimentally by examining the deviation of an equilibrium shape from a circle as,

$$\begin{aligned} R(\theta) &= R_0(1+\epsilon_4\cos 4\theta) && (3.72)\\ \theta &= \arctan\left(\frac{\phi_{\mathrm{y}}}{\phi_{\mathrm{x}}}\right) && (3.73) \end{aligned}$$

where R is the radial polar coordinate measured from a fixed origin, $\phi_{\mathrm{x}} = \dfrac{\partial\phi}{\partial \mathrm{x}}$ and $\phi_{\mathrm{y}} = \dfrac{\partial\phi}{\partial \mathrm{y}}$

The anisotropic 2-D Laplacian can thus be written as,

$$\nabla_{a2}^2\phi = \nabla\cdot[\varepsilon(\mathbf{n})^2\nabla\phi] + \frac{\partial}{\partial \mathrm{x}}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{\mathrm{x}})}\right) + \frac{\partial}{\partial \mathrm{y}}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{\mathrm{y}})}\right) \quad (3.74)$$

where $\varepsilon(\mathbf{n})$ is given by

$$\varepsilon(\mathbf{n}) = \varepsilon_0(1-3\epsilon_4)\left(1+\frac{4\epsilon_4}{1-3\epsilon_4}\frac{(\phi_{\mathrm{x}})^4+(\phi_{\mathrm{y}})^4}{|\nabla\phi|^4}\right) \quad (3.75)$$

where ε_0 denotes the isotropic gradient energy coefficient. The corresponding anisotropic Laplacian in 3-D is given by,

$$\begin{aligned} \nabla_{a3}^2\phi =& \nabla\cdot[\varepsilon(\mathbf{n})^2\nabla\phi] + \frac{\partial}{\partial \mathrm{x}}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{\mathrm{x}})}\right)\\ &+\frac{\partial}{\partial \mathrm{y}}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{\mathrm{y}})}\right) + \frac{\partial}{\partial \mathrm{z}}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{\mathrm{z}})}\right) \end{aligned} \quad (3.76)$$

where $\phi_{\mathrm{z}} = \dfrac{\partial\phi}{\partial \mathrm{z}}$.

$$\varepsilon(\mathbf{n}) = \varepsilon_0(1-3\epsilon_4)\left(1+\frac{4\epsilon_4}{1-3\epsilon_4}\frac{(\phi_{\mathrm{x}})^4+(\phi_{\mathrm{y}})^4+(\phi_{\mathrm{z}})^4}{|\nabla\phi|^4}\right) \quad (3.77)$$

Discretization of the 3-D anisotropic Laplacian was done by a convenient formulation as done by Singer *et al.* (2006),

$$\begin{aligned}\frac{\partial}{\partial r_i}\left(|\nabla\phi|^2\varepsilon(\mathbf{n})\frac{\partial\varepsilon(\mathbf{n})}{\partial(\phi_{r_i})}\right) =& \frac{\partial}{\partial r_i}\left(\varepsilon(\mathbf{n})16\gamma\left(\frac{\phi_{r_i}^2}{|\nabla\phi|^2}-\frac{\phi_{\mathrm{x}}^2+\phi_{\mathrm{y}}^2+\phi_{\mathrm{z}}^2}{|\nabla\phi|^4}\right)\frac{\partial\phi}{\partial r_i}\right)\\ =& \frac{\partial}{\partial r_i}\left(f(\phi_{\mathrm{x}},\phi_{\mathrm{y}},\phi_{\mathrm{z}})\frac{\partial\phi}{\partial r_i}\right)\end{aligned} \tag{3.78}$$

where $f(\phi_{\mathrm{x}},\phi_{\mathrm{y}},\phi_{\mathrm{z}}) = \varepsilon(\mathbf{n})16\gamma\left(\frac{\phi_{r_i}^2}{|\nabla\phi|^2}-\frac{\phi_{\mathrm{x}}^2+\phi_{\mathrm{y}}^2+\phi_{\mathrm{z}}^2}{|\nabla\phi|^4}\right)$ and $r_i = \mathrm{x},\mathrm{y},\mathrm{z}$

Thus, the governing equations for the evolution of phase field and composition are derived along with necessary auxiliary relations. The scope of this study has been confined to alloys in a binary system. The expressions are accordingly derived to be directly applicable to a binary system. The source code of the solver corresponds directly to most of the expressions used in this chapter. The application of realistic thermodynamic functions and the solution scheme to solve these equations are elaborated in the subsequent chapters.

CHAPTER 4

CALPHAD DESCRIPTION OF GIBBS ENERGIES

This chapter deals with the interfacing of the CALPHAD description of Gibbs energies with the phase-field model. To start with, a brief description of the Gibbs energy model that is used for the chosen systems, (Al-Cu and Al-Mg) is given.

4.1 Thermodynamic description and models

A thermodynamic description of a phase requires the assignment of Gibbs energy functions for each phase under consideration. In this context, CALPHAD provides a means to find required Gibbs energy functions through thermodynamic modelling. The CALPHAD method employs a variety of models to describe the temperature, pressure and composition dependencies of the Gibbs energy functions of various phases. In the present work there are two phases considered: liquid and fcc solid solution. Both the phases are modelled using random substitutional solution. For such phases, the molar Gibbs energy is given by

$$G_{\mathrm{m}} = {}^{\mathrm{ref}}G_{\mathrm{m}} + {}^{\mathrm{id}}G_{\mathrm{m}} + {}^{\mathrm{E}}G_{\mathrm{m}} \tag{4.1}$$

where ${}^{\mathrm{ref}}G_{\mathrm{m}}$ is the reference molar Gibbs energy of the elements in their pure state, ${}^{\mathrm{id}}G_{\mathrm{m}}$ the ideal Gibbs energy of mixing and ${}^{\mathrm{E}}G_{\mathrm{m}}$, the excess contribution to the Gibbs energy arising due to the non ideal interaction between the elements. The reference Gibbs energy (*i.e.* Gibbs energy of a mechanical mixture of components) is given by,

$${}^{\mathrm{ref}}G_{\mathrm{m}} = \sum_{i=1}^{2} x_{\mathrm{i}} {}^{\circ}G_{\mathrm{i}} \tag{4.2}$$

The Gibbs energy functions of the pure elements in their stable and metastable states are taken from the Scientific Group Thermodata Europe (SGTE) database (Dinsdale (1991)). They are

typically represented by a T dependent polynomial given by,

$$^{\circ}G_i - H_i^{\mathrm{SER}} = a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \tag{4.3}$$

where H_i^{SER} is the molar enthalpy of an element at 298.15 K and 1 bar in its standard element reference (SER) state; T is the absolute temperature; $^{\circ}G_{\mathrm{i}}(T)$ is the molar Gibbs energy of the element i the same structural state as the phase, which is referred to the enthalpy of its stable state at 298.15 K and 1 bar. The ideal contribution $^{\mathrm{id}}G_{\mathrm{m}} = -TS^{\mathrm{conf}}$ is based on the configurational entropy of a phase and it arises from the number of possible arrangements of the components in the phase. In the context of this study, the configuration part is assumed to be purely random. In a substitutional solution, the constituents have the same probability of occupying any site in the unit cell.

4.1.1 Models for excess Gibbs energy

Excess Gibbs energy ($^{\mathrm{E}}G_{\mathrm{m}}$) accounts for deviation from ideal behaviour of a solution. For a random substitutional solution it is given by the expression,

$$^{\mathrm{E}}G_{\mathrm{m}} = x_{\mathrm{A}}x_{\mathrm{B}}L_{\mathrm{A,B}} \tag{4.4}$$

where, $L_{\mathrm{A,B}}$ is the interaction term. From a physical point, for a strictly regular solution, the interaction parameter $L_{\mathrm{A,B}}$ can be regarded as the difference in the bond energy between like and unlike atoms.

4.1.2 Redlich-Kister binary excess model

In most cases as in Eq.4.4, it is often not sufficient to describe a system with a single interaction parameter with the available experimental data. In such cases, the binary regular solution term is more often extended as a polynomial in composition such as,

$$L_{\mathrm{A,B}} = \sum_{v=0}^{k}(x_{\mathrm{A}} - x_{\mathrm{B}})^v L_{\mathrm{A,B}}^v \tag{4.5}$$

This type of composition dependence of the binary interaction was proposed by Redlich and Kister (Redlich and Kister (1948)) and is more often known as the Redlich-Kister (RK) polynomial. Often one uses a linear temperature dependence,

$$L^{v}_{\mathrm{A,B}} = a^{v} + b^{v}T \tag{4.6}$$

Various a and b parameters are estimated by an optimization procedure that requires experimental data that can be related to Gibbs energy of the phase.

4.2 Relating phase-field and CALPHAD

In most phase-field formulations, volume free energy densities, f, are used to describe the free energy functional F, which could be either Helmholtz or Gibbs energies. However, to evaluate the chemical contribution of the free energies in using the thermodynamic functions obtained through CALPHAD, molar Gibbs energy densities, g, are preferred (Eiken *et al.* (2006)). Assuming molar volumes v_m to be the same for both phases, the volume free energy density f can be replaced with molar Gibbs energy densities g and the molar concentration c_i with x_i through,

$$x_i = \frac{n_i}{n_{tot}} \tag{4.7}$$

and

$$c_i = \frac{n_i}{V} = \frac{x_i}{\mathrm{V_m}} \tag{4.8}$$

where, n_i and x_i are the amount and the mole fraction of the solute i, respectively and V, the total volume of the system. The free energy density f, is related to Gibbs energy density g, by,

$$f(\phi, c) = \frac{1}{v_m} g(\phi, x_{\mathrm{B}}) \tag{4.9}$$

Reformulating the phase-field and concentration equations incorporating this change of variables from c to x_B and f to g in Eq.3.19 and Eq.3.23, we get,

$$\frac{1}{M_\phi}\frac{\partial \phi}{\partial t} = \varepsilon^2 \nabla^2 \phi - W \frac{dg_w(\phi)}{d\phi} - \frac{dp(\phi)}{d\phi}\left(g^{\mathrm{S}} - g^{\mathrm{L}} + \mu(x^{\mathrm{S}}_{\mathrm{B}} - x^{\mathrm{L}}_{\mathrm{B}})\right) \tag{4.10}$$

$$\frac{\partial x_{\mathrm{B}}}{\partial t} = \nabla\cdot\left[(1-h(\phi))D_{\mathrm{L}}\nabla x_{\mathrm{B}}^{\mathrm{L}}\right] + \nabla\cdot\left(\alpha\frac{\partial\phi}{\partial t}\frac{\nabla\phi}{|\nabla\phi|}\right) \tag{4.11}$$

along with the subsequent changes in Eq.3.66 and Eq.3.17.

4.3 Thermodynamic description of the binaries: Al-Cu and Al-Mg

The molar Gibbs energy description of a phase for a two component system in a random substitutional regular solution can be expressed as,

$$G_{\mathrm{m}} = x_{\mathrm{A}}{}^{\circ}G_{\mathrm{A}} + x_{\mathrm{B}}{}^{\circ}G_{\mathrm{B}} + RT(x_{\mathrm{A}}\ln x_{\mathrm{A}} + x_{\mathrm{B}}\ln x_{\mathrm{B}}) + x_{\mathrm{A}}x_{\mathrm{B}}\sum_{v=0}^{n}{}^{v}L_{\mathrm{AB}}(x_{\mathrm{A}}-x_{\mathrm{B}})^{v} \tag{4.12}$$

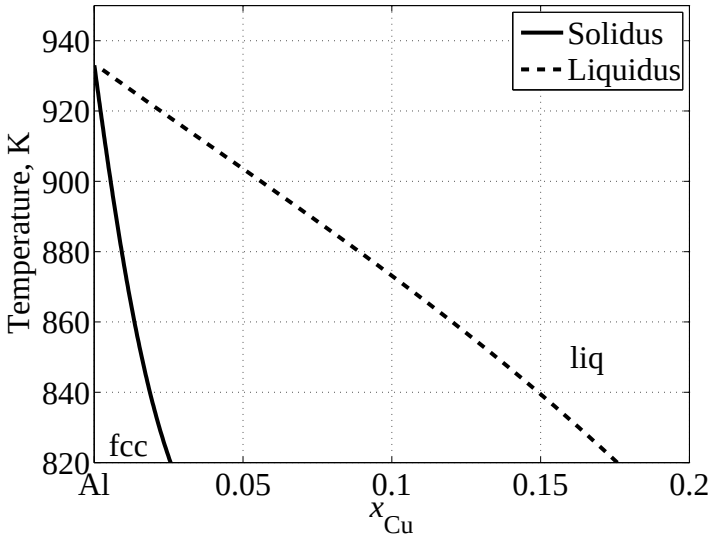
The systems considered in the present work are Al-Cu assessed by Witusiewicz *et al.* (2004) and Al-Mg by Zhong *et al.* (2005). In both cases, the phases considered were that of liquid and solid (fcc) which are modelled using Eq.4.12.

The phase boundaries of the liquidus ($x_{\mathrm{B}}^{\mathrm{L}}$) and solidus ($x_{\mathrm{B}}^{\mathrm{S}}$) for the Al-rich corner of Al-Cu and Al-Mg are calculated using Gibbs energy functions of the phases in question through the equilibrium condition at a given T and P,

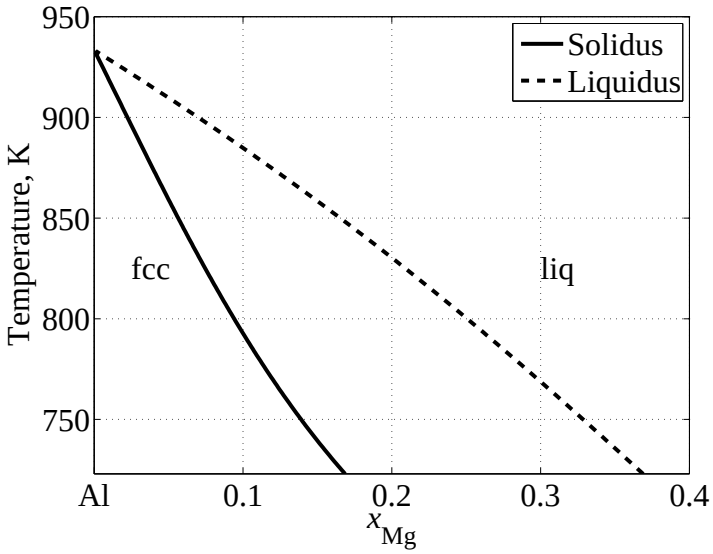
$$\begin{aligned}\mu_{\mathrm{A}}^{\mathrm{L}} &= \mu_{\mathrm{A}}^{\mathrm{S}}\\ \mu_{\mathrm{B}}^{\mathrm{L}} &= \mu_{\mathrm{B}}^{\mathrm{S}}\end{aligned} \tag{4.13}$$

The phase diagrams for the Al-rich corner of Al-Cu and Al-Mg are shown in Fig.4.1(a) and Fig.4.1(b), respectively.

The Gibbs energy functions for the elements are taken from the SGTE compilation by Dinsdale (1991). These are reproduced in Table.4.1. The relevant solution parameters for Al-Cu and Al-Mg systems are listed in Table 4.2 and Table 4.3, respectively.



(a)



(b)

Figure 4.1: Al-rich corner of (a) Al-Cu and (b) Al-Mg, computed using CALPHAD description of Gibbs energies.

Table 4.1: Gibbs energy functions of Al, Cu and Mg in J/ mol for the phases solid(fcc) and liquid (liq) (Dinsdale (1991))

Data in the form of G - HSER	
Aluminium	
fcc	
$-7976.15 + 137.093038T - 24.3671976T \ln T - 0.001884662T^2 - 8.77664 \times 10^{-7}T^3 + 74092T^{-1}$	$(298.15 < T < 700)$
$-11276.24 + 223.048446T - 38.5844296T \ln T + 0.018531982T^2 - 5.764227 \times 10^{-6}T^3 + 74092T^{-1}$	$(700 < T < 933.47)$
$-11278.38 + 188.68415T - 31.74819T \ln T - 1.231 \times 1028T^{-9}$	$(933.47 < T < 2900)$
liq	
$3028.879 + 125.251171T - 24.3671976T \ln T - 0.001884662T^2 - 8.77664 \times 10^{-7}T^3 + 74092T^{-1} + 7.934 \times 10^{-20}T^7$	$(298.15 < T < 700)$
$-271.21 + 211.206579T - 38.5844296T \ln T + 0.018531982T^2 - 5.764227 \times 10^{-6}T^3 + 74092T^{-1} + 7.934 \times 10^{-20}T^7$	$(700 < T < 933.47)$
$-795.996 + 177.430178T - 31.748192T \ln T$	$(933.47 < T < 2900)$
Copper	
fcc	
$-7770.458 + 130.485235T - 24.112392T \ln T - 0.00265684T^2 + 1.29223 \times 10^{-7}T^3 + 52478T^{-1}$	$(298.15 < T < 1357.8)$
$-13542.03 + 183.80383T - 31.38T \ln T + 3.642 \times 10^{29}T^{-9}$	$(1357.8 < T < 3200)$
liq	
$5194.277 + 120.973331T - 24.112392T \ln T - 0.00265684T^2 + 1.29223 \times 10^{-7}T^3 + 52478T^{-1} - 5.849 \times 10^{-21}T^7$	$(298.15 < T < 1357.8)$
$-46.545 + 173.881484 - 31.38T \ln T$	$(1357.8 < T < 3200)$
Magnesium	
fcc	
$-5767.34 + 142.775547T - 26.1849782T \ln T + 0.4858 \times 10^{-3}T^2 - 1.393669 \times 10^{-6}T^3 + 78950T^{-1}$	$(298.15 < T < 923)$
$-11530.185 + 203816215T - 34.3088T \ln T + 1.0382 \times 10^{28}T^{-9}$	$(923 < T < 3000)$
liq	
$-165.097 + 134.838617T - 26.1849782T \ln T + 0.4858 \times 10^{-3}T^2 - 1.393669 \times 10^{-6}T^3 + 78950T^{-1} - 8.0176 \times 10^{-20}T^7$	$(298.15 < T < 923)$
$-5439.869 + 195.324057T - 34.3088T \ln T$	$(923 < T < 3000)$

Table 4.2: Redlich-Kister coefficients for the Al-Cu system in J/mol (Witusiewicz *et al.* (2004))

	v	a_v^ϕ	b_v^ϕ
ϕ = liq	0	-67094	8.555
	1	32148	-7.118
	2	5915	-5.889
	3	-8175	6.049
ϕ = fcc	0	-53520	2
	1	38590	-2
	2	1170	0

Table 4.3: Redlich-Kister coefficients for the Al-Mg system in J/mol (Zhong *et al.* (2005))

	v	a_v^ϕ	b_v^ϕ
ϕ = liq	0	-9019	4.794
	1	-1093	1.412
	2	494	0
ϕ = fcc	0	1593	2.149
	1	1014	-0.660
	2	-673	0

The application of realistic thermodynamic functions is now possible by incorporating the above mentioned polynomial expressions for Gibbs energy as free energy density functions in the formulation. The simple nature of these functions enables a fast computation of realistic thermodynamic data as compared to the API route. The polynomial expressions in the Table 4.1, Table 4.2 and Table 4.3 are directly hard-coded into the program. The functions that contain these polynomials are kept modular for each system. The advantage of the entire source code being developed as part of this thesis enables hard-coding of these functions to gain compile time optimizations by the compiler and lead to faster computation. This is important as the compilation time is negligible compared to the execution time for these programs.

CHAPTER 5

NUMERICAL IMPLEMENTATION

This chapter focusses on the numerical and programming approaches used for solving the phase-field equations. Issues pertaining to the portability of the code across different hardware platforms *viz.* CPU and GPU, along with the programming strategies used are discussed.

5.1 Solution Scheme

An explicit Euler scheme was used for time discretization. Although, implicit approaches do take away the restriction on the time interval, parallelization of implicit schemes are often not straightforward. Explicit schemes, on the other hand, only require the previous time step values to calculate the present. This makes the explicit scheme to be executed with simpler algorithms. Also, use of explicit schemes renders the problem to be easily parallelizable across massively parallel architectures.

5.1.1 Gradients and Laplacians

A finite-difference approach was used to discretize the governing equations. An explicit forward difference in time and a first order central difference in space for the gradients (Eq.(5.1)) was used to discretize the equations. The Laplacians were discretized using a second order finite difference scheme.

$$\nabla\phi|_{i,j,k} = \frac{\phi_{i+1,j,k} - \phi_{i-1,j,k}}{2\Delta x} + \frac{\phi_{i,j+1,k} - \phi_{i,j-1,k}}{2\Delta y} + \frac{\phi_{i,j,k+1} - \phi_{i,j,k-1}}{2\Delta z} \tag{5.1}$$

In most conventional finite-differences used for the discretization of PDEs, an anisotropy is introduced into the numerical scheme. This comes from the directional bias of the error terms in the discretization. In order to preserve the symmetry in the simulation and mitigate the effect

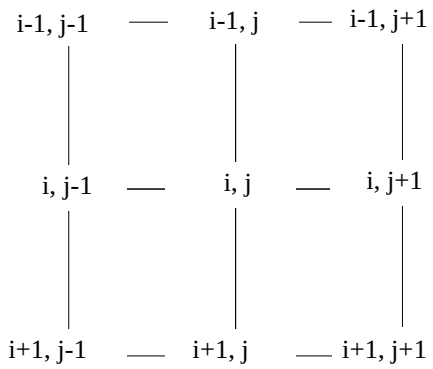


Figure 5.1: A 9-point stencil configuration for 2-D array computations

of grid anisotropy, a more generalised isotropic scheme for calculating the Laplacian was used, Kumar (2004). A 9-point Laplacian for 2-D and a 27-point Laplacian for 3-D were utilised (Fig. 5.1 and Fig. 5.2, respectively). Their corresponding discretization schemes are given by Eq.(5.2) and Eq.(5.3), respectively. Computation of Laplacian using this scheme produces both faster as well as more rotationally symmetric results.

$$\begin{aligned}\nabla^2_\phi = \frac{1}{6\Delta x^2}[(\phi_{i-1,j+1} + \phi_{i+1,j+1} + \phi_{i+1,j-1} + \phi_{i-1,j-1}) + \\ 4(\phi_{i-1,j} + \phi_{i+1,j} + \phi_{i,j+1} + \phi_{i,j-1}) - 20\phi_{i,j}]\end{aligned} \tag{5.2}$$

A 27- point Laplacian (O'Reilly and Beck (2006)), though discernibly large, can be expressed in a more lucid manner by splitting it in terms of the positions where the points lie on the grid as shown in Fig. 5.2. The 27 points can be split up as those that occupy the corners(**8**), edge centers (**12**), face centers (**6**)and body center (**1**).

$$\begin{aligned}\nabla^2_{27}\phi_i =& \frac{3}{13\Delta x^2}\left[\sum_{j\in N_f}(\phi_j - \phi_i) + \frac{1}{2}\sum_{j\in N_e}(\phi_j - \phi_i) + \frac{1}{3}\sum_{j\in N_c}(\phi_j - \phi_i)\right] \\ =& \frac{3}{13\Delta x^2}\left[\sum_{j\in N_f}\phi_j + \frac{1}{2}\sum_{j\in N_e}\phi_j + \frac{1}{3}\sum_{j\in N_c}\phi_j - \frac{44}{3}\phi_i\right]\end{aligned} \tag{5.3}$$

where, N_f are the face points on the cube, N_c are the corner points, N_e are the edge centered points and ϕ_i being the body centered point.

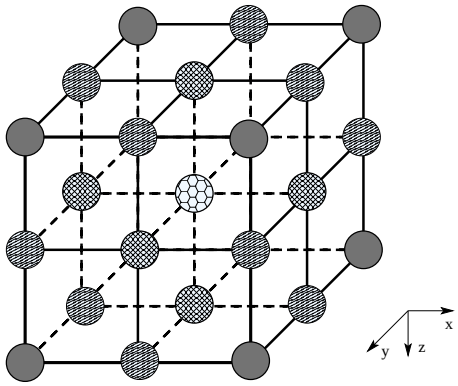


Figure 5.2: A 27-point stencil configuration for 3-D array computations

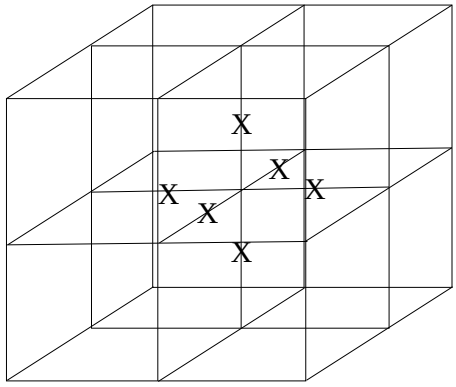


Figure 5.3: A 27-point stencil configuration showing the location of off-grid points marked as **X**

5.1.2 Dicretization of the anti-trapping current and off-grid locations

Discretization of the anti-trapping current was done along similar lines of Choudhury (2012). Rewriting the anti-trapping current from Eq.(3.23),

$$j_{at} = \alpha \frac{\partial \phi}{\partial t} \frac{\nabla \phi}{|\nabla \phi|} \tag{5.4}$$

The diffusion equation with the anti-trapping current is given by,

$$\frac{\partial c}{\partial t} = \nabla \cdot \left[(1 - h(\phi)) D_{\mathrm{L}} \nabla c_{\mathrm{L}} \right] + \nabla \cdot j_{at} \tag{5.5}$$

To calculate all the quantities at each grid point i, j, k with second order accuracy in space, it is required to calculate the gradients at the off-grid locations *i.e.* at half the cell size in each direction shown in Fig.5.3 as **X**. The values of $h(\phi)$ are determined at the off-grid positions by

averaging the values of the neighbouring nodes in each direction.

$$[(1-h(\phi))\nabla_{\mathrm{x}} c_{\mathrm{L}}]^{i+1/2,j,k} = \left[\frac{(1-h(\phi))^{i,j,k} + (1-h(\phi))^{i+1,j,k}}{2}\right]\left[\frac{c_{\mathrm{L}}^{i+1,j,k} - c_{\mathrm{L}}^{i,j,k}}{\Delta \mathrm{x}}\right] \tag{5.6}$$

Similarly, the anti-trapping current at the off-grid positions are also determined.

$$j_{at}^{i+1/2,j,k} = \frac{1}{2}\left([\alpha_{i,j,k} + \alpha_{i+1,j,k}]\left[\left(\frac{\partial \phi}{\partial t}\right)_{i,j,k} + \left(\frac{\partial \phi}{\partial t}\right)_{i+1,j,k}\right]\right)\frac{(\nabla_x \phi)_{i+1/2,j,k}}{|\nabla \phi|_{i+1/2,j,k}} \tag{5.7}$$

The discretization of the magnitude of the gradients at the off-grid positions are done by,

$$|\nabla \phi|_{i+1/2,j,k} = \sqrt{(\nabla_x \phi)^2_{i+1/2,j,k} + (\nabla_y \phi)^2_{i+1/2,j,k} + (\nabla_z \phi)^2_{i+1/2,j,k}} \tag{5.8}$$

To calculate the gradients $(\nabla_x \phi)_{i+1/2,j,k}$, $(\nabla_y \phi)_{i+1/2,j,k}$ and $(\nabla_z \phi)_{i+1/2,j,k}$, it is necessary to obtain the gradients in y and z directions respectively.

$$(\nabla_x \phi)_{i+1/2,j,k} = \frac{\phi_{i+1,j,k} - \phi_{i,j,k}}{\Delta \mathrm{x}} \tag{5.9}$$

$$(\nabla_y \phi)_{i+1/2,j,k} = \frac{(\nabla_y \phi)_{i+1,j,k} + (\nabla_y \phi)_{i,j,k}}{2} \tag{5.10}$$

$$(\nabla_z \phi)_{i+1/2,j,k} = \frac{(\nabla_z \phi)_{i+1,j,k} + (\nabla_z \phi)_{i,j,k}}{2} \tag{5.11}$$

To calculate $(\nabla_y \phi)$ and $(\nabla_z \phi)$ at i and $i+1$ positions, the following discretization is followed

$$(\nabla_y \phi)_{i,j,k} = \frac{\phi_{i,j+1,k} - \phi_{i,j-1,k}}{2\Delta \mathrm{y}} \tag{5.12}$$

$$(\nabla_y \phi)_{i+1,j,k} = \frac{\phi_{i+1,j+1,k} - \phi_{i+1,j-1,k}}{2\Delta \mathrm{y}} \tag{5.13}$$

$$(\nabla_z \phi)_{i,j,k} = \frac{\phi_{i,j,k+1} - \phi_{i,j,k-1}}{2\Delta \mathrm{z}} \tag{5.14}$$

$$(\nabla_z \phi)_{i+1,j,k} = \frac{\phi_{i+1,j,k+1} - \phi_{i+1,j,k-1}}{2\Delta \mathrm{z}} \tag{5.15}$$

In a similar manner, the gradients are calculated for other off-grid positions *viz.*, $(i-1/2, j, k)$, $(i, j+1/2, k)$, $(i, j-1/2, k)$, $(i, j, k+1/2)$ and $(i, j, k-1/2)$, which are elucidated in more detail in Choudhury (2012).

5.1.3 Stencils

The discretization of first and second order differentials and the requirement of gradient terms at the staggered grid locations reminds one of a similar method used for solving convection-diffusion problems Patankar (2007). The numerical techniques elucidated in Patankar's celebrated book Patankar (2007) to solve problems of this kind include the application of Tri-Diagonal Matrix Algorithm (TDMA), Alternate Direct Implicit (ADI) method and iterative schemes to evaluate nodal and staggered grid values. Unlike in the past, the numerical computation can be partitioned between the Central Processing Unit (CPU) and the Graphic Processing Unit (GPU) - both of which are usually available for a workstation. GPUs usually have a large number of compute cores that operate in parallel, a small memory and have a high latency to fetch data from the main memory. Therefore, a matrix based solver that runs on GPU would require frequent fetching of data over high latency paths. In the present study, a different numerical approach has been followed with a specific aim of harnessing the parallel processing potential of GPUs.

Instead of solving the problem by taking the entire array as input, a stencil based solver is employed. A stencil is created at every node point in the domain depending on the domain used. A 9-point stencil is created for 2-D (Fig.5.1) and a 27-point stencil for 3-D (Fig.5.2). The advantage of using a stencil based solver is that it can be ported across different platforms. The difference between solving on a CPU and a GPU is that CPU executes the program sequentially whereas the GPU executes concurrently. In both cases, the stencil based calculation is most effective. Also, all the desired points for the gradients and the Laplacian are present within the stencil and as a result all the calculations can be performed within the stencil for that particular node point (i, j, k).

In order to accommodate the points required for the gradients and the Laplacian, a 'Moore' neighbourhood stencil is chosen, where the number of points in the stencil grid is 9 for 2-D and 27 for 3-D, shown in Fig.5.1 and Fig.5.2, respectively. A stencil computation is characterized by updating each point in a structured grid by an expression, whose values depend on a fixed geometrical structure of neighbouring grid points. In our context, a rectilinear grid is assumed, which from a hardware architecture point of view is favoured due to the regularity of the data access pattern. This allows using the hardware prefetcher in an optimal way and streaming the data from the main memory to the compute elements Christen *et al.* (2012).

5.2 Seeding Configurations

Microstructure evolution occurs by the minimization of a free energy functional in phase-field simulations. Since the formation of a nucleus requires an initial increase in the free energy, nucleation cannot be treated implicit using this approach. Nucleation is generally not a part of phase-field formulation. There are a limited number of models (Gránásy *et al.* (2006), Eiken (2010)) that take nucleation into account as part of the phase-field formulation. The emphasis of this study is to analyse morphological patterns during growth and thermal cycling. Hence, the nucleation event has been considered as part of the initial condition.

In this study, seeds of nuclei are introduced by hand as part of initial condition of the domain. Different seeding configurations are chosen keeping in mind the symmetry of the problem and the computatational redundancy. The seeding configurations used for both 2-D and 3-D are shown in Fig.5.4.

a. The planar seed is used to perform equilibration studies to obtain a correlation between the gradient energy term and the solid-liquid surface energy.

b. The corner seed in 2-D is used to grow dendrites along the body diagonal of the square domain.

c. The central seed in 2-D is used to grow free dendrites with a known anisotropy and symmetry.

d. The corner seed in 3-D is used to grow one octant of a dendrite that could be mirrored along each face of the cubic domain to obtain the complete dendrite.

e. The array of three seeds at the bottom face of a 3-D domain is used to grow multiple dendrites to study the interaction between the dendrites during growth and coalescence.

f. The central seed in 3-D domain is used to ascertain the known anisotropy and symmetry of the free dendrite.

5.3 Program details

The target platforms were chosen based on the programming paradigms on which the simulations were executed were chosen to simulate dendritic growth. Two different computing paradigms were envisaged *viz.*, CPU and GPU and their architecture details are presented

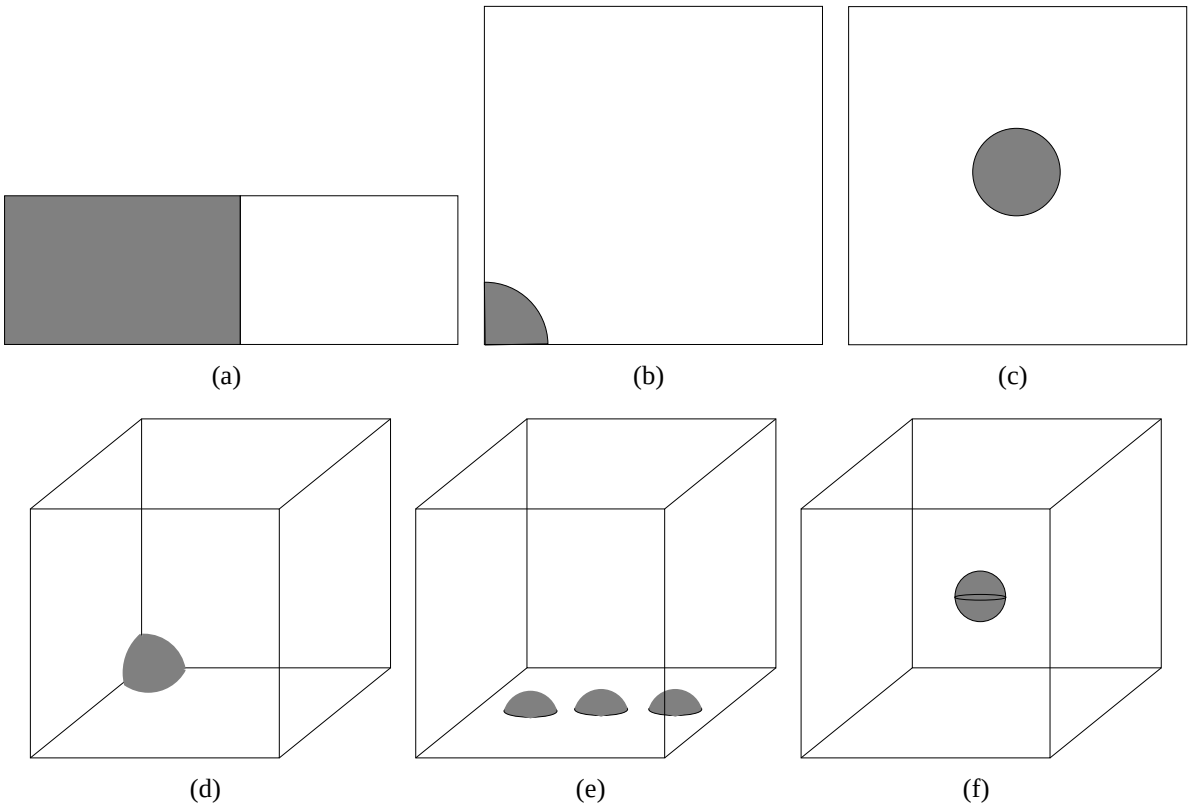


Figure 5.4: Different seeding configurations for 2-D: (a) planar, (b) corner, (c) center and 3-D: (d) corner of an octant, (e) base plane and (f) center. Solid is represented as dark regions and Liquid as white regions

Table 5.1: Cluster computing platform

System	IBM x iDataPlex dx360 M4
Chip architecture	Intel Xeon E5-2670
Type	superscalar, out of order
Clock	2.6 GHz
Sockets per node	2
Cores per socket	16
Compute nodes	292
Threads per socket	32
DRAM capacity per node	64 GB
DRAM pin bandwidth	51.2 GB/s

Table 5.2: GPU platforms

	NVIDIA M2070 (Tesla)	ATI Radeon HD5870 (Cypress)
Compute units	14	20
Processing elements	448	1600
Core clock frequency	1.147 GHz	850 MHz
Local Memory	48 KB	32 KB
Global Memory	6 GB	1 GB
Global Memory Bandwidth	148 GB/s	153.6 GB/s
Peak performance		
(float)	1.03 TFlops	2.72 TFlops
(double)	515.2 GFlops	544 GFlops

in Table 5.1 for the CPU cluster and Table 5.2 for the GPU cluster. The solvers were written in Fortran for the CPU and OpenCL for the GPU. Initial solvers were written as serial codes, but as phase-field computations are known for their computationally intensive nature, it is best to parallelize the solvers. Distributed computing using MPI routines were used for inter-processor communication and massively parallel computing using OpenCL. Implementation of the solvers on the CPU and the GPU are given in Algorithms 1, 2 and 3.

5.4 Non-dimensionalisation of the governing equations

Non-dimensionalisation or scaling is normalising of the governing equations by suitable substitution of variables for numerical convenience and computational stability. This is mainly done to simplify and parameterize problems and also to best understand or study quantities relative to an appropriate unit. In the models used, three different scales are used *viz.*, length, time and

Algorithm 1: CPU implementation: main program

Data: Structure $p_o(\phi, c)$,$p_n(\phi, c)$,$c_{\mathrm{SL}}(c_{\mathrm{S}}, c_{\mathrm{L}})$
Input: p_o
Output: p_n
MPI initialization: 1-D cartesian topology;
Domain initialization and decomposition;
foreach *timestep t* **do**
  Call MPI exchange: p_o;
  Calculate c_{S} and c_{L} using Eq.3.17 and Eq.3.18;
  Calculate $\frac{\partial \phi}{\partial t}$ from **Algorithm 2**;
  Apply boundary conditions;
  Call MPI exchange: $\frac{\partial \phi}{\partial t}$;
  Calculate ϕ_n and c_n from **Algorithm 2**;
  if *t*> *ntimesteps* **then**
    Call noise (random fluctuations) for secondary arms;
  end
  Apply boundary conditions;
  Update: $p_o \longleftarrow p_n$;
end

energy. Length is scaled on the capillary length $\tilde{l} = l/d_0$.

$$d_0 = \frac{(\sigma + \sigma_{\theta\theta})}{\frac{\partial^2 f_{\mathrm{L}}}{\partial c_{\mathrm{L}}^2}(c_{\mathrm{L}}^e - c_{\mathrm{S}}^e)} \tag{5.16}$$

where σ is the surface tension and $\sigma_{\theta\theta}$ is the stiffness. Time is scaled upon the diffusive time $\tilde{t} = tD_L/d_0^2$ and energy is scaled as $\tilde{f} = f/(RT^*/v_m)$ where R is the universal gas constant R=8.314 $\mathrm{Jmol^{-1}K^{-1}}$ and T^* is the reference temperature.

Substituting these scaling units in Eq.3.21 and replacing c with x_{B}/v_m

$$\frac{1}{v_m}\frac{\partial x_B}{\partial\left(\tilde{t}\frac{d_0^2}{D_L}\right)} = \frac{1}{v_m}\frac{\tilde{\nabla}}{d_0}\left[h(\phi)D_S\frac{\tilde{\nabla}}{d_0}x_B^S\right] + \frac{1}{v_m}\frac{\tilde{\nabla}}{d_0}\left[(1-h(\phi))D_L\frac{\tilde{\nabla}}{d_0}x_B^L\right] \tag{5.17}$$

where $\tilde{\nabla} = \frac{\nabla}{1/d_0}$

Rearranging the terms will give,

$$\frac{\partial x_B}{\partial \tilde{t}} = \frac{d_0^2}{D_L}\left(\frac{\tilde{\nabla}}{d_0}\left[h(\phi)D_S\frac{\tilde{\nabla}}{d_0}x_B^S\right] + \frac{\tilde{\nabla}}{d_0}\left[(1-h(\phi))D_L\frac{\tilde{\nabla}}{d_0}x_B^L\right]\right) \tag{5.18}$$

Algorithm 2: CPU implementation: Solvers

Data: Structure $p_o(\phi, c)$,$p(\phi, c)$,$c_{\mathrm{SL}}(c_{\mathrm{S}}, c_{\mathrm{L}})$
Solver: $\frac{\partial \phi}{\partial t}$, $p_n(\phi, c)$;
Input: p_o,c_{SL}
Output: $\frac{\partial \phi}{\partial t}$
foreach Element *in i,j* **do**
    Create: Stencil(p_o,c_{SL});
    function $\frac{\partial \phi}{\partial t}$ solver
        Calculate: $\nabla^2\phi$,$\nabla\phi$;
        Solve: RHS of Eq.3.19;
    end
end
Input: p_o,c_{SL},$\frac{\partial \phi}{\partial t}$
Output: $p_n(\phi, c)$
foreach Element *in i,j* **do**
    Create: Stencil(p_o,c_{SL},$\frac{\partial \phi}{\partial t}$);
    function $[\phi, c]$ solver
        Solve: for ϕ and c in Eq.3.19 and Eq.3.23;
    end
end

Algorithm 3: GPU: Solver kernels

Data: Structure $p_o(\phi, c)$,$p(\phi, c)$,$c_{\mathrm{SL}}(c_{\mathrm{S}}, c_{\mathrm{L}})$
Input: p_o
Output: p_n
foreach *timestep t* **do**
    foreach Element *in grid* **do**
        Call c_{SL} kernel;
        foreach Element *in grid excluding boundaries* **do**
            Call $\frac{\partial \phi}{\partial t}$ kernel;
            Map $\frac{\partial \phi}{\partial t}$ boundaries ;
            Call (ϕ, c) kernel;
            Map $p[\phi, c]$ boundaries ;
        end
        Call noise kernel;
    end
    Swap buffers $p_0 \longleftarrow p_n$;
end

$$\frac{\partial x_B}{\partial \tilde{t}} = \tilde{\nabla}\left[h(\phi)\frac{D_S}{D_L}\tilde{\nabla}x_B^S\right] + \tilde{\nabla}\left[(1-h(\phi))\frac{D_L}{D_L}\tilde{\nabla}x_B^L\right] \tag{5.19}$$

Similar substitutions can be made for the phase-field equation, Eq.(3.19).

5.5 Dendrite tip radius calculations

Validating dendritic growth using LGK analysis (Lipton *et al.* (1984)) requires the determination of the tip velocity and radius. This is done by extracting the phase-field contours at ϕ=0.5, which is usually taken as the interface. Finding the maximum point of the curve is straightforward in the case of a dendrite oriented along the axis. In cases, where the dendrite is oriented at an angle to the axes, the maximum point is obtained using Algorithm 4.

Algorithm 4: Determination of tip

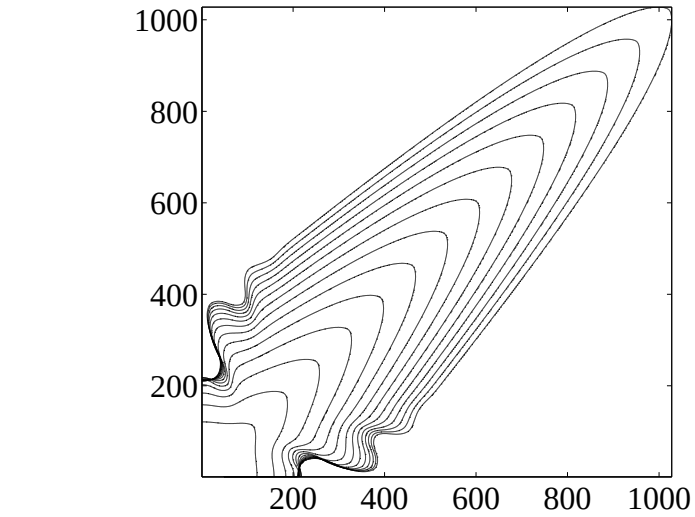
```
[x_max,x_max_count] = max(x);
[y_max,y_max_count] = max(y);
minimum_distance = x_max^2 + y_max^2;
foreach i = x_max_count : y_max_count do
    if [x(i)-x_max]^2 + [y(i)-y_max]^2 < minimum_distance then
        count_tip = i;
        minimum_distance=[x(i)-x_max]^2 + [y(i)-y_max]^2;
    end
end
tip=x(count_tip),y(count_tip);
```

The dendrite, not being a perfect parabola, a second order polynomial will not provide a good fit to the profile as shown in Fig.5.5(b). Hence, higher ordered polynomials are used for the fit. To do this, from either side of the tip, 8 to 12 points are chosen and the polynomial is fit. In our case, a fourth order polynomial was used to fit the profile. The calculations are made simpler by rotating and translating the dendrite tip to the origin. This simplifies the equation to determine the curvature from Eq.(5.20) to Eq.(5.21), where a_3 and a_4 are the coefficients of the polynomial.

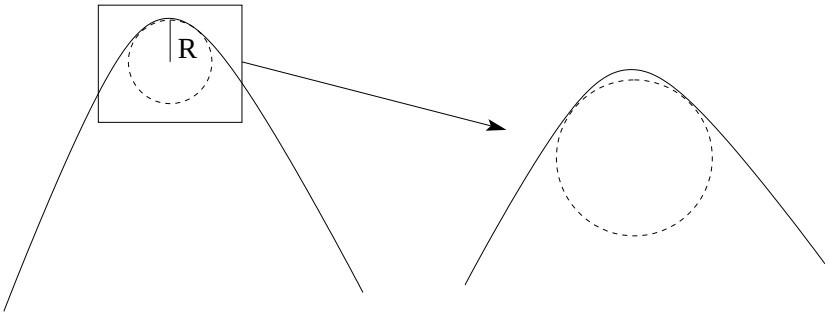
$$\kappa = \frac{d^2y/dx^2}{\left[1+(dy/dx)^2\right]^{3/2}} \tag{5.20}$$

$$\kappa = \frac{2a_3}{\left[1+(a_4)^2\right]^{3/2}} \tag{5.21}$$

The velocity of the tip is calculated by storing the tip values at each iteration. A steady state



(a)



(b)

Figure 5.5: (a) Phase-field contours showing dendrite evolution. The dendrite is grown from a seed placed in the corner of the computational box. The contours are taken at ϕ=0.5. (b) Inset shows that a second order polynomial will not provide a good fit for the dendrite tip

growth velocity will be obtained similar to one shown in Fig.5.8.

5.6 Parallel implementation

Parallel computing is based on the principle that a larger problem can be broken down into smaller ones, which are then solved concurrently. Parallel computation is necessary to distribute the load on several processors and more often to bring down the computational time. There are two types of memory models for achieving parallelism *viz.*, shared memory and distributed memory. Shared memory model is one in which the entire global memory is available for all the processors to view, whereas in a distributed memory model, the global memory is

not available, but have access to local memory. The Message Passing Interface (MPI), which comes under the distributed memory model, is a standardized and portable message passing system. The standard defines the syntax and semantics of a core of library routines in writing portable message passing programs.

5.6.1 Domain Decomposition

A one dimensional domain decomposition was done for 2-D and 3-D and is shown in Fig.5.6. It is necessary to perform a domain decomposition depending on how the compiler scans an array. Since a Fortran compiler scans the array in column major order, domain decomposition was done column-wise with the ghost cells along left and right. In the case of 3-D, the domain was decomposed along the depth, with ghost cell planes on the top and bottom as shown in Fig.5.6. In order to avoid the hassle of explicitly defining which processor should host which part of the domain, it is more convenient to split the global domain based on a virtual topology and let the MPI routines take care of the tasks. The mpi_cart routines supplied by MPI allow efficient management of virtual topologies, making the use of 2D and 3D decompositions easy and efficient. The global domain is decomposed into individual domains across column-wise direction in 2D and depth-wise (Z direction) in 3D. All ghost exchanges between the processors are done with a non-blocking send (mpi_isend) and a blocking receive (mpi_recv) which provides a better time optimization than the blocked communication (Jacobsen and Senocak (2013)).

5.6.2 Datatype creation

In order to send and receive data, the MPI routines first packs the data and then sends the packet. The padding used to round off the size of the packet depends on the library. This implies that a structure with a mix of datatypes could lead to different offsets required for each variable within a structure across libraries. In order to ensure that the structure data is sent and received with the same convention, it is important to convey the datatype structure of all derived datatypes to the MPI routines. For example, mpi_send or mpi_receive needs to know what type of data requires to be packed *viz.* integer, float or double. This requires that a separate datatype to be created by the MPI to hold this structure/ derived datatype, depending on the

number variables present in a structure. Below is an example code that shows how to create a MPI derived datatype with two elements in the structure.

```
offsets(1)     = 0
types(1)       = mpi_real
blockcounts(1) = 2

call mpi_type_extent(mpi_real, extent, ierr)

offsets(2)     = 2*extent
types(2)       = mpi_ub
blockcounts(2) = 1

call mpi_type_struct(2, blockcounts, offsets, types, nodeDataMPI, ierr)
call mpi_type_commit(nodeDataMPI, ierr)
```

where nodeDataMPI is the mpi_datatype to be created which holds a structure comprising of two float(C) or real (Fortran) elements. mpi_type_extent calls the size of the datatype real and mpi_ub is the upper bound of the derived datatype and seals the mpi_datatype.

5.6.3 Cartesian Topology of nodes

In order avoid the hassle of explicitly defining which processor should host which part of the domain, it is more convenient to split the global domain based on a virtual topology and let the MPI routines take care of the tasks. A 1- D cartesian topology suggests that the domain be split up along columns or rows and since Fortran is the programing language used, we split the domain along columns as shown in Fig. 5.6. Once done, we can assign node names accordingly *viz.*, the current node as me and the adjacent nodes left and right respectively.

```
call mpi_cart_create(oldcomm, ndims, dimsize, periods, reorder, newcomm, ierr)
```

Here, oldcomm is the initialized communicator handle, mpi_comm_world; ndims is the number of dimensions as (1,2,3); dimsize is the total number of processors/ nodes for the computation; periods, to determine whether the end nodes are periodic or not; reorder, to allow MPI to reorder the nodes and all this information is contained in the new handle newcomm which is the cartesian communicator whose topology is along one dimension and the number of domains being the number of processors.

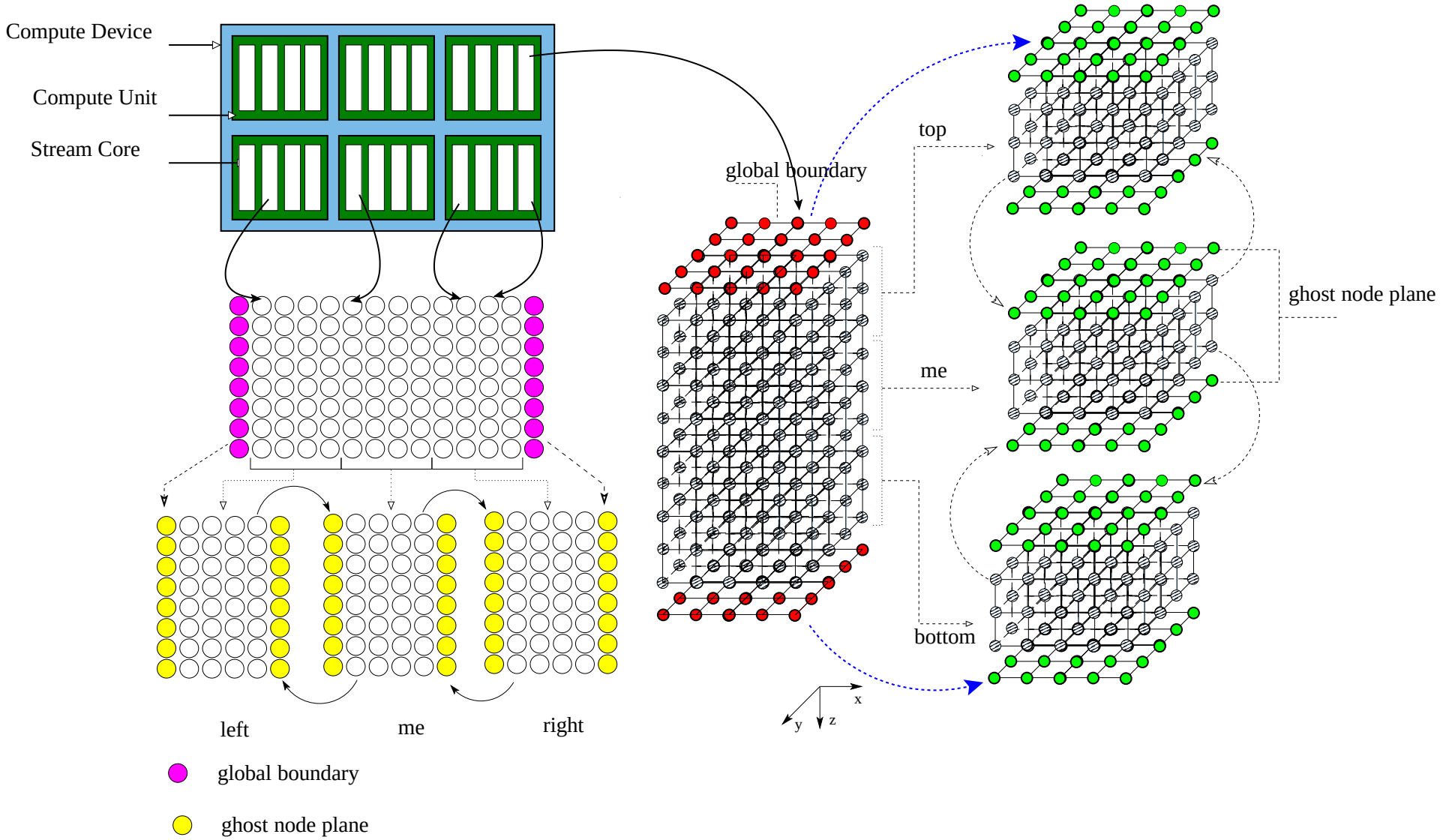


Figure 5.6: Implementation on a CPU and a GPU. A 1-D domain decomposition is used for both 2-D as well as 3-D arrays. Also shown is the GPU architecture and how the domain is solved on a GPU

5.6.4 Domain splitting

After creating the cartesian topology, the global domain is mapped on to local domains by identifying the processor rank and automatically assigning the slices as per the processor ranks. This avoids the traditional usage of slicing and sending the elements through a master and slave communication.

5.6.5 MPI communication and ghost nodes

Having created the slices alone is not sufficient to proceed, since there are issues in solving for the gradients and Laplacians at the end nodes. These elements require the adjacent nodes for the gradient calculations which is available only in the next processor. For example, to calculate the gradients at the rightmost column of each processor, we need the first column of elements from the next processor to the right. In order to effect this, we need to communicate those particular elements to be sent and received in the corresponding processors. But to achieve this, it is necessary to have an empty column to host these elements in the respective local domains. Hence, in creating the local domain, it is necessary to allocate the left and right columns on either side of the domain to be reserved for the transfer. These reserved nodes are called *ghost nodes*. To facilitate this transfer, two paired communications $\mathrm{mpi_send}$ and $\mathrm{mpi_recv}$ favour this exchange. The boundary nodes are also exchanged with the global boundaries. Example calls to relevant library functions are given below.

```
call mpi_send(data(index_send), count, MPI_datatype, destination, tag, communicator_handle,
ierr)
call mpi_recv(data(index_receive), count, MPI_datatype, source, tag, communicator_handle,
status, ierr)
```

Here $\mathrm{data(index_send)}$ is the starting index value of the transfer and count is the number of elements to be transferred from the index start value, $\mathrm{destination}$ is the processor to which it is to be sent, the tag is used for identifying who is sending whom and the $\mathrm{communicator_handle}$ is the cartesian topology communicator handle which in our case is the $\mathrm{newcomm}$.

MPI data receive is the reverse of send; it specifies the allocation point, $\mathrm{data(index_receive)}$ and the number of elements required for the receive i.e. count. In order for the transfer between send and receive to be effective, it is necessary to know which processor is sending to whom

and the subsequent receive. This is made possible by the tag which ensures that the right processors are involved in the exchange.

5.6.6 Domain decomposition for a 3- D array

In the case of 3-D, $\mathrm{Fortran}$ scans through the array in a layer wise fashion, where the order of fastest moving indices are column, row and then plane respectively as indicated in Fig. 5.6. The communication will be in a top to bottom fashion. Here instead of 1-D ghost array column, we have Nx$\times$Ny ghost array planes which receive and store the end points. Here the slices are cubic chunks of elements passed to different processors.

In the case of parallelization of the diffusion equation with the anti-trapping current, $\frac{\partial \phi}{\partial t}$ is required at every node point. For this, a two stage parallelization is done; the first step involves communicating the ϕ and x_B values for calculating $\frac{\partial \phi}{\partial t}$ and in the second step communicating $\frac{\partial \phi}{\partial t}$ with the adjacent processors. This process though involves double communication, is a fairly simple exchange keeping the number of ghost cells to just a singular array/ plane.

5.6.7 Parallel implementation on a GPU

Modern processor architectures are now moving to the era of parallelism for increased performance. CPUs improve their performance by adding multiple cores but faced with technical challenges as higher clock speeds in a fixed power envelope. Graphics Processing Units (GPUs) have also evolved from fixed function rendering devices into programmable parallel processors. CPU based parallel programming models are based on standards but usually assume a shared address space and do not generally encompass vector operations.

Implementation on a GPU proceeds in a rather straight forward manner, where the entire computational domain is processed by stream cores or the processing elements.

Each node/ grid point in the domain is processed individually by a stream core as opposed to the CPU where a chunk of a domain is evaluated by each processor/ core. For example, if it takes to 1 second for a calculation on a node, then for a domain chunk of 10$\times$10 will take 1$\times$10$\times$10 = 100 seconds; whereas the 10$\times$10 domain will be solved in 1 second on a GPU, the scalability depending on the number of stream cores present.

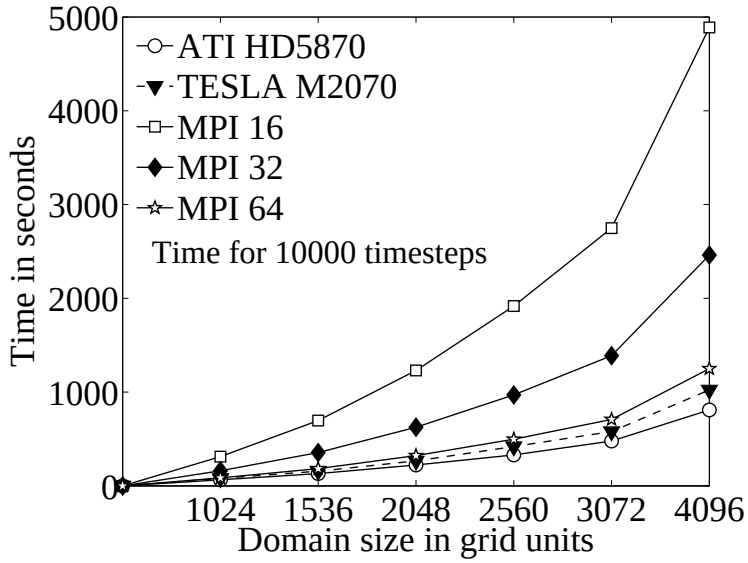


Figure 5.7: Performance benchmark for a parallel implementation of the solver across multiple CPUs and a single GPU

The performance benchmark between CPU and GPU for solving the governing equations incorporating the anti-trapping current was done and the scalability of the domain is well highlighted in Fig.5.7. It can be seen that the scalability issues are well addressed in the case of a GPU. However, to efficiently reap the benefits of a GPU, it is necessary to address the issues pertaining to memory usage since the programming paradigm on a GPU is purely memory based. Memory issues do not pop up in 2-D simulations when attempting a naive parallelism, but they become painfully huge when dealing with 3-D arrays. Efficient algorithms are required to move through the domain by keeping the memory usage at a minimum.

Since our attempts at GPU programming are still in its infancy, we have stuck to the naive parallelism method and for 3-D domains due to memory restrictions on the present GPU, domain sizes greater than 512^3 were unable to be processed. To check the convergence of the results on the CPU and the GPU, dendrite tip velocities were plotted for two sets of interface widths. The invariance of the simulation results are presented in Fig.5.8.

5.6.8 Memory model of a GPU

The memory model of a GPU depends on its architecture as shown in Fig.5.9. Following is a basic outlook of the programming methodology as adopted by the author on a GPU using OpenCL as the programming language. A naive parallelism approach method is followed

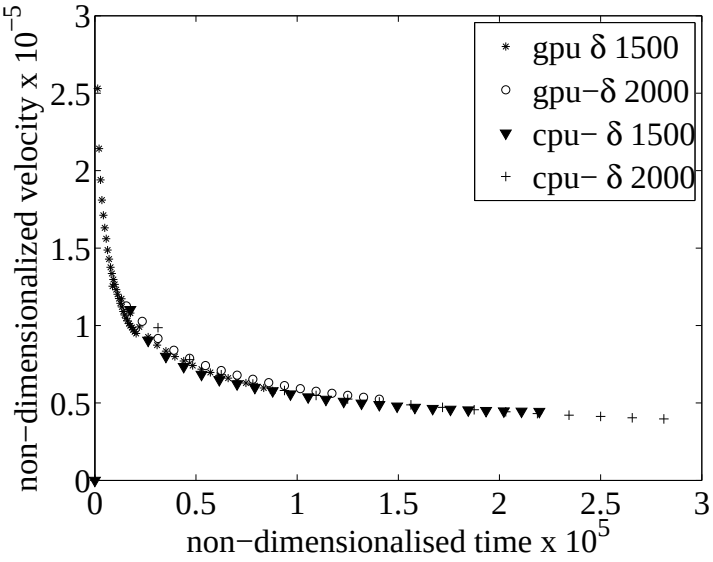


Figure 5.8: Convergence of dendrite interface velocities from numerical codes executed on a GPU (OpenCL - NVIDIA Tesla M2070T) and CPU (Fortran MPI - 64 processors)

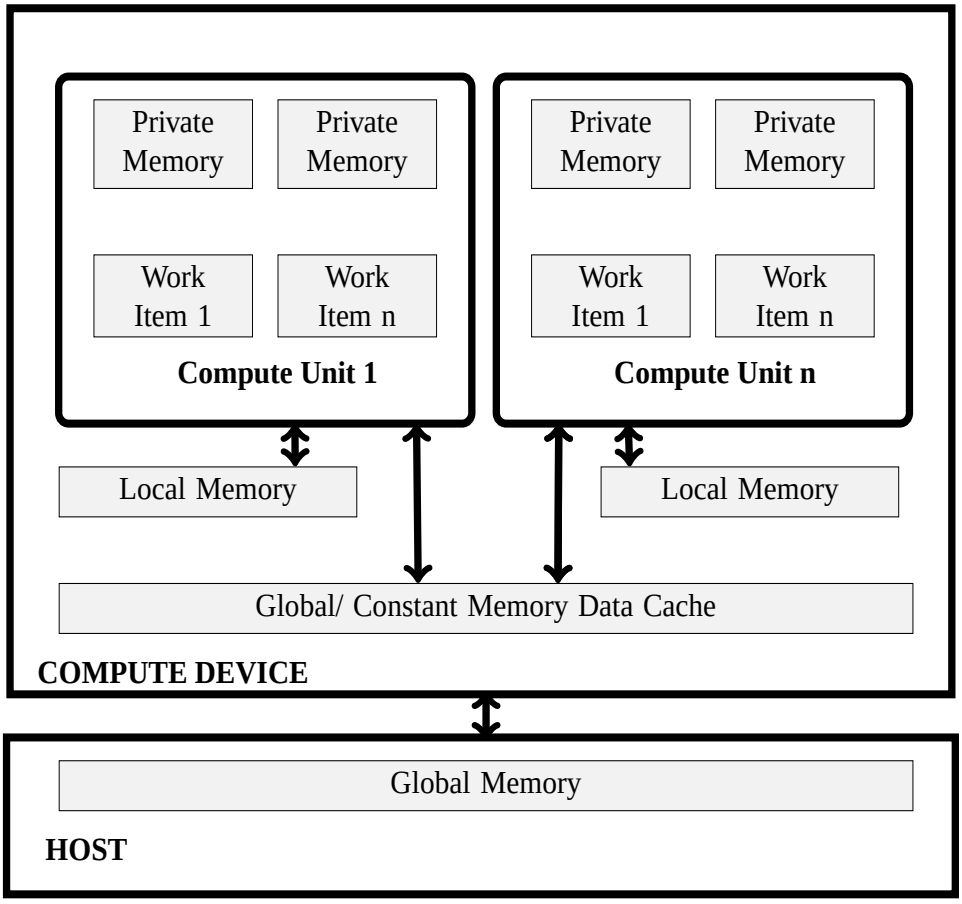


Figure 5.9: Memory model of a typical GPU with reference to OpenCL terminologies

which involves accessing the global memory cache and OpenCL being a low level programming language, it ends with the instruction set being quite large.

- Initialize the domain. Allocate memory for the required data structures.
- Identify the compute platforms where the processing needs to be done (CPU,GPU) and get their respective device ids.
- Create a context under which the instructions to be processed are executed. Different contexts can be created on different devices or the same context can be extended to different devices.
- Inside the context, a command queue is created in which the prioritization of instructions are input.
- Memory buffers are then created which hold the respective structures (phase-field, concentration, c_{S}, c_{L}), which can be designated as read only, write only and read write.
- The buffers are then queued up on the device memory.
  The sample code below demonstrates the querying of platforms, the devices present, creation of contexts, command_queues, buffers and writing buffers on to the device memory.

```
/* ------------------------------------------------------------------ */
/* Create context and command_queue on the device */
/* ------------------------------------------------------------------*/
clGetPlatformIDs(1, &platform_id, &return_num_platforms);
clGetDeviceIDs(platform_id, CL_DEVICE_TYPE_GPU, no_of_gpus, device_id,
&ret_num_devices);

context=clCreateContext(NULL, 1, &device_id, NULL, NULL, &err);
command_queue = clCreateCommandQueue(context, device_id, 0, &err);

device_buffer = clCreateBuffer(context,
            CL_MEM_READ_WRITE | CL_MEM_COPY_HOST_PTR,
            array_size*sizeof(data_type), array_pointer, &err);

clEnqueueWriteBuffer( command_queue, device_buffer, CL_TRUE, 0,
                array_size*sizeof(data_type), array_pointer, 0, NULL, NULL );
```

- In OpenCL, the .cl kernel file is read only during execution and hence the program has to be built during runtime and a buffer file has to be created to hold the errors, if any, which pop up during the compilation of the kernel file during runtime. This is shown below as a snippet.

```
#define MAX_SOURCE_SIZE (0x100000)
FILE *fp;
const char fileName[] = "solver_kernel.cl";
size_t source_size;
char *source_str;

/************* Load kernel source file ******************/

fp = fopen(fileName, "rb");
source_str = (char *)malloc(MAX_SOURCE_SIZE);
source_size = fread(source_str, 1, MAX_SOURCE_SIZE, fp);
fclose(fp);

/************* Create kernel from source ******************/

program = clCreateProgramWithSource(context, 1,
(const char **)&source_str, (const size_t *)&source_size, &err);

/************* Build the program for each device ***********/

clBuildProgram(program, 1, &device_id, NULL, NULL, NULL);
char buildString[100000];
clGetProgramBuildInfo(program, device_id,
CL_PROGRAM_BUILD_LOG, sizeof(char) * 100000, buildString, NULL);
```

- The kernel files are created which are nothing but the functions that are to be called from the host code and all the kernel arguments are also needed to be set as shown below.

```
/* ------------------------------------------------------------*/
/* Create kernels */
/* ------------------------------------------------------------*/

kernel[0] = clCreateKernel(program, "solver_1", &err);
kernel[1] = clCreateKernel(program, "solver_2", &err);
.
.
.
kernel[n-1] = clCreateKernel(program, "solver_n-1", &err);
/* ------------------------------------------------------------*/
/* Set kernel arguments */
/* ------------------------------------------------------------*/
clSetKernelArg(kernel[0], 1, sizeof(int), &nx);
clSetKernelArg(kernel[0], 2, sizeof(int), &ny);
clSetKernelArg(kernel[0], 3, sizeof(int), &nz);
.
.
```

- Execution of the program start here and the host calls the required kernels and instructs the compute device to execute the instruction set in the respective order. Kernels are submitted for execution using clEnqueueNDRangeKernel specifying the dimensions *viz.*, workdim = 2,3 for 2-D or 3-D, respectively.
- All kernel calls are set to execute with the command clFush and clFinish.

```
for(t=1;t<=ntimesteps;t++){
    err=clEnqueueNDRangeKernel( command_queue, kernel[0], workdim, NULL,
    array_size, NULL, 0, NULL, NULL );
    err=clEnqueueNDRangeKernel( command_queue, kernel[2], workdim, NULL,
    array_size, NULL, 0, NULL, NULL );
    .
    .
    err=clEnqueueNDRangeKernel( command_queue, kernel[n-1], workdim, NULL,
    array_size, NULL, 0, NULL, NULL );
    /* ------------------------------------------------*/
    /* Read buffer after specified time interval
    /* ------------------------------------------------*/
    if((t%saveInterval)==0){
        err=clEnqueueReadBuffer ( command_queue, buffer_on_device, CL_TRUE, 0,
        array_size*sizeof(data_type), buffer_to_be_read_into, 0, NULL, NULL );
    }
}
err=clFlush(command_queue);
err=clFinish(command_queue);
```

5.7 Parallel implementation across multiple GPUs

For simulating large domains both in 2-D and 3-D, it is often necessary to decompose the domain in order to bring down the memory overload and also the computation time as discussed earlier. To avoid this, a simpler version would be to distribute the load across multiple GPUs. There are many ways of splitting the data across the nodes. One is to create a single context and use different buffers on each of the GPUs in a compute node. This method cannot to scaled as it is restricted by the number of GPUs in a compute node. The other method is to tag each CPU to a GPU by combining the methodologies in Section 5.6.3 and Section 5.6.7. A MPI virtual topology is created and a 1-D domain decomposition is done as shown in Fig.5.10. Depending on the number of GPUs present in a compute node, the tagging of GPU to a CPU is done as described in Algorithm 5. Each MPI process is given a part of the domain with each MPI process to hold two additional arrays functioning as ghost nodes. Before the start of each iteration, the boundary layers need to be transferred to adjacent nodes. This is done by a memory transfer from device to host using clEnqueueReadBuffer to a 1-D array and then sending that array to the adjacent node. The adjacent MPI process receives the array on to a buffer and then writes the buffer to the main array through clEnqueueWriteBuffer. The extreme MPI processes map the boundary elements depending on the type of boundary condition. (n-1)th row elements are mapped to nth row elements in a no-flux boundary condition.

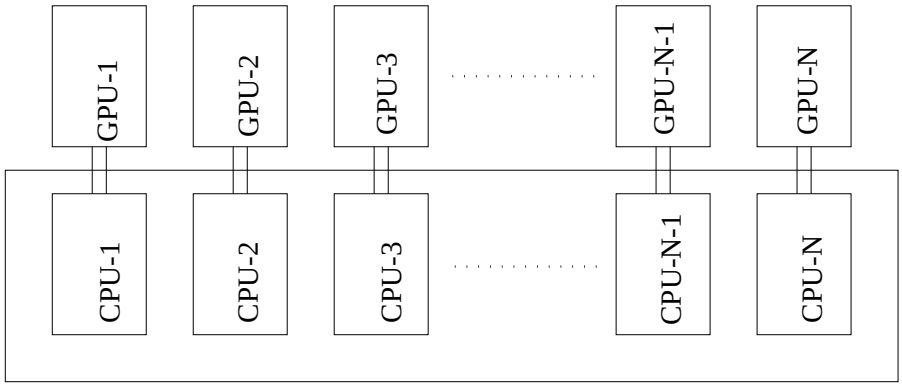


Figure 5.10: Parallel implementation across multiple GPUs using MPI+OpenCL

Algorithm 5: Host mapping to GPU where n is the total number of GPUs present in each compute node

```
if mpi_rank%ngpus = 0 then
    device_id=0;
else if mpi_rank%ngpus = n-2 then
    device_id=n-2;
else
    device_id=n-1;
end
```

5.8 Data handling and visualization

The individual chunks of output data from each processor are collected as unformatted binary files from the respective processors. Data stitching is done by reading the individual .bin files and writing them into a single (.mat) file, using MatIO, a library for writing and reading .mat files from C or Fortran, (Hulbert (2013)). For 2-D, the output .mat files are visualized in Matlab®.

For 3-D, however, Matlab®required a huge memory overhead for reading the data and it was found that **MayaVi** offeres a much simpler and an easier interface for importing and visualizing 3-D data. MayaVi, sits on top of VTK (Visualization Toolkit, Schroeder *et al.* (1993–)) an open-source, freely available software system for 3-D computer graphics, image processing and visualization.

VTK consists of a C++ class library and several interpreted interface layers including Tcl/Tk, Java, and Python. MavaVi, a package in Python, proved versatile since data files in any format can be imported. More references to **MayaVi** can be found at (Ramachandran (2005*a*,*b*);

Ramachandran and Varoquaux (2008*a*, 2005–, 2008*b*)).

The native format of VTK (.vtk) can be processed through any application which runs on VTK. In this context, ParaView (Sandia National Laboratory *et al.* (2002–)), an open-source, multi-platform data analysis and visualization application, and known for quickly building visualizations was used to analyze data using qualitative and quantitative techniques. The data exploration can be done interactively in 3D or programmatically using ParaView's batch processing capabilities. ParaView was developed to analyze extremely large datasets using distributed memory computing resources and can be run on supercomputers to analyze datasets of terascale as well as on laptops for smaller data.

5.9 Parameter optimisation

In the case where numerical simulations are concerned, it boils down to one simple fact *"garbage in = garbage out"*. If system parameters are not chosen properly then we may end up with unrealistic microstructural features as depicted in Fig.5.11. Hence, careful choice of the parameters *viz.*, $\Delta \mathrm{x}, \Delta \mathrm{t}, \delta$ is necessary to obtain both qualitative and quantitative information from the profiles. The artefacts shown in Fig.5.11 are a result of the effect of increasing $\Delta \mathrm{x}$. Although it is safer to continue the simulations with a smaller $\Delta \mathrm{x}$, thereby avoiding the hassle of parameter optimisation, it is well known that phase-field simulations are computationally intensive and a smaller $\Delta \mathrm{x}$ would mean a larger domain size, which invariably leads to increased domain and computational times as $\Delta \mathrm{t}$ is also dependent on $\Delta \mathrm{x}$. To reduce the computational time, $\Delta \mathrm{t}$ also has to be increased, but the cap is the Neumann stability requirement, namely, $\Delta \mathrm{t} < \Delta \mathrm{x}^2/2$. In our simulations, which deals with stiff differential equations and also due to the CALPHAD coupling, in most cases, this stability limit has to be kept well below the cap, which in our case $\Delta \mathrm{t} < \Delta \mathrm{x}^2/5$.

One more restriction, is the choice of interface width δ where it is quite well known that though we can go for larger δ in the presence of the anti-trapping current, it also has a limit on the δ as a function of undercooling. In the limit of lower undercooling, the domain sizes have to be increased for simulating secondary arms which in turn increases the computational time. For higher undercoolings, though $\Delta \mathrm{x}$ and δ have to be reduced, evolution time is greatly reduced because of higher supersaturation, but $\Delta \mathrm{x}$ needs to be reduced accordingly. Fig.5.11 depicts the

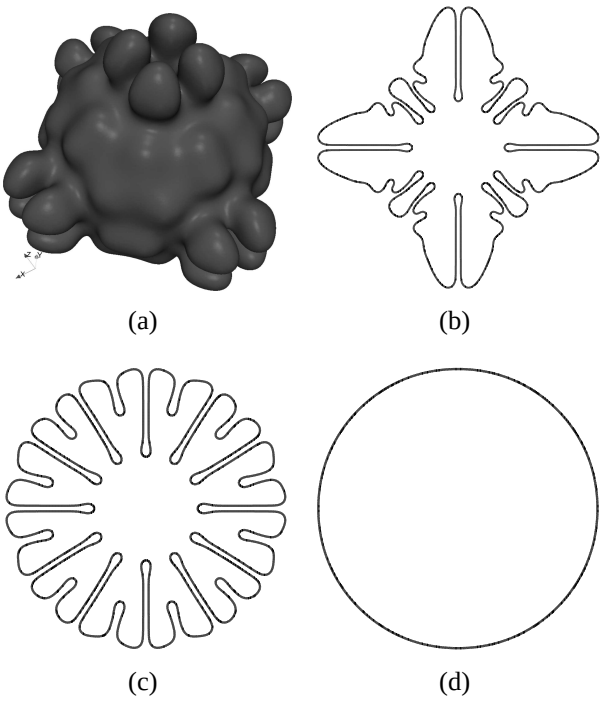


Figure 5.11: Tip splitting (a),(b) and (c) is an isotropic surface energy growth profile without any noise, as an effect of larger dx values, whereas an isotropic growth profile in the computational domain should produce a cleaner representation as in (d) thereby on the need for a finer grid resolution

issues encountered when choosing larger Δx values leading to tip splitting artefact. A simpler check of the choice of Δx is to grow an isotropic seed which should grow into a circle in 2-D or a sphere in 3-D as shown in Fig.5.11(d). Simulations executed with these large values gives a microstructural pattern with artefacts that are not supposed to be present.

5.10 Noise

During actual growth conditions, due to thermal/ solutal fluctuations in the solidifying melt, perturbations set in along the primary arm of the dendrite. These perturbations if favoured, get amplified and grow into secondary arms/ branches. To simulate the effect of perturbations, numerical noise is introduced along the liquid side of the interface ($0.1 < \phi < 0.5$), shown in Fig.5.12.

Introducing noise in the CPU calculations is quite straightforward. A random number is generated between -1 and 1 and then added to the phase-field value ϕ. It is necessary to check that the noise introduced does not affect the tip velocity in comparison with the primary growth.

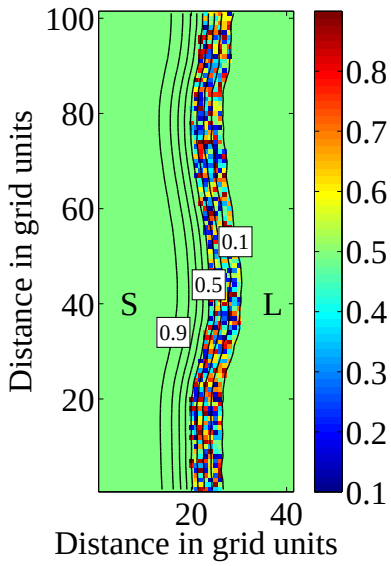


Figure 5.12: Noise introduced along the liquid side of the interface given by the phase-field contours ($0.1 < \phi < 0.5$). S represent Solid and L, Liquid

This is done by summing up the entire noise and averaging it out. This is then added to each point along the liquid side of the interface. The amplitude of the noise also plays an important role in defining the pattern formation. The noise amplitude needs to be kept well within 1% so as not to amplify the tip velocity.

In OpenCL, there are no native random number libraries and hence, noise has to be written as separate kernels. Noise was generated by using a XOR shift random number generator written as a kernel. They generate the next number in their sequence by repeatedly taking the exclusive OR of a number with a bit shifted version of itself which makes them extremely fast on modern computer architectures. Two constant random integers are initialized within the kernel and a seed is generated using global_id(0). The parameters for every thread were obtained by using two random numbers which remain the same initialized in the kernel, the global ID of the thread and a heuristically chosen 32bit number and then reduced to a float type number between -1 and 1.

The numerical techniques described in this chapter enabled simulation of 3-D dendrites on a desktop computer and a set of 3-D dendrites on a GPU cluster within reasonable wall-time durations. The role of numerical optimizations to achieve realistic morphological simulations cannot be underestimated.

CHAPTER 6

CASE STUDIES IN 2-D

In this chapter, we study the dendrite fragmentation and evolution of a fine globular microstructure that takes place during a semi-solid processing technique such as electromagnetic stirring in a rheocasting setup through phase-field modelling. Before proceeding, it is necessary to validate the simulations in order to quantify the morphological features. Apart from numerical parameter stabilization, a thermodynamic validation of the morphological features is required. To proceed with the simulations, it is required to arrive at the choice of the model parameters used *i.e.*, W and ε against the interface width δ chosen.

6.1 Determination of W and ε

To connect the equilibrium properties with the model parameters W and ε for a particular choice of the interface width δ and surface tension σ_{SL}, which is a physical quantity, a planar growth is simulated by splitting the computational domain into two halves. Values for the surface tension σ_{SL} for Al-Mg and Al-Cu binary alloys are taken from Gunduz and Hunt (1985). The left half being solid and the right being liquid, in their respective equilibrium compositions, shown in Fig.5.4(a). The system is allowed to equilibrate by setting the diffusivities of both solid and liquid to be the same ($D_{\mathrm{S}} = D_{\mathrm{L}}$). At equilibrium, the driving force for phase transformation $\left(f^{\mathrm{S}} - f^{\mathrm{L}} + \mu(c_{\mathrm{S}} - c_{\mathrm{L}})\right) = 0$. The equilibrium phase-field profile ϕ_0, is the solution to the phase-field equation, Eq.(3.19) at steady state $\dfrac{\partial \phi}{\partial t} = 0$ which gives,

$$\varepsilon^2 \frac{d^2\phi_0}{d\mathrm{x}^2} = W \frac{dg_w(\phi_0)}{d\phi_0} \tag{6.1}$$

With $g_w(\phi) = \phi^2(1-\phi)^2$, under the conditions of $\phi_0 = 1$(solid) at $x \to -\infty$ and $\phi_0 = 0$(liquid) at $x \to \infty$, the equilibrium phase-field profile is given by,

$$\phi_0(\mathrm{x}) = \frac{1}{2}\left[1 - \tanh\left(\frac{\sqrt{W}}{\sqrt{2\varepsilon}}\right)\right] \tag{6.2}$$

The equilibrium concentration profile is then given by

$$c_0(\mathrm{x}) = h(\phi_0(\mathrm{x}))c_\mathrm{S}^e + (1 - h(\phi_0(\mathrm{x})))c_\mathrm{L}^e \tag{6.3}$$

where $h(\phi_0) = \phi_0^3(6\phi_0^2 - 15\phi_0 + 10)$. At equilibrium, the surface tension (σ_{SL}) is also retrieved as shown in Fig.6.3(a) for Al-Cu and Fig.6.3(b) for Al-Mg. The simulated and sharp-interface profiles of phase-field and concentration are shown in Fig.6.3 and Fig.6.2 for Al-Cu and Al-Mg, respectively.

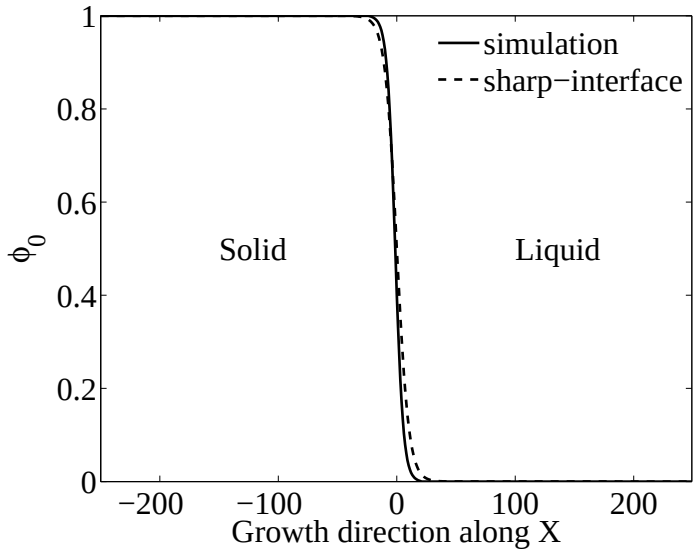
The model parameters having fixed, benchmark studies need to be performed to validate dendritic growth. Two studies are necessary to benchmark the simulations

- validation of the model with the anti-trapping current *viz.*, velocity invariance and Gibbs-Thomson condition
- validation of the thermodynamic data with Lipton *et al.* (1984) analysis of dendritic growth

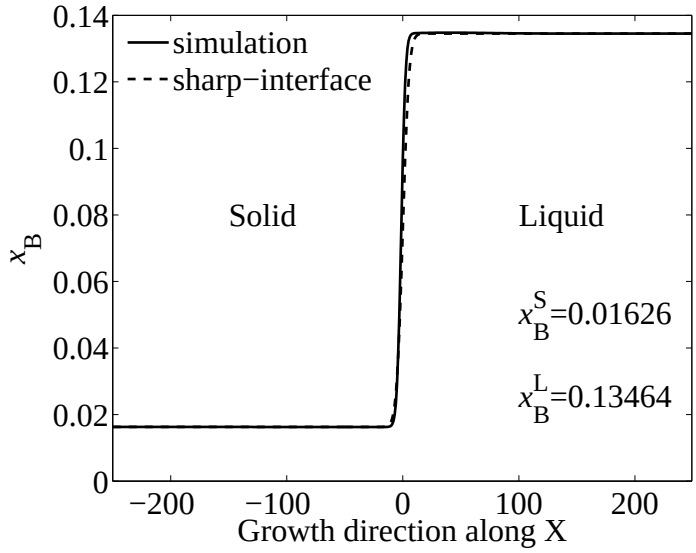
6.2 Validation of dendritic growth

The simulation setup involves a free dendrite growing into an undercooled melt. Since an isothermal phase-field model is used, the temperature of the domain remains constant throughout the simulations. The isothermal condition allows the freedom that the temperature dependent Gibbs energy parameters do not change with neither time nor space and hence they can be calculated before the start of the run. The individual compositions of the solid and the liquid phases, on the other hand, need to be calculated for each iteration at every grid point. The anti-trapping current is included in the diffusion equation to negate the effect of chemical potential jump across the solid-liquid interface.

One possibility is to check the invariance of tip velocity as a function of varying interface widths, shown in Fig.6.4(a) and Fig.6.5(a), which proves the existence of a range of interface widths for which the velocity remains invariant. The other is the convergence of solid composition at the interface with the Gibbs-Thomson curvature corrected composition. Fig.6.4(b) and Fig.6.5(b) show the composition profiles obtained, which are compared with the equilibrium composition and curvature corrected composition.

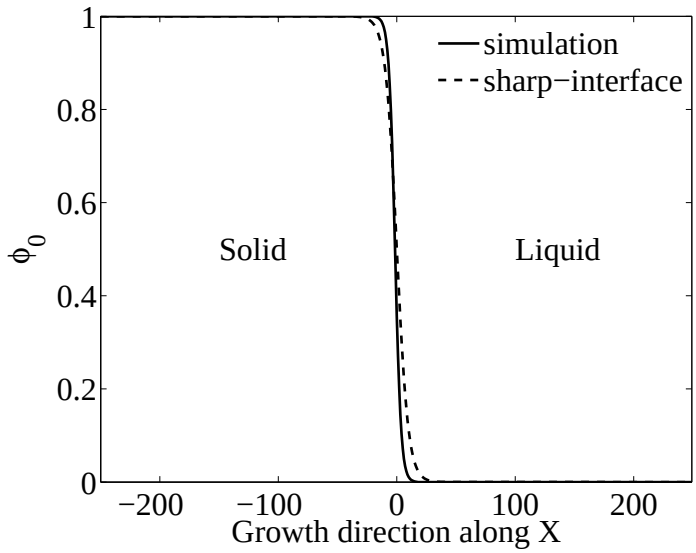


(a)

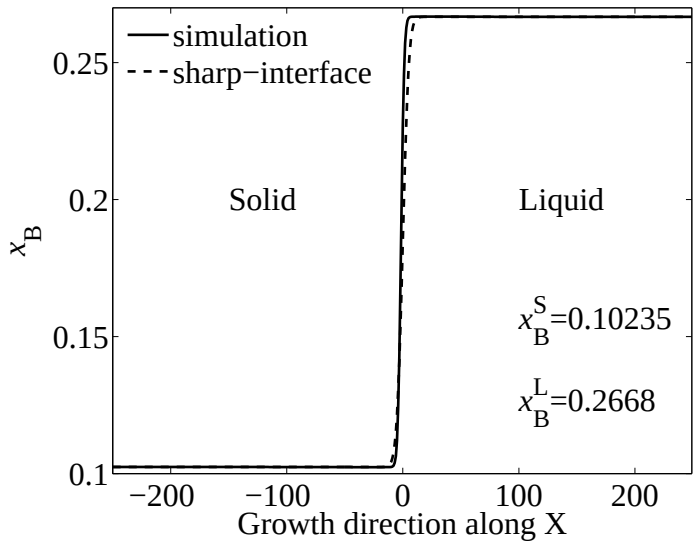


(b)

Figure 6.1: Simulated and sharp-interface (a) phase-field and (b) composition profiles at equilibrium for Al-Cu

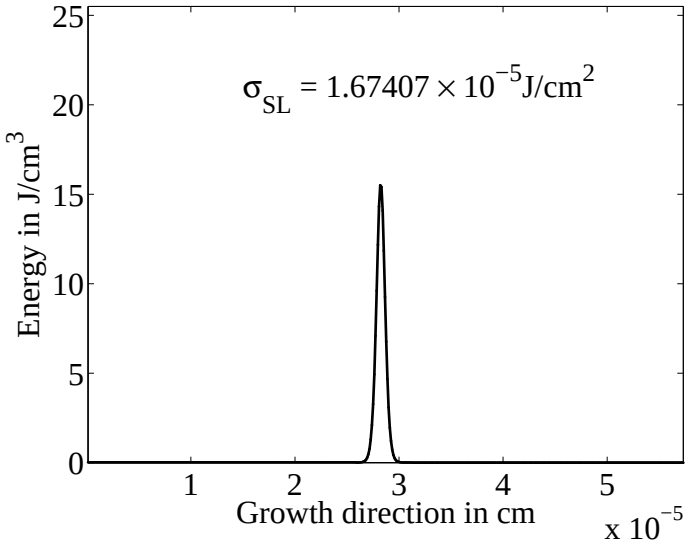


(a)

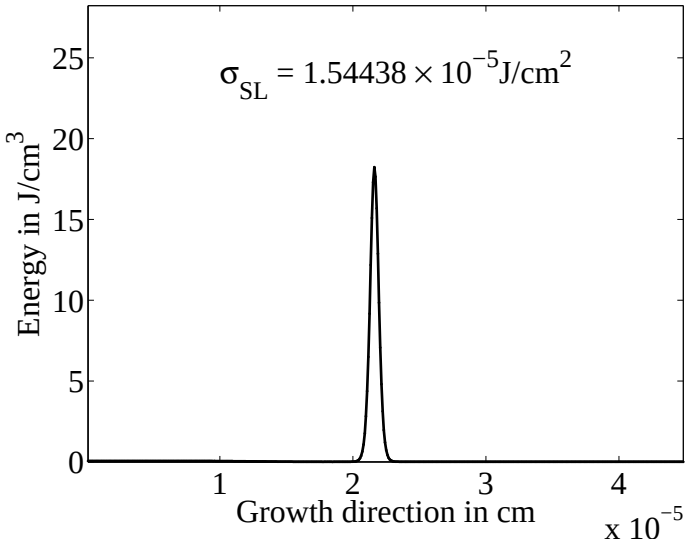


(b)

Figure 6.2: Simulated and sharp-interface (a) phase-field and (b) composition profiles at equilibrium for Al-Mg

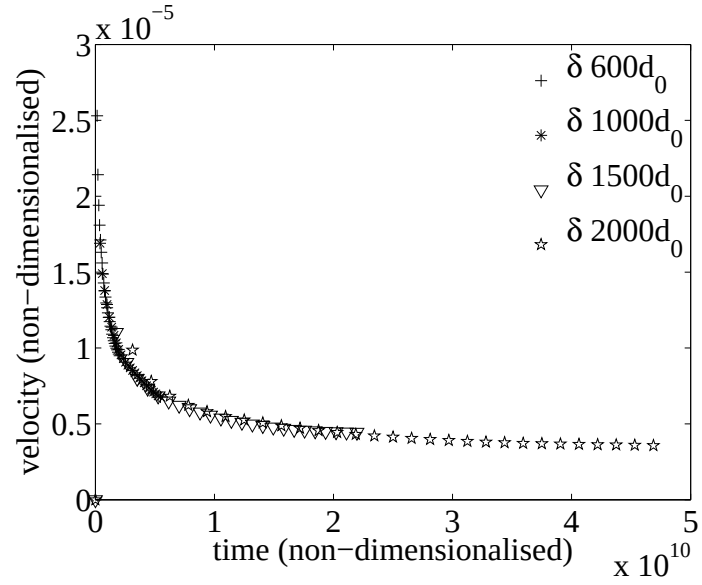


(a)

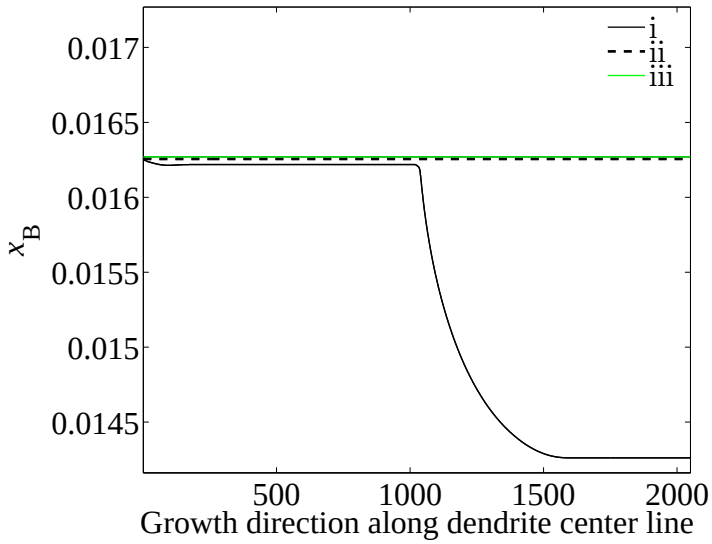


(b)

Figure 6.3: Surface tension getting retrieved after having attained equilibrium for the system (a) Al-Cu at T_0=850 K; $x_{\mathrm{B}}^{\mathrm{S}}$ = 0.01627, $x_{\mathrm{B}}^{\mathrm{L}}$=0.1345 and (b) Al-Mg at T_0=790 K; $x_{\mathrm{B}}^{\mathrm{S}}$ = 0.1024, $x_{\mathrm{B}}^{\mathrm{L}}$=0.2667

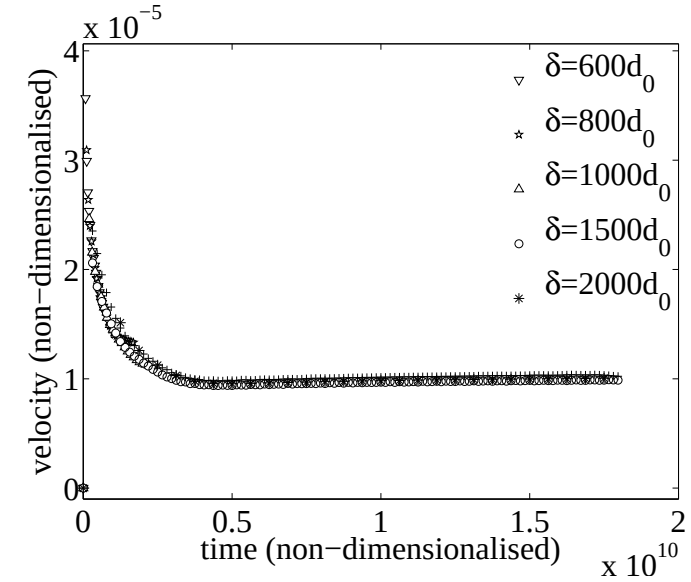


(a)

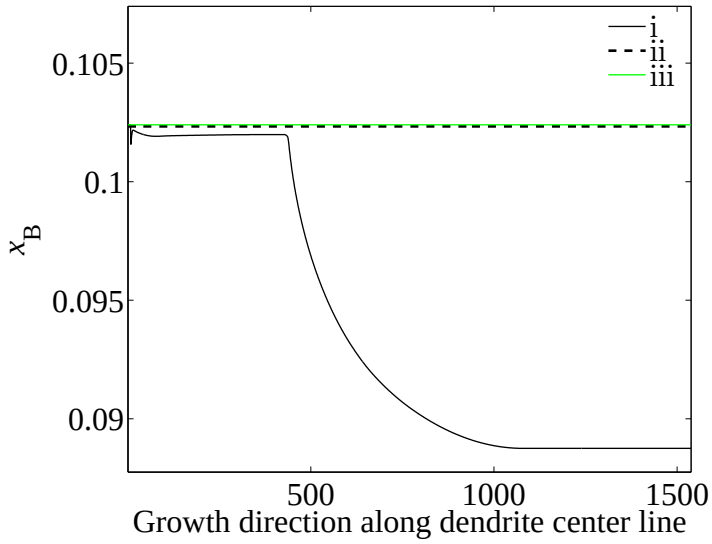


(b)

Figure 6.4: (a) Invariance of tip velocity for different interface widths δ plotted as a function of time for Al-Cu. (b) Comparison of the (i) simulated solid composition profile, $x_{\mathrm{B}}^{\mathrm{S}}$, along the centre line of the dendrite with the (ii) curvature (κ) corrected composition through the Gibbs-Thomson relation, $x_{\mathrm{B}}^{\mathrm{S},e} + \frac{(\sigma+\sigma_{\theta\theta})}{P}\kappa$, where, $P = \dfrac{\partial^2 g^{\mathrm{S}}}{\partial {x_{\mathrm{B}}^{\mathrm{S}}}^2}(x_{\mathrm{B}}^{\mathrm{S}} - x_{\mathrm{B}}^{\mathrm{L}})$ and the (iii) equilibrium composition $x_{\mathrm{B}}^{\mathrm{S},e}$ for Al-Cu



(a)



(b)

Figure 6.5: (a) Invariance of tip velocity for different interface widths δ plotted as a function of time for Al-Mg. (b) Comparison of the (i) simulated solid composition profile, $x_{\mathrm{B}}^{\mathrm{S}}$, along the centre line of the dendrite with the (ii) curvature (κ) corrected composition through the Gibbs-Thomson relation, $x_{\mathrm{B}}^{\mathrm{S},e} + \frac{(\sigma+\sigma_{\theta\theta})}{P}\kappa$, where, $P = \dfrac{\partial^2 g^{\mathrm{S}}}{\partial {x_{\mathrm{B}}^{\mathrm{S}}}^2}(x_{\mathrm{B}}^{\mathrm{S}} - x_{\mathrm{B}}^{\mathrm{L}})$ and the (iii) equilibrium composition $x_{\mathrm{B}}^{\mathrm{S},e}$ for Al-Mg

Dendritic growth is characterized by an unique velocity and radius of the tip for a given undercooling and constant initial composition of the melt far away from the dendrite tip. The model assumes local equilibrium condition with no interface kinetic effect. A more appropriate approach of validating dendritic growth would be to employ the microsolvability criterion. The focus of the present work was, however, on the overall morphological transitions due to continuous thermal cycling and hence LGK analysis (Lipton *et al.* (1984)) is adequate to validate dendrite growth.

The simulated features were compared with the predictions of the LGK theory (Lipton *et al.* (1984)), which predicts a unique velocity and radius of the tip at a given melt undercooling and constant initial composition of the melt far away from the dendrite tip. The model assumes local equilibrium condition with no interface kinetic effect. The issues to be concerned are the constant undercooling of the melt far away from the tip and composition increase due to absence of flux at the boundaries. These are circumvented by choosing a computational domain approximately 3 times larger than the expected dendrite size. The dendrite was nucleated at the left corner and was tilted at 45° to the axis. The initial undercooling was set at a supersaturated initial composition which is related to the undercooling by $m_L(x_{\mathrm{B}}^0 - x_{\mathrm{B}}^{\mathrm{L}})$. To solve the second issue, of composition getting enriched within the domain, it is necessary that the dendrite tip reaches a steady state before composition reflection at the boundaries actually affect the steady state tip. LGK theory tries to solve two simultaneous equations in Peclet number (P) and tip radius (R) as shown in Eq.(6.4) and Eq.(6.5)

$$\begin{aligned}\Delta T =& \Delta T_T + \Delta T_c + \Delta T_R \\ =& \frac{\Delta H_f}{c(p)} I(P_T) + mC_0 \left[1 - \frac{1}{1-(1-k)I(P_c)}\right] + \frac{\Gamma}{R}\end{aligned} \tag{6.4}$$

where

$$I(P) = \sqrt{\pi P} e^P erfc(\sqrt{P})$$

$$R = \frac{\Gamma/\sigma^*}{\left[\frac{\Delta H_f}{c(p)} I(P_T) - P_c \frac{mC_0(1-k)}{1-(1-k)I(P_c)}\right]} \tag{6.5}$$

where P_T, P_C are the respective thermal and solutal Peclet numbers and $I(P)$ is the Ivantsov solution for a parabola in 2-D . Considering it to be a solutal dendrite, the thermal terms are neglected from which the modified equations without the thermal terms become,

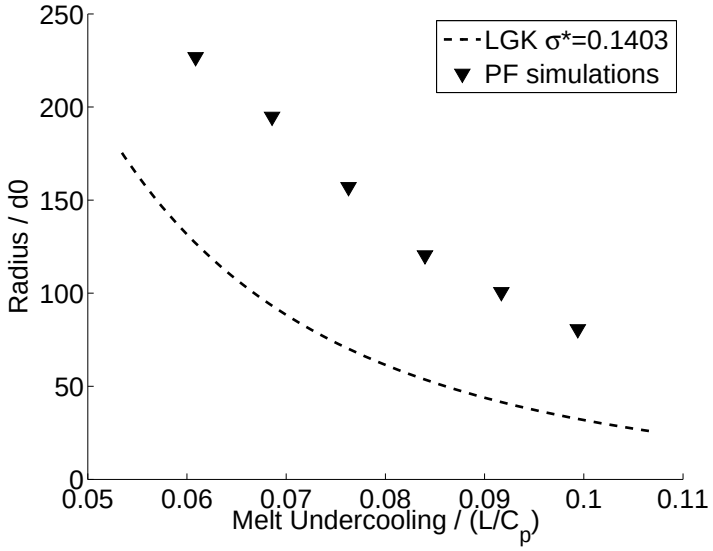
$$\Delta T = mC_0\left[1 - \frac{1}{1-(1-k)I(P_c)}\right] + \frac{\Gamma}{R} \tag{6.6}$$

$$R = \frac{\Gamma/\sigma^*}{\left[-P_c\dfrac{mC_0(1-k)}{1-(1-k)I(P_c)}\right]} \tag{6.7}$$

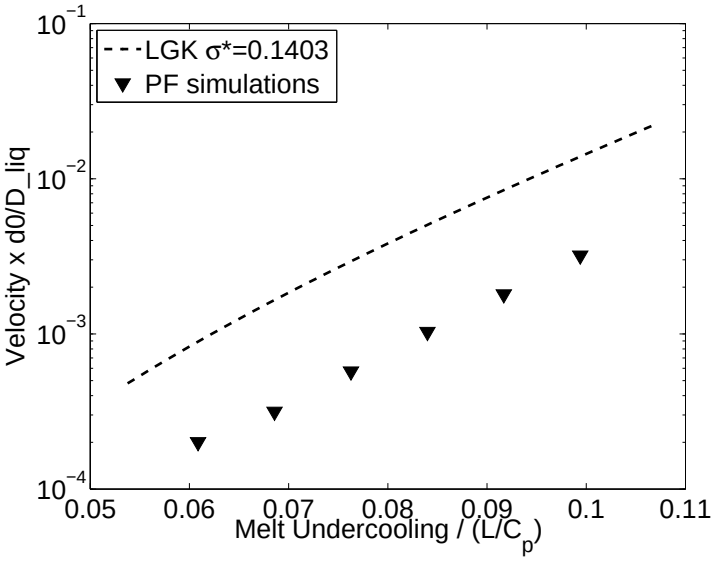
An important parameter which comes in the matching is the stability parameter σ^*, which determines the steady state tip radius and velocity. Though LGK theory proposes that the value of σ^* for a 3-D dendrite growth to be 0.025, it varies with the anisotropy of the solid-liquid interface. There are two ways to determine the parameter σ^*, one is through the microsolvability theory by Barbieri and Langer (1989). Second is through employing techniques like phase-field simulations to determine the dendrite tip radius and velocity which are a result of the dynamic minimization of the Gibbs energy and σ^* is therefore an output of these simulations. From the phase-field simulations, the tip radius and velocity are calculated by fitting a fourth order polynomial to the phase-field contour at ϕ = 0.5 and determining the values and repeating for various undercooling thereby getting an array of tip radius and velocities. σ^* is thus calculated as $\frac{2d_0D_{\mathrm{L}}}{VR^2}$, where d_0 is the capillary length given by $d_0 = \frac{\Gamma}{m\Delta c}$. Γ is the Gibbs-Thomson coefficient, m, the liquidus slope and Δc, the magnitude of solute rejection at the interface.

The value of σ^* was 0.1403 for Al-Cu and 0.1449 for Al-Mg for the capillary anisotropy, ϵ_c of 0.04 and 0.05, respectively, calculated as an average value from the undercoolings considered. The comparisons are shown in Fig.6.6 and Fig.6.7. The simulations at each undercooling were performed at interface widths which are within the range, where the results are invariant of this parameter. Though, a reasonable agreement can be found between the LGK model and our simulations, it should be noted that LGK model is not exact and assumes a parabolic tip which is usually not the case in the case of dendrites (Eshraghi *et al.* (2012)) and also the deviations found in the comparisons are due to the capillary anisotropy, where the assumptions are for a needle crystal in the case of LGK theory.

To study the morphological changes occurring during thermal cycling, the phase-field simulations are performed to mimic a linear electromagnetic stirring setup (LEMS) developed with a

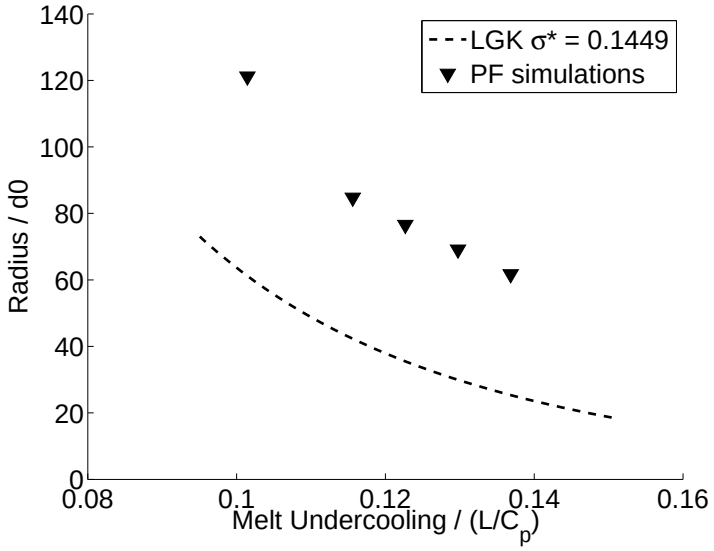


(a)

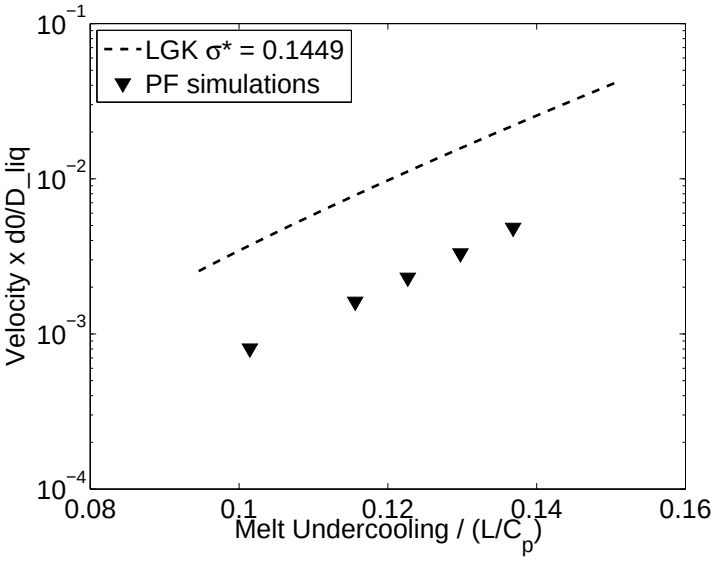


(b)

Figure 6.6: Comparison of LGK with (a) dendritic tip velocity and (b) tip radius as a function of undercooling for ϵ_c=0.04 for the system Al-Cu



(a)



(b)

Figure 6.7: Comparison of LGK with (a) dendritic tip velocity and (b) tip radius as a function of undercooling for ϵ_c=0.05 for the system Al-Mg

rheocasting unit. The description of LEMS can be found in detail in Kumar (2008) along with experimental results and parametric variation. To illustrate the globularisation phenomena, a brief description of the electromagnetic stirring setup is presented here.

6.3 Linear Electromagnetic Stirrer (LEMS)

The linear electromagnetic stirrer (LEMS) shown in Fig.6.8 and described in more detail in Kumar (2008), consists of set of six primary coils which are excited by a three-phase supply to produce an axially travelling magnetic field. The melt to be stirred is placed in the annular space of the stirrer. The travelling field induces secondary current in the melt, whereupon the current and the magnetic fields interact to generate Lorentz forces in the melt which serves as the net mechanical force to stir the melt in the mould. The mould design and the cooling arrangement are such that solidification in the mould is primarily unidirectional. The heat from the solidifying metal is extracted at the bottom of the mould such that the axisymmetric electromagnetic stirring effectively shears the dendrites formed at the solid-liquid interface. The outer surface of the mould is cooled with water jets which provides a high heat transfer coefficient and wider range of cooling rates in the metal. The temperature profiles show a vertical thermal gradient inside the melt. Ingots are solidified by extracting heat at the bottom of the rheocasting setup leading to ingots that are nearly 30 cm long and have effectively undergone directional solidification in the presence of electromagnetic stirring. It can be noted that the electromagnetic field in the mould creates a vortex that makes the fluid circulate between the colder bottom region and the hotter top region within the mould.

Fig.6.9(a) shows a schematic phase diagram depicting rheocasting. The schematic of the rheocasting mould and the forced convection pattern due to electromagnetic stirring is shown in Fig.6.9(b). Superimposed on a stream line is a schematic of how a free floating dendrite would experience the surrounding temperature field (higher temperature at top and lower at bottom of the mould). It is well established that in a directional solidification setup there is a layer of free floating dendrites just above the mushy zone. We can assume that the thermal profile experienced by a dendrite advected by the forced convection present in the mould would be fluctuating between two values *viz.*, (a) a lower temperature when the dendrite is close to the bottom of the melt and (b) a higher temperature when the dendrite is far away from the solid-



Figure 6.8: Linear Electromagnetic Stirrer (LEMS) (Kumar (2008))

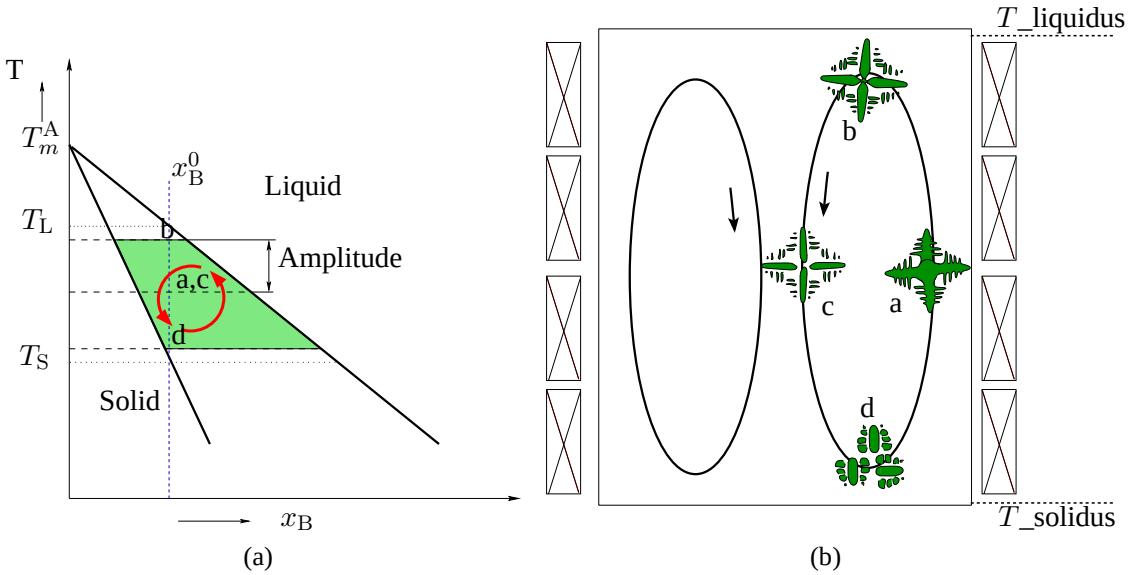


Figure 6.9: (a) Schematic phase diagram illustrating a rheocasting technique, (b) Schematic of a rheocasting setup

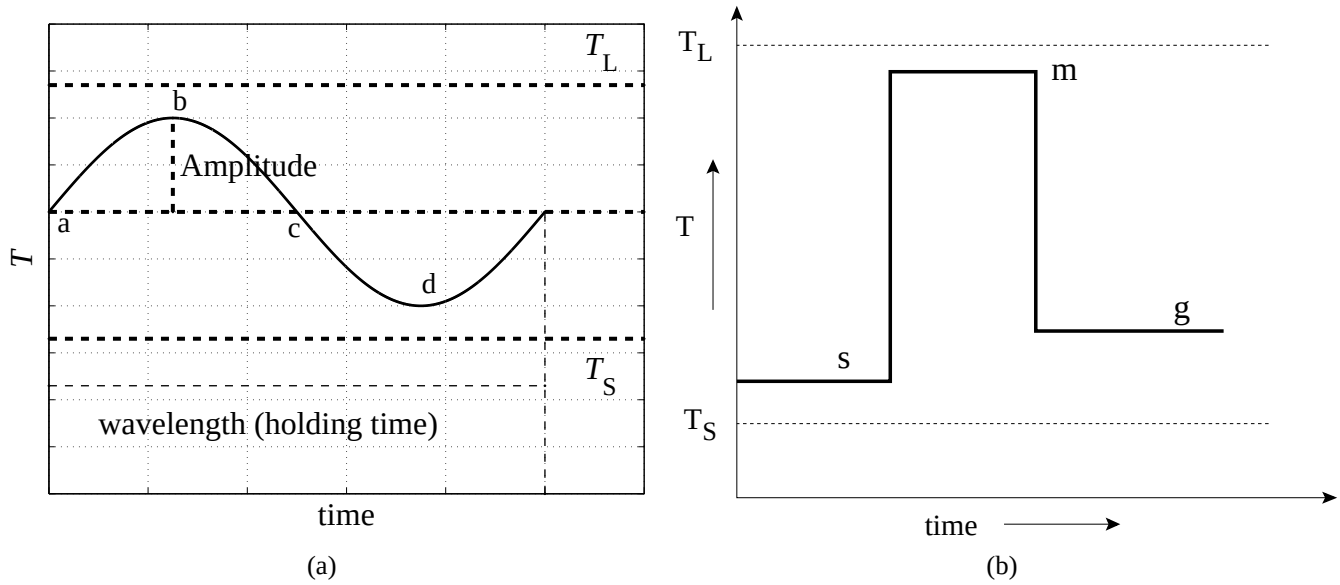


Figure 6.10: (a) Thermal profile experienced by a dendrite under linear stirring, (b) Discrete thermal profile

liquid interface. A more precise approach would be to take a Lagrangian approach by tracking the dendrite within the melt and applying the thermal conditions as experienced by it during the forced convection.

Kumar (2008) had also put forth that electromagnetic stirring generally produces a uniform temperature distribution in the melt. A higher stirring current coupled with lower excitation frequencies produce a more globular microstructure. Higher frequencies enhance induction heating, whereas lower frequencies produce more uniform distribution of force-field. Higher stirring current maybe related to the amplitude of the thermal cycle shown in Fig.6.10(a). Higher frequencies maybe related with the wavelength of the cycle. The thermal gradient imposed during the DC casting coupled with LEMS can be related to the amplitude of the thermal cycle used in the numerical simulations.

In order to simulate this process using a computational approach, it is necessary to consider the effects that are required to be envisaged. In this study, the effect of thermal and solutal gradients are mainly studied neglecting the effects of convection. Although it should be noted that convective forces during stirring nullifies the solutal gradients across the melt, it is safe to assume that in the vicinity of the dendrite the effect of convective forces are less pronounced (due to the tortuous liquid channels that lead to low permeability). Also, it may be realistic to assume that there is no relative velocity between the solid dendrite and the liquid immediately surrounding it during the advection process. Thus, a static domain with a dendrite growing

within it and a thermal cycle imposed on the entire domain could simulate the excursion of a free dendrite along the path in the stirred zone. If we were to take a Lagrangian approach and travel with a dendrite, the thermal profile taken by the free floating dendrite would resemble a sinusoidal waveform as shown in Fig.6.10(a). A much more simpler description would involve using a two-step approach as shown in Fig.6.10(b)

6.4 Numerical Simulations

The simulations performed will be outlined based on the phase-field models used and the type of thermal profiles employed. The first set of simulations involved studying the effect of gradients on a single dendrite which involved using both a discrete profile as well as a single profile. The model used was the isothermal binary alloy solidification by Wheeler *et al.* (1992) with the model free energy data considering the properties of the Ni-Cu system. The second set of simulations involved capturing the effect of the thermal and solutal gradients utilising actual thermodynamic coupling using CAPLHAD description of Gibbs energies as discussed in Chapter 4. The anti-trapping phase-field model by Kim (2007) with one-sided diffusion (no diffusion in solid) was used. The systems considered were Al-Cu and Al-Mg, since both possess broad freezing ranges along the Al-rich side.

Alloy solidification presents a rather complex scenario than that of pure components since phase transformation kinetics involves both heat and mass transfer. However, since these two processes are occurring at two different time scales, in most cases, it is sufficient enough to consider the slower of the two, namely, mass transfer. Heat transfer, taking place on a faster scale, (nearly two orders of magnitude larger than mass transfer), can usually be treated as frozen or isothermal as temperature evolves so rapidly compared to solute redistribution, that the process can be assumed to take place in a quasi-steady state. In most cases, the ratio of thermal (D_T) and solutal (D) diffusivities (Lewis number=D_T/D) ranges between 10^4 to 10^2. The coupled nature of the evolution of phase-field and solute distribution profiles is favoured for study over the coupling between solutal and thermal fields. It is deemed that this approach will highlight the role of curvatures and solute gradients on the evolution of the microstructure.

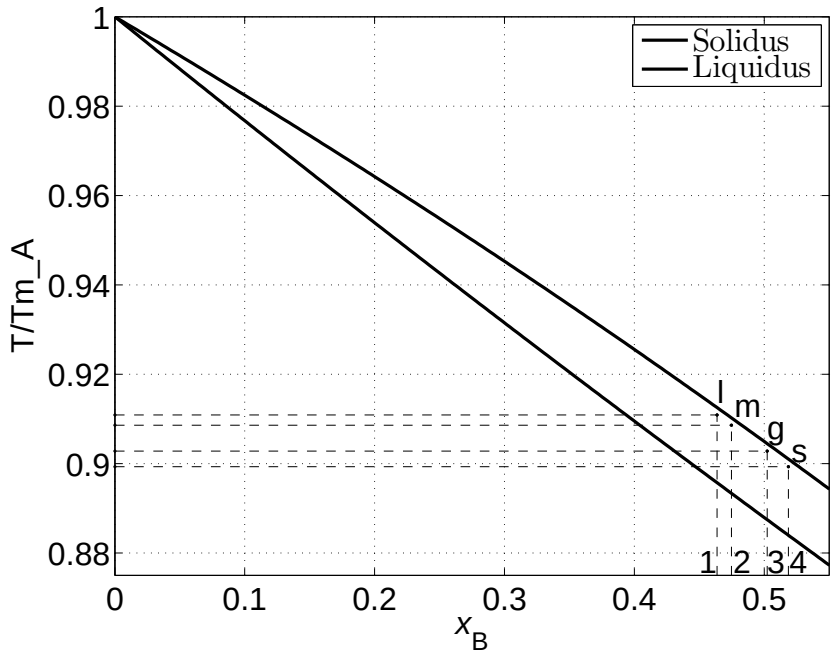
6.5 Discrete thermal profile

The governing equations were solved using an explicit finite-difference technique in a simulation domain 1500×1000. Ni-Cu was chosen as the alloy system for the study. Numerical stability was ensured by maintaining $\Delta \mathrm{t} \leqq \frac{\Delta \mathrm{x}^2}{D_\mathrm{L}}$. $\Delta \mathrm{t}$ is the time step size used for the calculations, $\Delta \mathrm{x}$ the grid size and D_L the diffusivity of the solute in the liquid. No flux conditions ($\partial\phi/\partial t = 0$, $\partial c/\partial t = 0$) at the boundaries were applied both for phase and concentration fields. A square seed was nucleated at the left boundary of the domain where the initial concentration of the melt was at **1** at the liquidus temperature denoted by point **l** as shown in Fig.6.11(a). The melt is initially undercooled to 20 K below the liquidus temperature to **s**. Due to undercooling the dendrite starts to evolve and after a few timesteps when the dendrite has traversed the domain, the temperature of the whole domain is now increased to point **m** (4 K below the liquidus) and held there for a duration sufficient to allow for the remelting process to be significant.

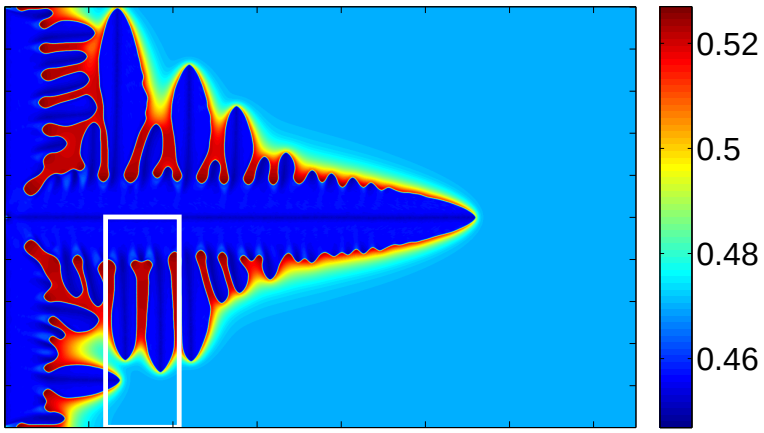
It is possible to perform detailed calibration studies to tune this holding time. This holding time represents the effective time spent by the dendrite at a given temperature. This can be determined by the actual thermal profile at the location of the freely floating dendrite by integrating the calculated velocities. This particular study is done to illustrate the solute segregation induced remelting qualitatively. Root remelting occurs at this higher temperature and the secondary arms get detached from the primary dendrite. The local concentration now tries to stabilize to the equilibrium concentration at **m**. After holding at **m**, the temperature of the domain is now decreased to point **g** (14 K below liquidus) and held for a similar duration as in the heating stage. During this stage, the fragments grow in a globular manner. The temperature profile for the whole simulation is shown in Fig. 6.10(b).

This treatment can be described as an isothermal analysis of soaking and subsequent coarsening.

The phase diagram in Fig.6.11(a) shows that the initial concentration of the melt is at **1** which corresponds to $0.463 x_\mathrm{B}$. As the melt is solidified at 20 K below the liquidus temperature, the equilibrium concentration at **s** is found to be $0.52 x_\mathrm{B}$. Hence in Fig. 6.11(b), we can see that the local equilibrium concentration at the solid-liquid interface to be $0.52 x_\mathrm{B}$ whereas the far field composition is still at $0.463 x_\mathrm{B}$. Now, when the temperature of the melt is suddenly increased to **m**, the solute concentration prevailing between the secondary arm roots are now



(a)



(b)

Figure 6.11: An isothermal Ni-Cu dendrite grown with respect to the parameters provided in Table 6.1

Table 6.1: Parameters used for the discrete thermal profile

Scaling parameters		
Reference length	d_0	7.1×10^{-8} cm
Diffusivity in liquid	D_{L}	1×10^{-5} $\mathrm{cm^2 s^{-1}}$
Reference temperature	T_r	1728 K
Energy scale	RT_r/v_m	1941.43 $\mathrm{Jcm^{-3}}$
Material parameters		
Melting point Ni	T_m^{A}	1728 K
Melting point Cu	T_m^{B}	1358 K
Molar volume	v_m	7.42 $\mathrm{cm^3 mol^{-1}}$
Simulation parameters		
Interface width	δ	5×10^{-6} cm
Grid size	Nx $\times$ Ny	1500x1000
Spatial resolution	Δx	2.5×10^{-6} cm
Time resolution	Δt	$0.1\Delta \mathrm{x}^2$

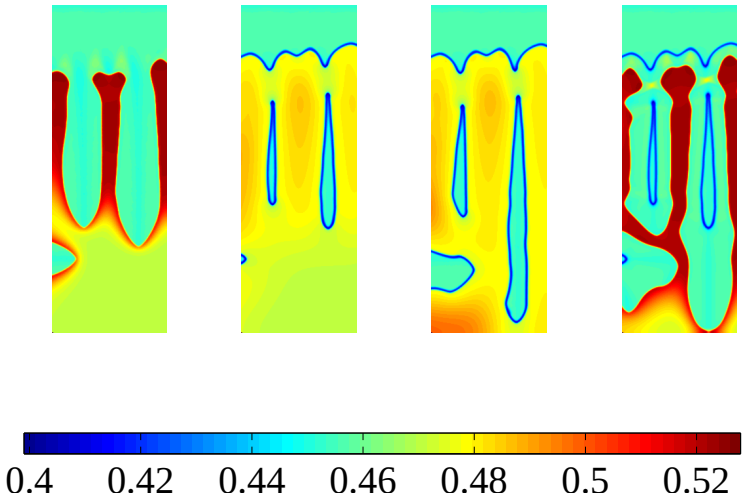


Figure 6.12: Growth and remelting depicting the detachment of the secondary arms from the primary and subsequent coarsening of the detached arms

at a much higher concentration as opposed to the equilibrium concentration of $0.474x_{\mathrm{B}}$. As a result of this higher concentration, the secondary arm roots start to disengage themselves from the primary stem. The temperature is then decreased to point **g**, which causes the pinched off arms to coarsen. Globularization of individual fragments now occurs due to the movement of the solute flux from areas of high curvature to areas of low curvature, which is evident from Fig.6.12 according to Fan (2002). If an advection were to be superimposed on the domain, we could expect these dendrite fragments to get transported away. Such fragments would then evolve in their morphology depending on the local thermal and solutal conditions. It is deemed that a demonstration of fragmentation and a tendency of the fragments to evolve towards a globular morphology is adequate to conclude that the process would lead to a fine globular microstructure.

From Fig. 6.12 we can conclude that the pinch-off of side arms takes place due to melting at the root induced by a lower liquidus at the local composition as corroborated experimentally in another system using direct visualization experiments (Ruvalcaba *et al.* (2007)). It is well established that the segregation of solute is expected at regions of high curvature due to Gibbs-Thomson effect. The root melting observed in the simulation study presented can be attributed to the lower local liquidus temperatures of these regions. As the solidification proceeds from the bottom of the ingot towards the top, more and more of such dendrite fragmentation is expected to take place leading to finer and more globular microstructure towards the end.

Finer and more globular microstructure towards the end of a directional solidification in a rheocasting setup in the presence of electromagnetic stirring could be explained by a repeated process of root melting in free floating dendrites induced by fluctuating thermal conditions.

6.6 Continuous thermal profile

To simulate the process of repeated melting, a continuous thermal profile, shown in Fig.6.10(a) was utilised as justified earlier. The simulations were done in two stages: an initial study, employing a single cycle to capture the effects of thermal and solutal gradients on a single dendrite and then moving further to the case of multiple cycles and multiple dendrites. The case of multiple dendrites is studied with Al-Cu and Al-Mg binaries incorporating CALPHAD description of Gibbs energies.

Table 6.2: Parameters used for a single sinusoidal thermal cycle

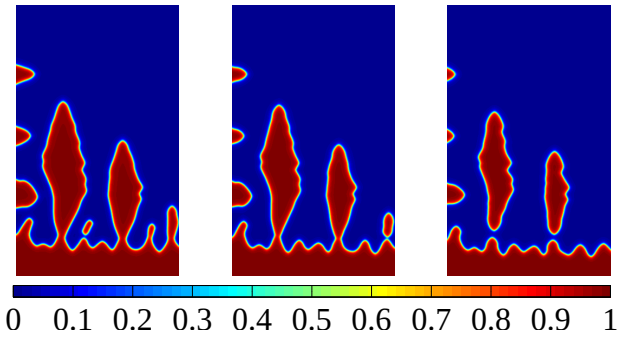
Scaling parameters		
Reference length	d_0	$7.1\text{x}10^{-8}$ cm
Diffusivity in liquid	D_L	$1\text{x}10^{-5}$ cm^2s^{-1}
Reference temperature	T_r	1728 K
Energy scale	RT_r/v_m	1941.43 Jcm^{-3}
Grid parameters		
Grid size	Nx = Ny	1000
Spatial resolution	Δx	100
Time resolution	Δt	$0.002\Delta\text{x}^2$
Material parameters		
Melting point A	T_m^A	1728 K
Melting point B	T_m^B	1358 K
Molar volume	v_m	7.42 $\text{cm}^3\text{mol}^{-1}$

Table 6.3: Wavelengths and cycle times used for the thermal profiles

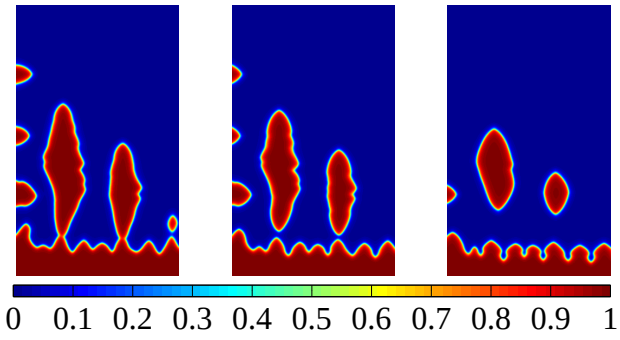
Amplitude		
ΔT_1	6K	
ΔT_2	10K	
Cycle time/	wavelength	Time in ms
τ_1	50000	0.5
τ_2	70000	0.7
τ_3	100000	1.0

Table 6.2 lists the various parameters used to simulate dendritic growth. Table 6.3 gives the simulation parameters used for the thermal cycling studies. A simple lens shaped phase diagram as shown in Fig.6.9(a) is used for the study. Fig.6.10(a) shows the schematic of the thermal profile used. The points in Fig.6.9(a) and Fig.6.10(a) *viz.*, **a, b, c, d** are all referenced the same. The amplitude and wavelength of the thermal cycle have a profound impact on the final microstructure. The final morphological changes for the different thermal cycles used are given in Fig.6.13. The corresponding conditions are provided in Table 6.3. From the thermal profile in Fig. 6.10(a), when the temperature of the domain is at point **b**, the higher temperature causes the rejection of the solute from the solid as well as the adjoining liquid. As this solute is rejected, lateral transfer of the solute towards the root takes place. This solute pileup brings about a decrease in the liquidus temperature locally.

Even though there is a decrease in the temperature at point **c**, the solute-enriched regions in the vicinity of the roots starts to remelt and gets detached from the primary arm. This process of melting effected by solute enrichment is referred to as *solutal remelting* (Rettenmayr *et al.* (2001)). Further decrease in the temperature to point **d**, causes an influx of the solute



(a)



(b)

Figure 6.13: Phase-field profiles showing the morphology at the end of each thermal cycle for two different amplitudes (a) ΔT_1=6 K and (b) ΔT_2=10 K and wavelengths, τ_1, τ_2, τ_3 as in Table 6.3

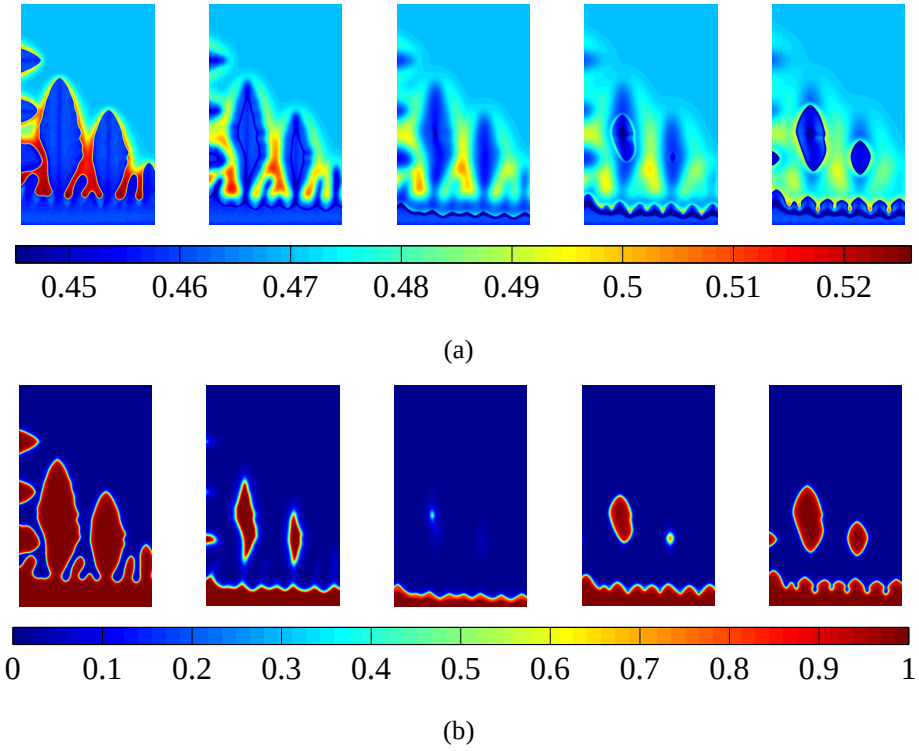


Figure 6.14: (a) Concentration and (b) phase-field profiles showing the fragmentation and subsequent coarsening for a single thermal cycle whose schematic is shown in Fig. 6.10(a) and its sequence being a-b-c-d-e. The third figure in the profiles show the solute induced remelting or solutal remelting i.e. point **c** in Fig. 6.10(a).

atoms to enter the solid masses that have survived root remelting. Since in this context, morphological changes due to thermal fluctuations are being studied, at the coarsening stage, there is a tendency for the solid masses to grow in an elongated manner due to the surface energy anisotropy. Although this situation exists, further branching of the solid is prevented as the temperature increases in a cyclic manner. The concentration and phase-field maps illustrating the morphological change for a single thermal cycle are shown in Fig.6.14.

6.7 Multiple dendrites and multiple cycling

The effect of thermal and solutal gradients were studied in the context of a single dendrite and how these effects substantially aid the dendritic breakdown leading to subsequent coarsening in a globular fashion. Extending the study further, to understand the effect of these gradients across multiple dendrites, two systems of technical relevance *viz.*, Al-Cu and Al-Mg along the Al-rich corner were chosen for the study. The phases considered were solid (α) and liquid,

whose Gibbs energies were obtained from Witusiewicz *et al.* (2004) and Zhong *et al.* (2005) for Al-Cu and Al-Mg, respectively.

The modified anti-trapping model of Kim (2007) reduced for two components was used with the relevant Gibbs energies incorporated as outlined in Chapter 4. This model which assumes no diffusion in the solid, was used for studying the effect of thermal cycling. A one-sided diffusion model under the influence of thermal cycling generally introduces artefacts on the solid side of the interface. In this case, it was necessary to ascertain the effect of including diffusion in the solid and to observe any significant changes in the overall morphology. This prompted us to do a preliminary study to understand the role of diffusion related kinetics in the solid as a function of periodic melting and coarsening. For this initial case, Kim *et al.* (1999) (KKS) model was used and the Gibbs energies of solid and liquid phases for Al-Mg was incorporated. On a stricter note, the effect of anti-trapping current is less pronounced at higher undercoolings compared to lower undercoolings. This allows to set larger interface widths without affecting the kinetics. An important feature to be concerned is that using a model assuming no diffusion in solid will have the tendency to induce the onset of dendritic growth at early stages, which will otherwise be prolonged if diffusion in solid is present.

While the anti-trapping model would enable numerical computations in domains that are close to realistic dimensions, any possibility of remelting during the actual process precludes an artefact-free simulation. Solutal remelting during a free dendrite growth or directional solidification is a realistic possibility. The one-sided diffusion model leads to (local) sharp changes in the solute profile during the remelting part of the thermal cycle and these could lead to other artefacts. In such processes, a smaller interface width that does not require anti-trapping model would be recommended. While there are recent studies to develop an anti-trapping model using a two-sided diffusion Ohno and Matsuura (2009), this study highlights the artefacts that arise out of the usage of an anti-trapping model that uses a one-sided diffusion.

The parameters for the first set of simulations incorporating varied solid diffusivities are given in Table 6.4. The study on morphological changes by varying the stirring cycles are given in Table 6.5 for Al-Mg and Table 6.6 for Al-Cu.

Since phase-field equations evolve as a result of Gibbs energy minimization, nucleation cannot be treated implicit and hence nucleation sites have to be explicitly specified. The seeding can be random, in this case, specific sites were chosen, whose preference is purely arbitrary. Initial

Table 6.4: Simulation parameters for the study on the variation of solid diffusivities. Other material and simulation parameters are as provided in Table 6.5

Grid parameters		
Interface width	δ	$60\mathrm{d}_0$
Domain size	Nx=Ny	2048
Spatial resolution	$\Delta\mathrm{x}$	$\delta/10$
Time resolution	$\Delta\mathrm{t}$	$0.1\Delta\mathrm{x}^2$

Table 6.5: Parameters used for the growth of Al-Mg multiple dendrites

Scaling parameters		
Reference length	d_0	$1.345\mathrm{x}10^{-8}$ cm
Diffusivity in liquid	D_L	$3\mathrm{x}10^{-5}\ \mathrm{cm}^2\mathrm{s}^{-1}$
Reference temperature	T_r	933 K (Melting point of Al)
Energy scale	RT_r/v_m	729.7236 Jcm^{-3}
Grid parameters		
Interface width	δ	$100\mathrm{d}_0$
Domain size	Nx=Ny	2048
Spatial resolution	$\Delta\mathrm{x}$	$\delta/8$
Time resolution	$\Delta\mathrm{t}$	$0.2\Delta\mathrm{x}^2$
Material parameters		
Melting point Al	T_m^A	933 K
Melting point Mg	T_m^B	923 K
Molar volume	$\mathrm{v_m}$	10.63 $\mathrm{cm}^3\mathrm{mol}^{-1}$
Simulation parameters		
Initial temperature	T_0	770 K
Initial composition of the melt	x^0_Mg	0.165
Composition of solid	x^S_Mg	0.1024
Equilibrium Composition of liquid	$x^{\mathrm{L},e}_\mathrm{Mg}$	0.2667 at T=790 K
Equilibrium Composition of solid	$x^{\mathrm{S},e}_\mathrm{Mg}$	0.1024 at T=790 K
Liquidus slope	m_L	-628.93 K
Thermal undercooling	ΔT_t	20 K
Solutal undercooling	$\Delta T_c = m(x^0_\mathrm{Mg} - x^{L,e}_\mathrm{Mg})$	63.96 K
Total undercooling	$\Delta T = \Delta T_t + \Delta T_c$	83.96 K
Strength of capillary anisotropy	ϵ_c	0.01

Table 6.6: Parameters used for the growth of Al-Cu multiple dendrites

Scaling parameters		
Reference length	d_0	$1.7194\text{x}10^{-8}$ cm
Diffusivity in liquid	D_L	$3\text{x}10^{-5}\ \text{cm}^2\text{s}^{-1}$
Reference temperature	T_r	933 K (Melting point of Al)
Energy scale	RT_r/v_m	729.7236 Jcm^{-3}
Grid parameters		
Interface width	δ	100d_0
Domain size	Nx=Ny	2048
Spatial resolution	Δx	$\delta/8$
Time resolution	Δt	$0.1\Delta\text{x}^2$
Material parameters		
Melting point Al	T_m^A	933 K
Melting point Cu	T_m^B	1358 K
Molar volume	v_m	10.63 $\text{cm}^3\text{mol}^{-1}$
Simulation parameters		
Initial temperature	T_0	840 K
Initial composition of the melt	x_Cu^0	0.08
Composition of solid	x_Cu^S	0.01627
Equilibrium Composition of liquid	$x_\text{Cu}^{\text{L},e}$	0.1345 at T=850 K
Equilibrium Composition of solid	$x_\text{Cu}^{\text{S},e}$	0.01627 at T=850 K
Liquidus slope	m_L	-684.93 K
Thermal undercooling	ΔT_t	10K
Solutal undercooling	$\Delta T_c = m(x_\text{Cu}^0 - x_\text{Cu}^{L,e})$	37.32 K
Total undercooling	$\Delta T = \Delta T_t + \Delta T_c$	47.32 K
Strength of capillary anisotropy	ϵ_c	0.01

seeds nucleated at various positions into an uniformly undercooled melt grow into dendrites purely by supersaturation. Secondary arms are generated by introducing random fluctuations along the liquid side of the interface ($0<\phi<0.5$). The overall stochastic noise for the domain is ensured to be zero. Periodic boundary conditions were used for simulating dendritic growth in the case of both Al-Cu and Al-Mg.

Detailed numerical explanations are provided in Chapter 5. The fully developed dendritic morphology shown in Fig.6.15(a), is taken as the initial microstructure for further thermal cycling. Five different thermal cycles listed in Table 6.7 were employed with the same amplitude. The main focus of the present work was to study the effect of solute redistribution in the melt and how it effectuates the morphological transition when subjected to thermal gradients, in this case thermal cycling.

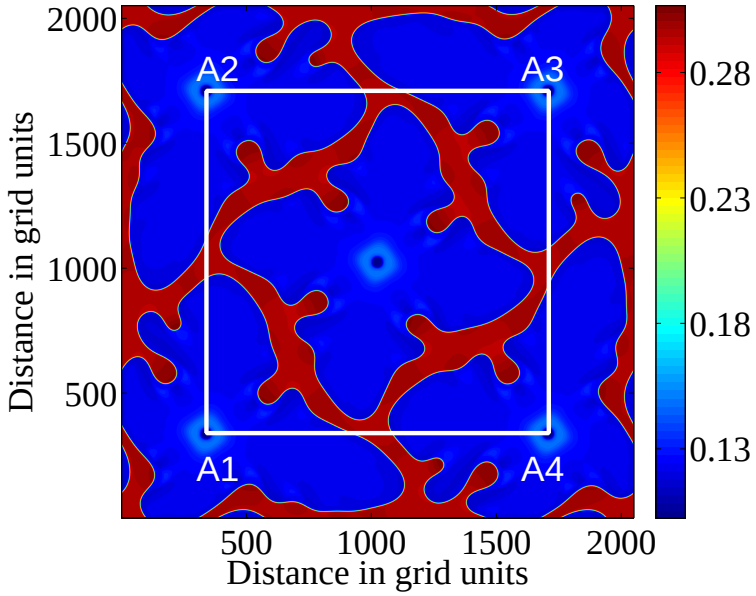
6.8 Simulations with varying solid diffusivities

The set of simulations with varying diffusivities were performed with Al-Mg as the chosen system. Since the studies pertaining to diffusion inside a solid is representative of any system, it was deemed sufficient to study with any one of the binaries and an arbitrary choice of Al-Mg binary was made.

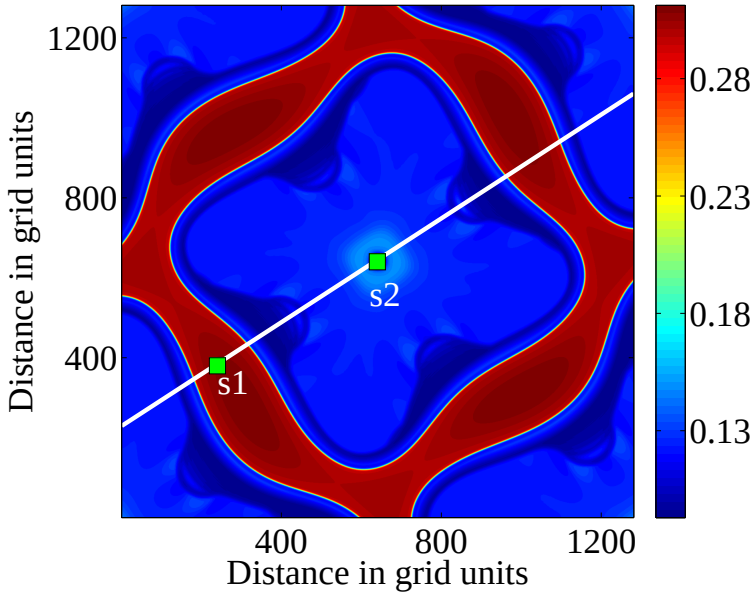
A solutal dendrite was grown by setting the initial composition (x_{B}^{0}) of the melt at $0.165x_{\mathrm{Mg}}$. The temperature corresponding to this composition is 790 K with equilibrium solid and liquid compositions to be at $0.1024x_{\mathrm{Mg}}$ and $0.2667x_{\mathrm{Mg}}$, respectively. Apart from this supersaturation, the initial temperature of the melt is taken down to 770 K with the total undercooling given by $\Delta T = \Delta T_t + m(x_{\mathrm{B}}^{0} - x_{\mathrm{B}}^{\mathrm{L}}) = 89$ K. The strength of the capillary anisotropy was taken as 0.01 referring from the work of Gunduz and Hunt (1985). The two sets of simulations differ here on the choice of solid diffusivities and hence on the limit of interface width leading to a smaller Δx and Δt. This fixes a cap on the simulation domain and the computation time, respectively. However, an anti-trapping formulation with no solid diffusion allowed for larger interface width and lesser time for growth. Since, there are no available experimental data on the kineic coefficient β, it is taken as 0. This condition relates to local equilibrium at the interface. The parameters used for both the simulations are put forth in Table 6.4 and Table 6.5.

Figure 6.15(a) is taken as the initial morphology to study the effect of thermal cycling as a function of different solid diffusivities. The simulations were performed by changing the diffusion rates in the solid *viz.*, D_S/D_L=0.1, 0.01, 0.0001, 0.000001 and 0.0. The morphological changes and the solute redistribution within the solid was brought about more succinctly by changing the diffusivity value in the solid. These changes are more clearly highlighted in Fig.6.16, where it is seen that as the ratio of D_S/D_L reduces, the solute segregation within the solid becomes more and more pronounced. This is observed by a band-like formation caused by the pile up of solute on the solid side of the SL interface. In the case of D_S/D_L=0, these bands do not diffuse back into the solid. This is more clearly indicated in Fig.6.17 by the composition profiles along the line s1-s2 for D_S/D_L=0 and D_S/D_L=0.0001, where the steep composition gradients (spikes) for zero solid diffusion is replaced by a smooth diffusion pattern which will eventually diffuse into the solid. This is not reflected in our simulations as the computational times were longer due to smaller Δt. This simulation was also done to underscore the effect of neglecting solid diffusion.

Although solid diffusion is treated finite, the morphological features exhibited in Fig.6.16 does not seem to affect the overall morphology and only the solute segregation profiles within the solid differ in the simulations. The morphological changes are much more clearly elucidated in Section 6.9.



(a)



(b)

Figure 6.15: (a) Al-Mg dendrite simulated using the conditions given in Table 6.5, (b) Region of interest A1-A2-A3-A4. All concentration profiles are taken along the line s1-s2

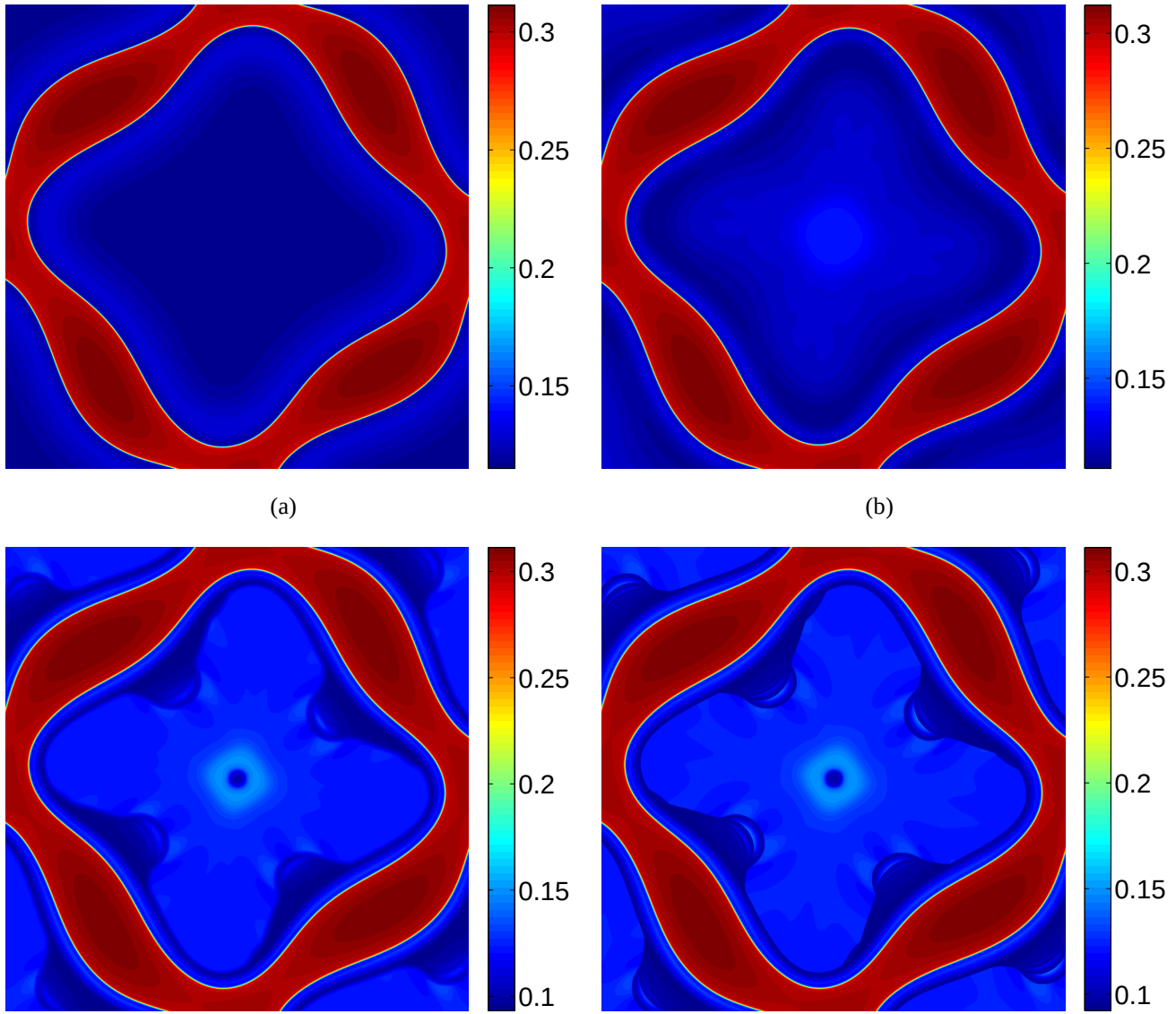


Figure 6.16: Concentration profiles with varying diffusivities in the solid (a) D_S/D_L = 0.1, (b) D_S/D_L = 0.01, (c) D_S/D_L = 0.0001, (d) D_S/D_L= 0.0

Table 6.7: Wavelengths used for the thermal profiles for amplitude ΔT=30K for Al-Mg. The total time is kept constant by considering the total number of timesteps which is 90×10^5 timesteps = 1.695 ms

Cycle time	wavelength	Time in ms
τ_1	200000	0.0377
τ_2	300000	0.0565
τ_3	400000	0.0754
τ_4	500000	0.0942
τ_5	600000	0.1131
τ_6	1000000	0.1884

6.9 Stirring cycles

Simulations concentrating more on the morphological changes, were done by assuming no diffusion in the solid. Going by the studies on the composition profiles studied in the previous section, it can be said, that by neglecting solid diffusion, no adverse changes are brought to the overall morphology. However, it was emphasized that solid diffusion does play a role in the solute redistribution within the solid.

Following by this assumption, the anti-trapping model was used for our study which takes into account only diffusion in the liquid and this section details more on changing the thermal gradients as a consequence of stirring frequencies. As stated earlier in Section 6.8, the amplitude of the sinusoidal wave corresponds to the intensity of the current required to propel the mush inside the stirrer and the wavelength is related to the stirring time/ frequency. A longer wavelength implies lower frequency or longer stirring time for a single cycle which also denotes a shallower thermal gradient. The thermal gradients can also be related in context to a cooling slope method of producing semi-solid slurries. A shallower slope connotes longer wavelengths and steeper gradients relate to shorter wavelengths. The amplitude and wavelengths used are given in Table 6.7 for Al-Mg and Table 6.8 for Al-Cu.

The simulation was done by placing seeds at defined locations in the melt and defining their compositions at their equilibrium solid value x^{α}_{Mg}=0.1024 at 770 K for Al-Mg and x^{α}_{Cu}=0.01627 at 840 K for Al-Cu. Since the phase-field simulations start with a step profile, very steep composition gradients persist at the solid and liquid side of the interface. Numerical instabilities set in since the anti-trapping function in the diffusion equation will not be able to resolve the interface. To circumvent this, the anti-trapping function is switched off for a few timesteps

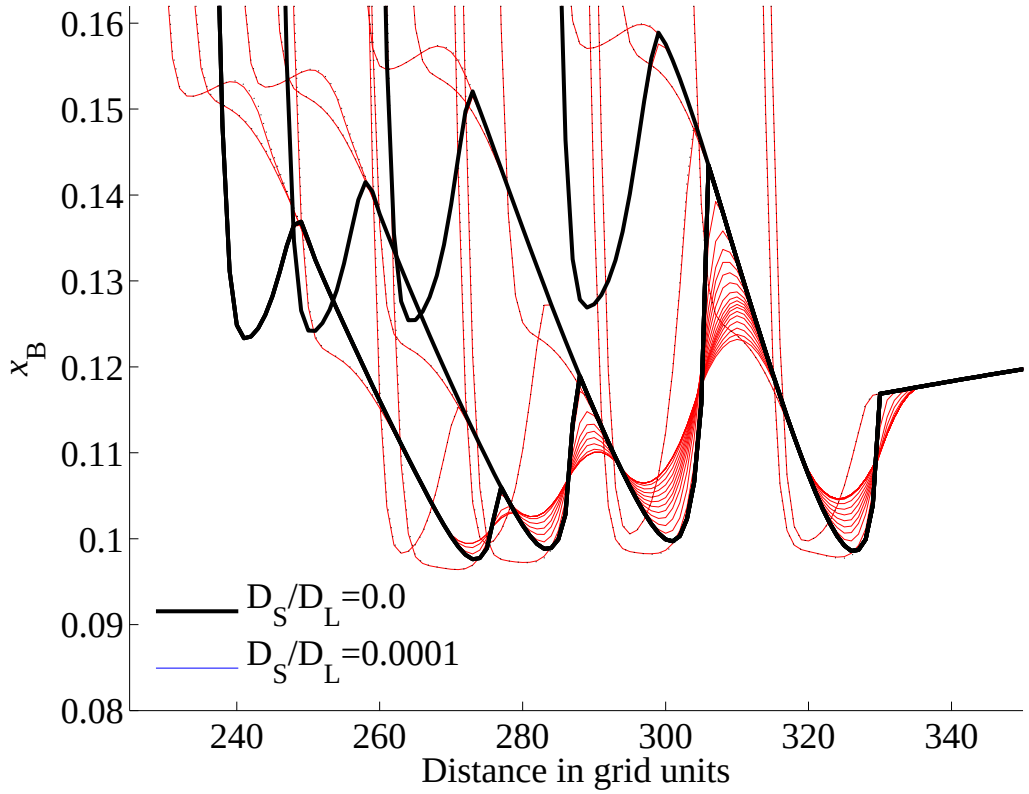


Figure 6.17: Solute pileup near the roots showing the effect of D_S; having a limited diffusivity causes the solute piled up near the surface to back diffuse into the solid (red lines). D_S=0 will not allow these solute to get diffused into the solid leading to the formation of bands, shown as spikes (black lines)

Table 6.8: Wavelengths used for the thermal profiles for amplitude ΔT=30K for Al-Cu. The total time is kept constant by considering the total number of timesteps which is 290×10^5 timesteps = 4.465 ms

Cycle time	wavelength	Time in ms
τ_1	200000	0.0308
τ_2	300000	0.0462
τ_3	400000	0.0616
τ_4	500000	0.0770
τ_5	600000	0.0924
τ_6	1000000	0.1540

until a smooth "S" type composition profile is obtained, that happens within a few steps since phase-field equations relax very fast. Numerical noise is introduced once the seed takes the initial four-fold shape. Periodic boundary conditions were assumed, to simulate the effect of continuous flow of solute in and out of the computational domain. The microstructure shown in Fig.6.18 is taken as the initial microstructure for subsequent simulations.

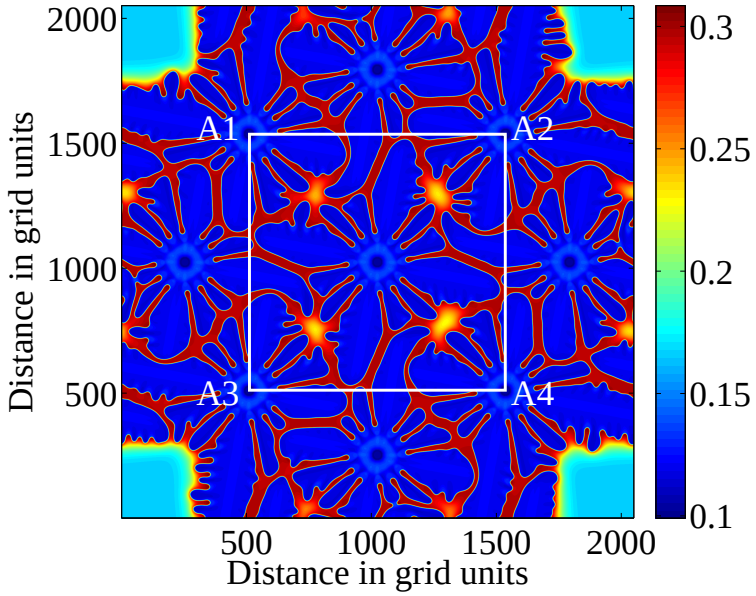
The thermal cycling simulations were done by assuming a condition of preheat, *i.e.*, by increasing the temperature of the simulation domain to 800 K, 30 K higher than the solidified temperature for Al-Mg system. For the case of Al-Cu, a preheat of 10 K was assumed and the thermal profiles as per Table 6.8 were used. The amplitude for the simulations was ΔT=30K and the cycling was purely an instantaneous discrete isothermal process. At every time step of the simulation, the temperature (T_c) of the entire domain is increased by,

$$T_c = T_0 + \Delta T \sin\left(\frac{2\pi}{\lambda}\mathrm{t}\right) \tag{6.8}$$

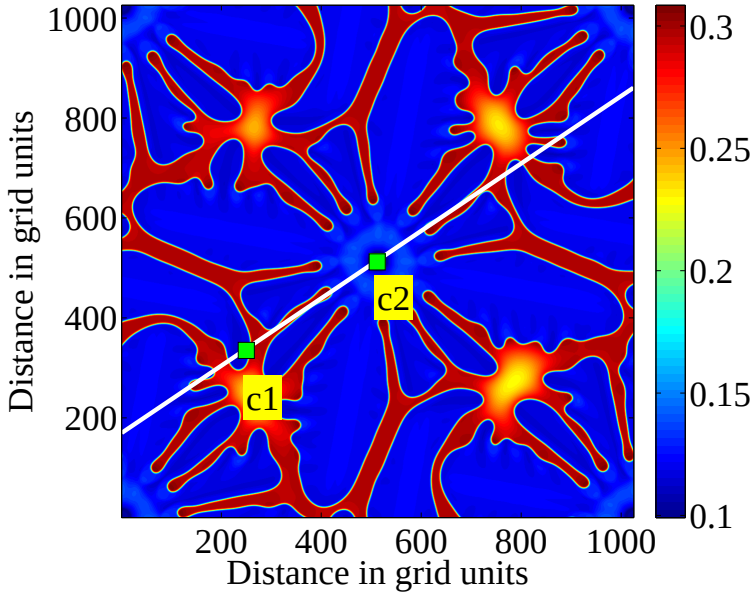
where T_c is the temperature at each discrete step in time t and T_0 is the initial temperature about which the thermal cycling is done.

The time step size t is given by $t = n\Delta t$ where n is any positive real number, say n=1 gives $t = \Delta t$. Finer value of n though gives a better resolution to the continuous profile, also increases the computational time. The thermal cycle follows the path shown in Fig.6.9 **a-b-c-d-a**. Two paths are present *viz.*, path **d-a-b**, melting, characterized by the shrinking of the dendrites and path **b-c-d**, resolidification, characterized by dendrite coarsening. It is required at this conjecture to point out that, neither melting nor coarsening initiated at the said points. It was observed that melting did not stop at **b** but rather continued to a point between **b-c**. This was attributed to solutal remelting as explained earlier in Section 6.14 and also the coarsening continued to a point between **d-a**.

As melting proceeds, an increase in temperature causes the solute composition to decrease as we move up the solidus and a decrease in temperature causes the solute to be pushed out of the solid. The extent of melting and coarsening was identified by the minimum and the maximum solute levels, respectively, reached. As mentioned in the previous section, an absence of diffusion condition in solid will have a tendency to produce bands seen as spikes in the composition profiles taken along "c1-c2", shown in Fig.6.19. Simulations performed on Al-



(a)



(b)

Figure 6.18: (a) Al-Mg dendrite simulated using the conditions given in Table 6.5, (b) Region of interest A1-A2-A3-A4. All concentration profiles are taken along the line c1-c2

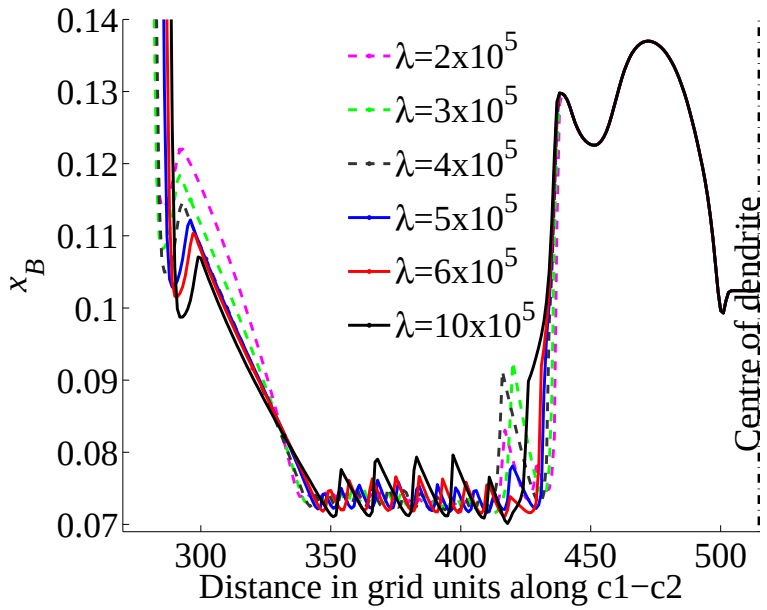


Figure 6.19: Composition profiles showing solute pile up near the roots along the path c1-c2, for Al-Mg

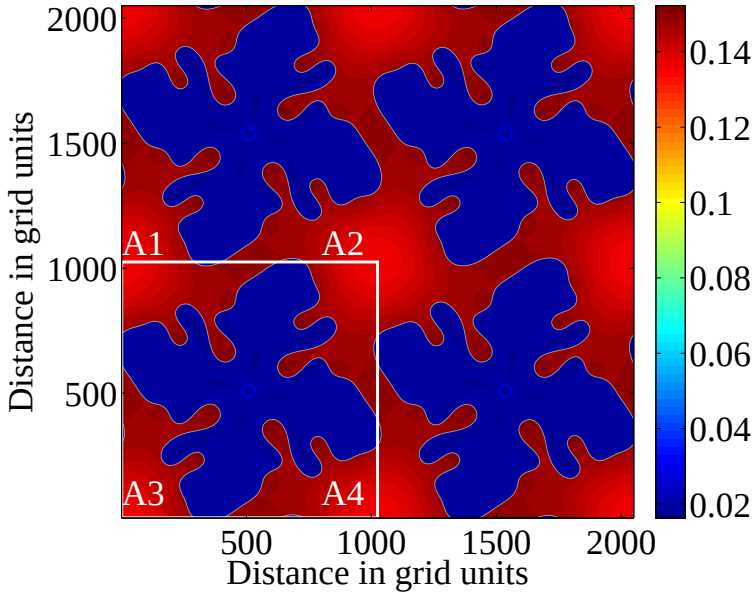
Cu showed the same trend, except that due to numerical instabilities, it was required to use lower Δx and hence lower Δt, which increased the computational time considerably. The compositional profiles corresponding to the dendritic morphology in Fig.6.20 are shown in Fig.6.21.

One more observation that can be made is that, as the holding time increased, for larger wavelengths, the amount of solute piled up near the interface is found to increase. This is attributed to the availability of more dwell time at every instantaneous discrete time step.

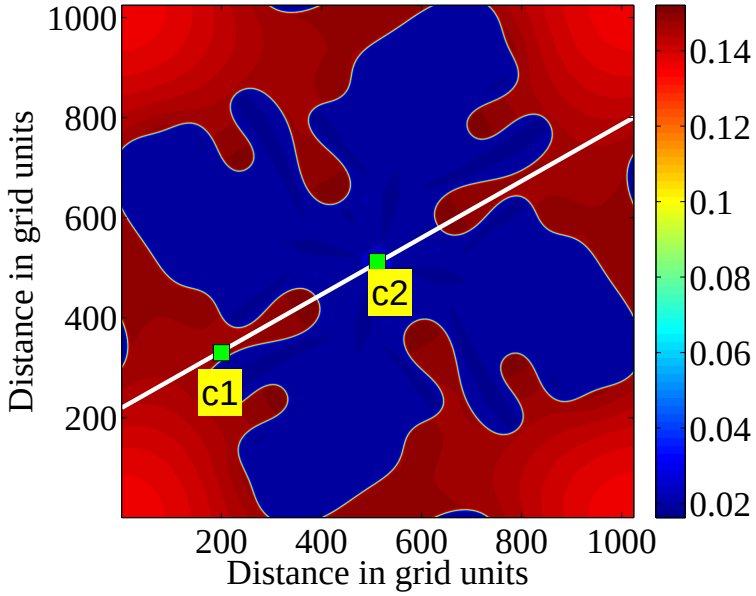
The overall morphological sequence, from dendritic to globular, brought about as a result of continuous cycling is shown for Al-Mg in Fig.6.22 and Fig.6.23 for Al-Cu. A more detailed quantitative analysis of the transition is studied using fractal dimensions, explained in detail in the following section.

6.10 Quantitative analysis of growth morphologies

A qualitative observation, though shows the morphological transition from dendritic to globular, quantifying the morphological features is necessary. Two different approaches to quantify the morphological features were attempted *viz.*, fractal approach and shape factor determina-



(a)



(b)

Figure 6.20: (a) Al-Cu dendrite simulated using the conditions given in Table 6.6, (b) Region of interest A1-A2-A3-A4. All concentration profiles are taken along the line c1-c2

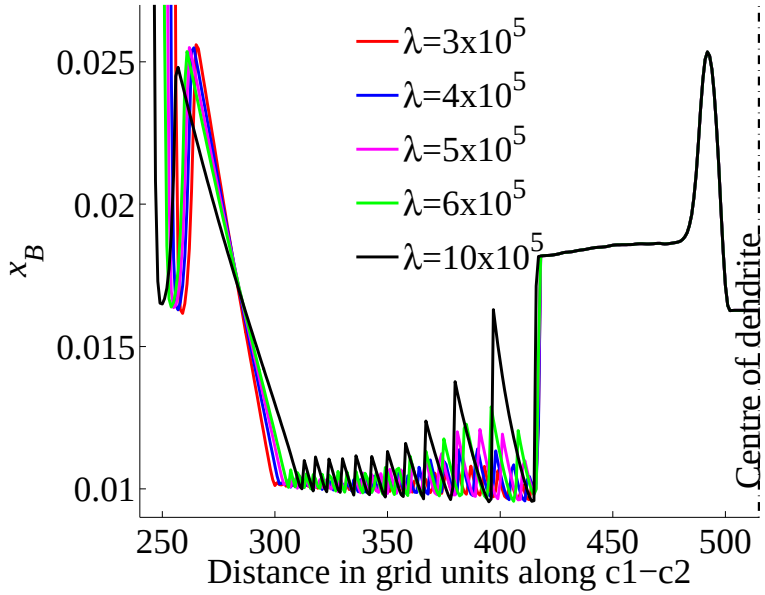


Figure 6.21: Composition profiles showing solute pile up near the roots along the path c1-c2 for Al-Cu

tion. The two approaches are discussed briefly below.

6.10.1 Fractal analysis

Fractals are self similar objects that possess a higher degree of complexity. The best way to measure the complexity is through fractal dimension. Fractal dimension allows one to measure the degree of complexity by comparing how detailed a pattern changes with the scale at which it is being measured. It is characterized by the space-filling capacity of a pattern. It differs from Euclidean dimension in that it can take non-integer values. A better way of explaining a fractal is based on lines and curves; a curve with a fractal dimension close to 1 say (1.05) behaves very much like a line, but on the contrary, a curve with a fractal dimension of 1.93 is much more convoluted and very much fills the surface. There are two classes of dimensions encountered when dealing with fractals. One is the self similar dimension which is widely used for exact or mathematical fractals that are self similar. The other is the box counting dimension to measure patterns that are not self similar, which is the case of most commonly occurring patterns in nature. In this context, dendrites, naturally occurring, do not possess self similarity, but are considered as tree-like topological spaces that has an inherent complexity and a pattern to it.

In the box counting method, we cover a shape or a pattern with boxes and find how the number

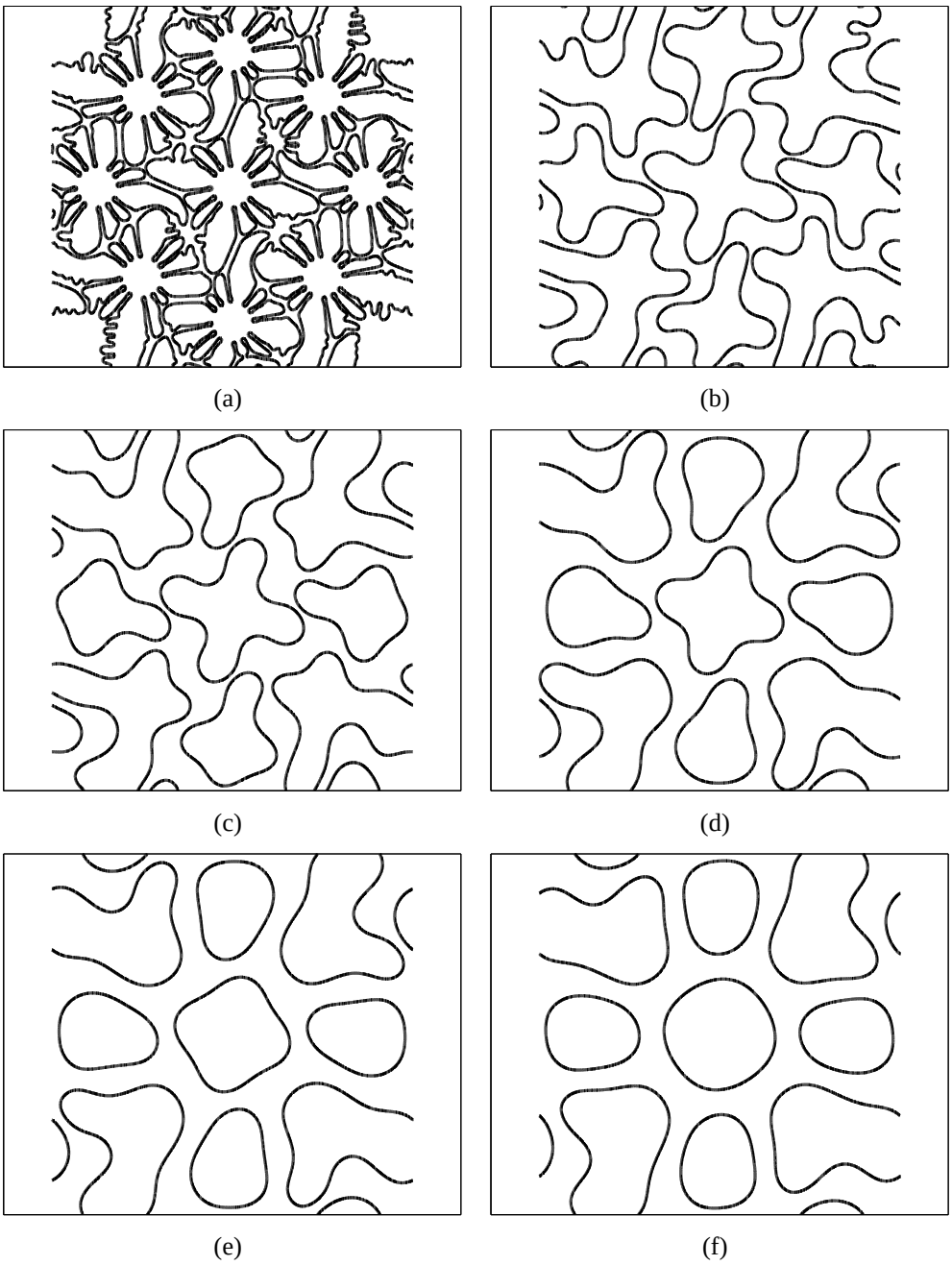


Figure 6.22: Evolution of globular morphology in Al-Mg through multiple cycling for $\lambda = 1x10^6$. The final morphology is after 10 cycles

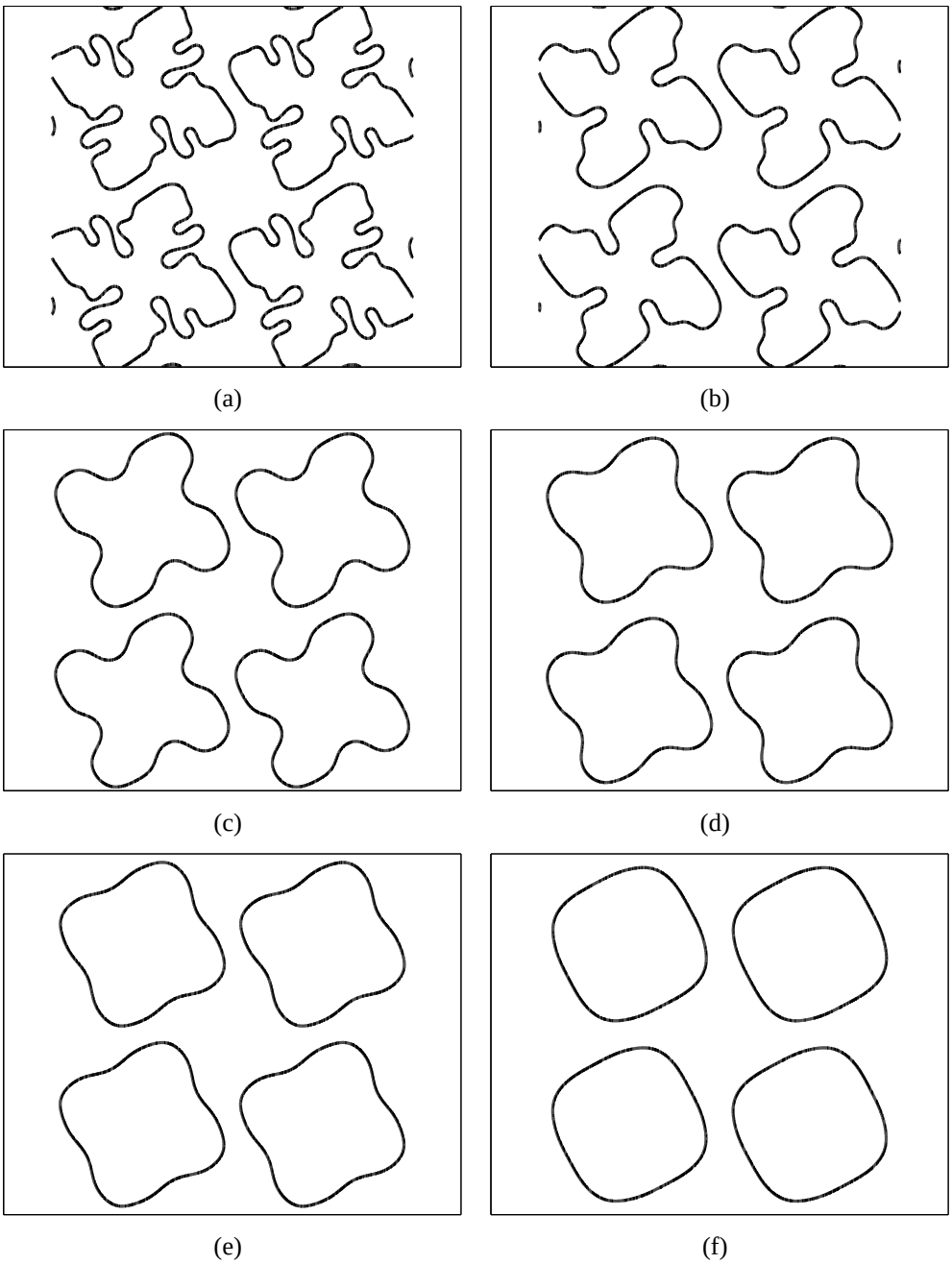


Figure 6.23: Evolution of globular morphology in Al-Cu through multiple cycling for $\lambda = 1x10^6$. The final morphology is after 29 cycles

of boxes N changes with the size of the boxes $1/r$. If we take an object in Euclidean dimension (D=1,2,3) and reduce its linear size by $1/r$ in each spatial direction, then its measure (length, area or volume) would increase to $N(r) = (1/r)^D$ times the original. Here 1/r is considered instead of r because, as the squares get smaller, more number of squares would be needed to cover the shape. For example, taking a line of unit length and dividing it into 3 equal segments ($r = 1/3$) would require 3 boxes to cover the line, which implies that after taking log, $D = \log(N(r))/\log(1/r) = 1$. It will turn out that D will be an integer in the Euclidean space and a non-integer in the fractal space. In the fractal space, *i.e.,* for more complicated profiles, the relation between $N(r)$ and $1/r$ may be a power law, $N(r) = k(1/r)^D$. To calculate the fractal dimension of a set(image) S, fill the space with boxes of side length $1/r$ and count the number of boxes which intercept the pattern outline. The box counting dimension or the Minkowski-Bouligand dimension (Falconer (1990)) is calculated by taking into account the change in this number as the box size goes finer. The fractal dimension is then given by,

$$D = \lim_{r \to 0} \frac{\log N(r)}{\log(1/r)} \tag{6.9}$$

If the limit exists, it is called the box counting dimension, D, of the shape or the pattern. The resulting plot will be a line whose slope gives the fractal dimension. Fig.6.24(a) is a dendrite image whose fractal dimension was calculated using the box counting method and Fig.6.24(b) shows the log-log plot for both the pattern (actual box count) as well as the entire surface (space-filling box count). The space-filling box count gives the Euclidean dimension and the actual box count gives the fractal dimension of the set.

A circle is a curve in 1-D space which takes on a value of 1 in the Euclidean dimension. In quantifying the fractal dimension, as we move from dendritic morphology to a globular morphology, this transition is reflected by the change of this fractal dimension given in Fig.6.25 from 1.74 to 1.54 for Al-Mg, a clear indication of the morphological change shown in Fig.6.22

6.10.2 Shape factor determination

A much more elegant way of addressing the globular transition is through the measurement of shape factor. Shape factor is a dimensionless quantity defining the shape of a particle, indepen-

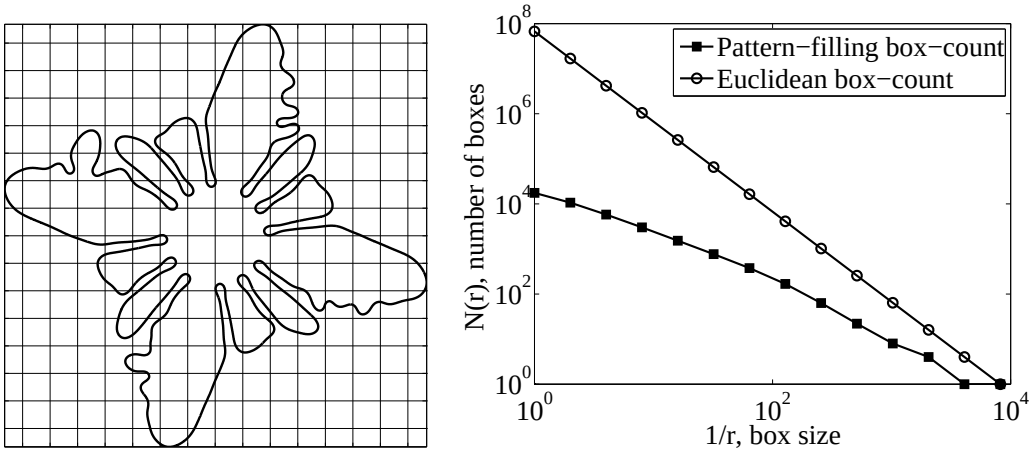


Figure 6.24: (a) Example of the box counting method, a number of boxes with varied sizes fill the domain and those boxes which intercept the dendrite pixels are counted as one. (b) log-log plot showing the power law relation between the box size and the number of boxes which had intercepted for a given box size. The squares covering the entire space gives the Euclidean space filling with a D value of 2 whereas the boxes which intercept the dendrite show a deviation from the straight line, clearly indicating the fractal nature of the dendrite.

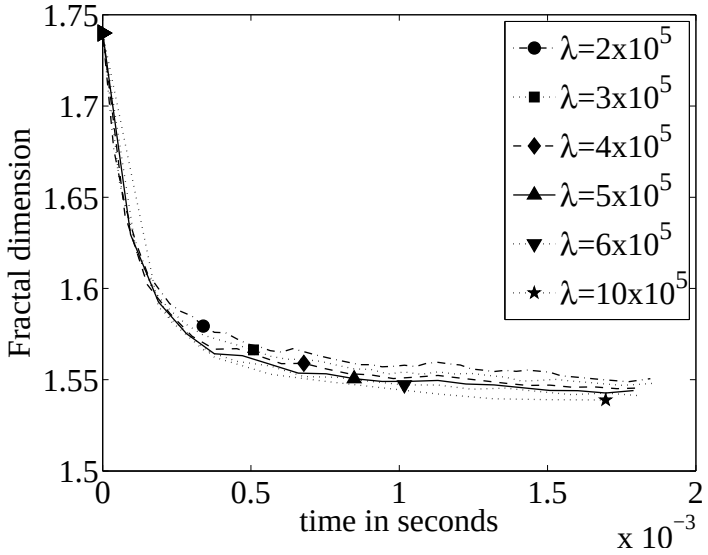


Figure 6.25: Fractal dimension depicting dendritic to globular transition in Al-Mg as a function of stirring cycles

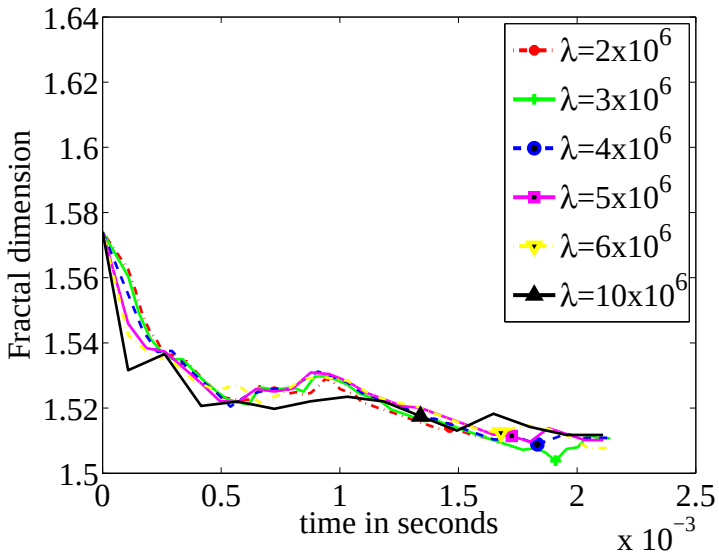


Figure 6.26: Fractal dimension depicting dendritic to globular transition in Al-Cu as a function of stirring cycles

dent of its size. It is determined by measuring the dimensional quantities like area, perimeter, ferret diameters, chord length, centroid, second moments, *etc.* Shape factor analysis allows one to compare the degree of deviation of particle shape from standard images like circle, square or equilateral polyhedron. This is characterized by determining the 2-D particle boundary using image analysis. The shape factor values are normalized ranging from 0 to 1. For example, shape factor for an ideal circle, square or cube equals 1.0 and any small deviation from ideal shapes will have a value less than 1.0. Standard shape factor measurements include aspect ratio, circularity or globularity, elongation shape factor, compactness shape factor and waviness shape factor.

Since our analysis is concerned with change in shape of grain from dendrite to globular due to thermal cycling, circularity or globularity measurement is carried out. The morphology of grains, globular or dendritic, was analyzed using the formula $(4\pi A)/P^2$. Image analysis was performed to determine the cross sectional area (A) and perimeter (P) of the grains. The boundary between solid and liquid phase was considered as a grain. For spherical/ globular structure the value is 1 and for a dendrite the value tends to 0 (Kirkwood *et al.* (2010); Wang *et al.* (2004)). The shape-factor measurements as a function of time are shown in Fig.6.27 and Fig.6.28 for Al-Mg and Al-Cu, respectively. The shape factor for Al-Mg was found to change from 0.043 to 0.584, clearly indicating the morphological transition from dendrite to globular. Similarly, the morphological transition is effectively captured in Al-Cu with shape-factor values

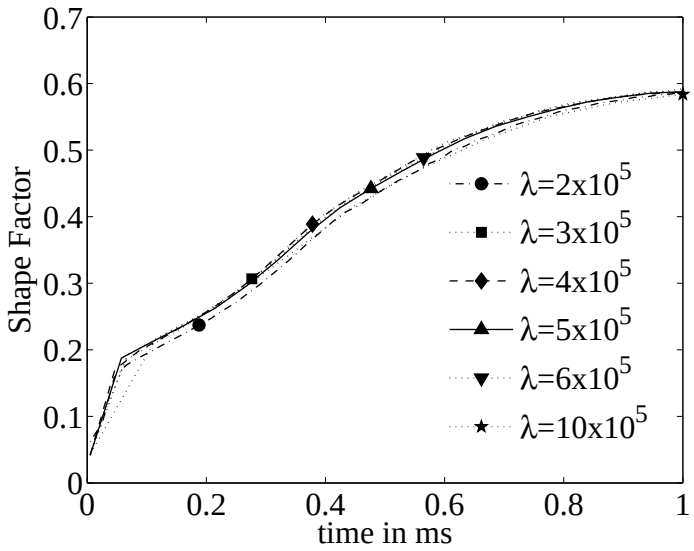


Figure 6.27: Shape factor as a function of stirring cycles for Al-Mg showing the dendritic to globular transition

varying from 0.188 to 0.646.

6.11 Discussion

Microstructural evolution during melting and resolidification in a thermal gradient was investigated by Buchmann and Rettenmayr (2005). As melting is assumed to start at regions of high surface energy, the high curvature regions in case of dendrites will be the first to melt. During solidification, the interface is in local equilibrium with the surrounding liquid. Concentration fluctuations at the interface would more often lead to a symmetry breakdown. This creates a concentration gradient leading to diffusion in the liquid. Diffusion of solute atoms away from the S-L interface leads to solidification, whereas diffusion towards the interface leads to melting. At the solidifying interface, during resolidification, solid with equilibrium composition lower than the supersaturated composition will precipitate. The liquid between the interfaces will move with the interface and grow in thickness due to the composition difference of melting and resolidifying material.

All morphological changes are driven by solute diffusion through the liquid matrix by gradients established by local surface curvature. Furthermore, the finer morphologies, such as dendritic arms and small particles will coarsen rapidly and disappear first because of their associated higher curvatures Limodin *et al.* (2007). This shape change can be approximated, as done by

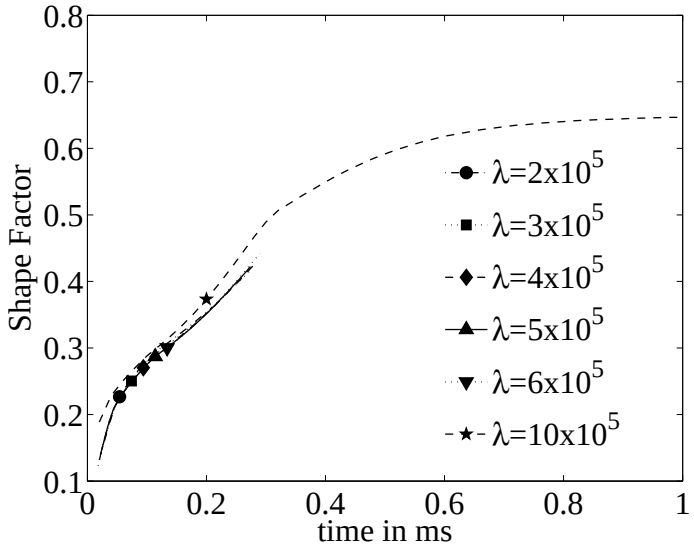


Figure 6.28: Shape factor as a function of stirring cycles for Al-Cu showing the dendritic to globular transition

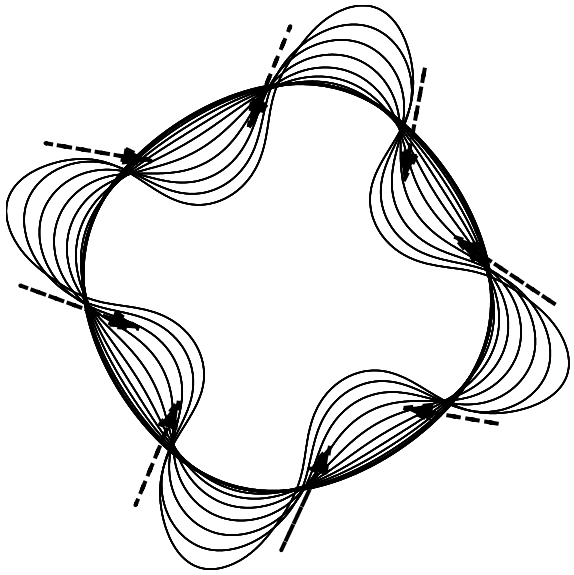


Figure 6.29: Phase-field profiles showing the change from a dendritic to globular morphology for Al-Mg. Surface diffusion occurs from regions of higher curvature to lower curvature regions *i.e.* regions of higher chemical potential to lower chemical potential as depicted by the arrows

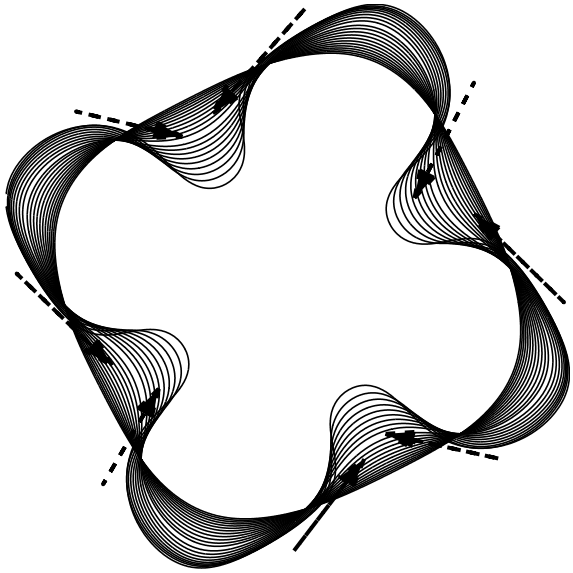


Figure 6.30: Phase-field profiles showing the change from a dendritic to globular morphology for Al-Cu. Surface diffusion occurs from regions of higher curvature to lower curvature regions *i.e.* regions of higher chemical potential to lower chemical potential as depicted by the arrows

Mullins (1959), by the principle of surface tension driven diffusion with particularly simple geometry *i.e.*, smoothing of a surface with a sinusoidal ripple in it. This analysis is so chosen as it is able to describe the relative contributions of the competing transport processes. The chemical potential of the solute atoms on the hills is higher ($\mu > 0$) than that beneath a flat surface (μ=0), while in the valleys it is lower (μ<0). The local rate of change of height of the surface at any point is given by the sum of the flux away from the surface into the solid and the divergence of the surface flux. This analysis is clearly seen by the evolution profiles in Fig.6.29 for Al-Mg and Fig.6.30 for Al-Cu.

CHAPTER 7

GROWTH OF ALLOY DENDRITES IN 3-D

Dendritic growth in 3-D was attempted to understand the morphological features and the mechanism of solutal flow. Modelling the solutal flow through the dendrites is important because all the morphological transitions take place together with solute redistribution in the vicinity of the dendrite. This chapter focuses on the growth of solutal dendrites in 3-D. Issues pertaining to 3-D simulation though earlier discussed in Chapter 5, some unique insights into the physical processes that take place during the growth of a 3-D dendrite are discussed in this chapter.

Simulation of 3-D dendrites using phase-field method was first attempted by Kobayashi (1992), which propelled the phase-field revolution. Many simulations on 3-D dendrites using the phase-field method were done, of which significant work was done by Karma and Rappel (1996*a*) into the quantitative aspects of 3-D dendrite growth morphology using the microscopic solvability theory. One of the early parallel phase-field simulations were performed by George and Warren (2002), who extended the work of Warren and Boettinger (1995) to 3-D.

In this study, two different sets of simulations were carried out.

- The first set of simulations were done using the isothermal binary alloy solidification model by Warren and Boettinger (1995) with model free energy data and the material properties of the Ni-Cu system. In this work, we further take up the study of not only the alloy growth in 3-D, but also the effect of thermal cycling on the fragmentation and the coarsening of dendrite arms.
- The second set of simulations involved incorporating Gibbs energy expressions of Al-Mg obtained using CALPHAD description, to study the effect of solute redistribution due to thermal cycling. The anti-trapping phase-field model by Kim (2007) with one-sided diffusion was also envisaged, but only the growth of a single dendrite was studied due to memory constraints and computational time.

The simulations were done on a structured computational grid. A second order finite-difference in space and first-order in time were adopted. The numerical approach and simulation schemes adopted were discussed in Chapter 5 on Numerical Implementation.

7.1 Dendrite growth in Ni-Cu Alloy

The isothermal binary alloy solidification model by Warren and Boettinger (1995) was extended to 3-D and the governing equations for the phase-field and concentration were derived from an entropy functional $S(\phi, c, T)$. Evolution of phase-field is given by,

$$\frac{\partial \phi}{\partial t} = M_\phi \left(\varepsilon^2 \nabla^2 \phi - (1-c) \frac{\partial f^{\mathrm{A}}}{\partial \phi} - c \frac{\partial f^{\mathrm{B}}}{\partial \phi} \right) \tag{7.1}$$

where, $M^\phi = (1-c)M^{\mathrm{A}} + cM^{\mathrm{B}}$, is the phase-field mobility given as a weighted sum of relative concentrations of the individual mobilities. The terms $\frac{\partial f^{\mathrm{A}}}{\partial \phi}$ and $\frac{\partial f^{\mathrm{B}}}{\partial \phi}$ are the derivatives of free energy with respect to ϕ and are given by,

$$\frac{\partial f^{\mathrm{A}}}{\partial \phi}(T, \phi) = W^{\mathrm{A}} g'_w(\phi) + 30 g_w(\phi) L^{\mathrm{A}} \left(\frac{1}{T} - \frac{1}{T_m^{\mathrm{A}}} \right) \tag{7.2}$$

$$\frac{\partial f^{\mathrm{B}}}{\partial \phi}(T, \phi) = W^{\mathrm{B}} g'_w(\phi) + 30 g_w(\phi) L^{\mathrm{B}} \left(\frac{1}{T} - \frac{1}{T_m^{\mathrm{B}}} \right) \tag{7.3}$$

W^{A} and W^{B} are free energy barrier heights, $g_w(\phi) = \phi^2(1-\phi)^2$ is the double well function and L^{A} and L^{B} are the latent heats of fusion of A and B. T is the temperature of the domain and $T_m^{\mathrm{A,B}}$ is the melting temperature of the pure elements A and B. The prime denotes differentiation with respect to ϕ. The governing equation for the diffusion of solute is given by,

$$\frac{\partial c}{\partial t} = \nabla \cdot D_c \left[\nabla c + c(1-c) \left(\frac{\partial f^{\mathrm{B}}}{\partial \phi} - \frac{\partial f^{\mathrm{A}}}{\partial \phi} \right) \nabla \phi \right] \tag{7.4}$$

and the diffusivity of the solute is given by the weighted average of diffusivity in solid and liquid, $D_c = D_{\mathrm{S}} + p(\phi)(D_{\mathrm{L}} - D_{\mathrm{S}})$. $p(\phi) = \phi^3(6\phi^2 + 15\phi - 10)$ is an odd-ordered polynomial which provides a smooth interpolating function between the two phases. The angular dependence of the surface energy σ is included in the gradient energy term as,

$$\varepsilon(\mathbf{n}) = \varepsilon_0(1 - 3\epsilon_4) \left(1 + \frac{4\epsilon_4}{1 - 3\epsilon_4} \frac{(\phi_x)^4 + (\phi_y)^4 + (\phi_z)^4}{|\nabla \phi|^4} \right) \tag{7.5}$$

where ϵ_4 is the strength of capillary anisotropy.

Table 7.1: Simulation parameters for the growth of Ni-Cu dendrite in a supersaturated melt. Other material and simulation parameters are as provided in Table 6.2

Grid parameters		
Interface width	δ	$500\mathrm{d}_0$
Spatial resolution	$\Delta\mathrm{x}$	$\delta/5$
Time resolution	$\Delta\mathrm{t}$	$0.2\Delta\mathrm{x}^2$
Volume per slice	nx$\times$ ny$\times$ nz	$18 \times 1024 \times 512$
No of slices/ CPUs	ncpus	64
Total domain size	Nx$\times$ Ny$\times$ Nz	$1024 \times 1024 \times 512$

7.1.1 Simulation parameters

The governing equations in Eq.(7.1) and Eq.(7.4) are solved using second-order finite-difference in space and explicit forward difference in time. Detailed algorithms and numerical issues are dealt in more detail in Chapter 5. Dendritic growth is simulated by nucleating a circular seed in the X-Y plane. After a few time steps, noise is introduced along the liquid side of the interface. This initiates the growth of secondary arms and the simulations are stopped once the primary arms reach the domain edges. Parallelization is achieved through a distributed programming model using mpi routines. A fully grown Ni-Cu dendrite grown across 64 CPUs is shown in Fig.7.1 and the corresponding simulation parameters are given in Table 7.1.

7.1.2 Morphological features

A most important aspect of solidifying microstructures is microsegregation, which results from the freezing of the solute enriched liquid in the interdendritic regions (Ghosh (2001)). Segregation causes rejection of solutes by the solid with consequent enrichment of the remaining liquid. This more often leads to the formation of oxides, intermetallics, etc. during solidification. These reactions have a strong effect on the segregation patterns and subsequently on the final morphology. Localized microsegregation leads to macrosegregation as a result of the physical movement of the solute in the interdendritic liquid. However, the flow of solute in the interdendritic space is difficult to visualize experimentally and very few data are available to even for qualitative assessment.

Availability of various models to study microstructure evolution have provided with insights to study deeper into these aspects of interdendritic segregation and solute enrichment. However, most of these model predictions are restricted to 2-D, that has its own inherent limitations. One

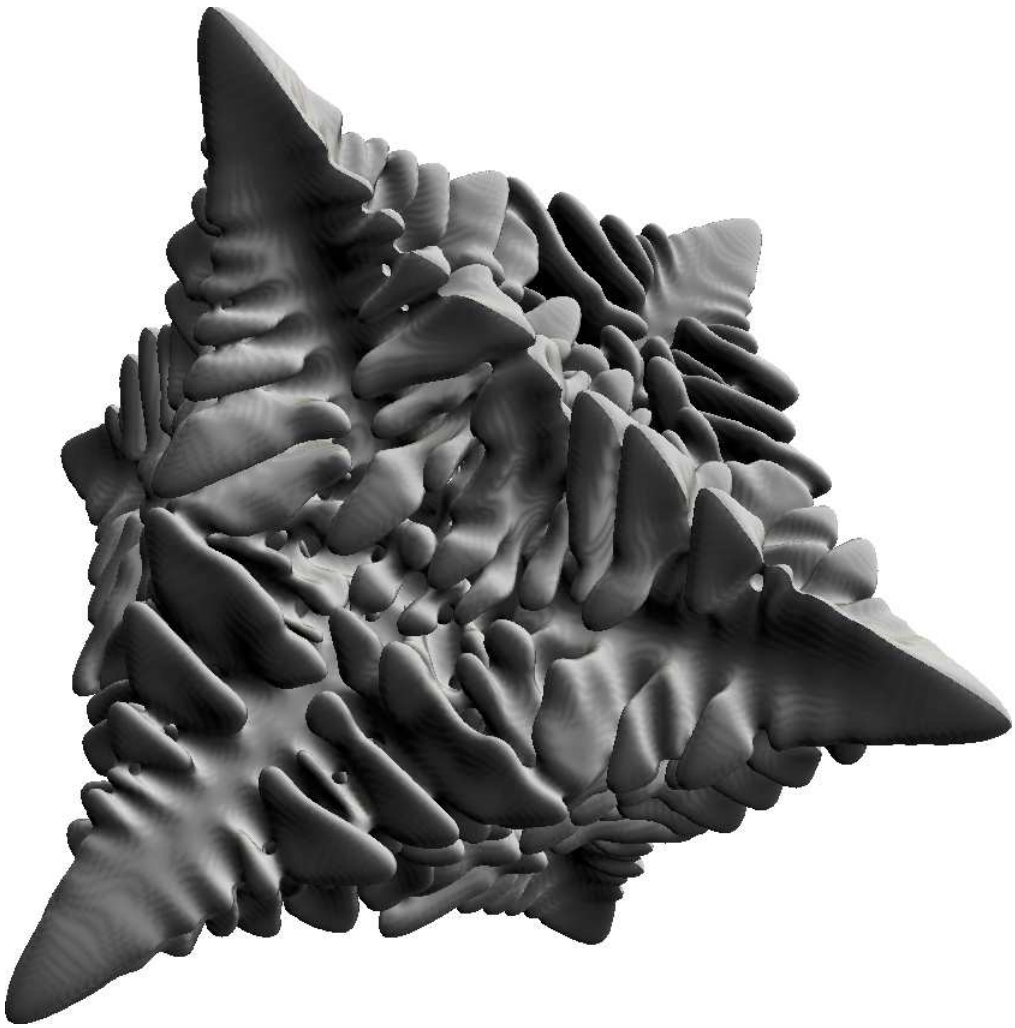


Figure 7.1: Phase-field profile taken at an isosurface value of ϕ =0.5 of a Ni-Cu dendrite grown in a CPU cluster across 64 processors. The dendrite is grown by placing a seed on the XY base plane and grown along Z. The entire dendrite is obtained by mirroring about the XY plane

major limitation encountered in 2-D, is the restriction of solute movement along 2 directions only. One more major growth limitation is in conceding one of the 2 curvatures, where curvature plays a key role in the solidification kinetics, thereby defining the growth morphology. Since microsegreation leads to macrosegregation in the interdendritic regions, it often becomes impossible to treat this phenomena in 2-D. Only large scale 3-D modelling techniques will be able to provide a deeper insight into this aspect. In today's scenario it is made possible by the availability of the ever increasing computing power in the simulation of microstructure evolution in 3-D.

It is possible that the artificial geometric confinement of the interdendritic liquid in a 2-D simulation (leading to increase in the solute concentration) could give numerically similar segregation patterns as in a 3-D simulation that considers the curvature effect in two directions (higher solute concentration via Gibbs-Thomson effect) and a solute diffusion path in three dimensions (leading to lower solute concentrations via solute diffusion through channels). However, it is obvious to note that the 3-D simulations take more physical phenomena into account and therefore are more realistic. This feature is of importance in predicting the solute segregation induced nucleation of intermetallic phases during dendritic solidification such as in thin-slab casting of multi-component alloys, solidification of multicomponent alloy braze fillers and so on.

A qualitative observation of the growth morphology of a 3-D dendrite shown in Fig.7.2, reveals that the solute flow in the interdendritic regions is continuous. This is opposed by the restriction of solute movement in 2-D, that constrains the solute and entraps them. Fig.7.3 shows a dendrite growth pattern in 2-D. The bright red spots near the roots of the primary are solute enriched zones. In 2-D, however, they are unable to flow out of the plane of the paper. The solute which has now become trapped within the cavities tend to modify the morphology of the surrounding regions. In reality, the solute is able to flow around the dendrite, which is clearly captured in 3-D simulations. The solutal flow around the dendrite is shown in Fig.7.4. The interconnected flow of solute is shown in Fig.7.5. Though the final morphological features observed in 3-D and 2-D plane sections tend to remain the same qualitatively, it should be noted that the principal curvatures κ_1 and κ_2 have a significant role in the final morphological shaping.

It is also observed that as the strength of the capillary anisotropy is increased, branching is suppressed (Fig.7.2(a)) as opposed to the more branched Fig.7.2(b). Lower anisotropies creates

Table 7.2: Wavelength and cycle time used for the thermal cycle

Amplitude	ΔT	10 K
wavelength	λ	50000 (0.78 ms)

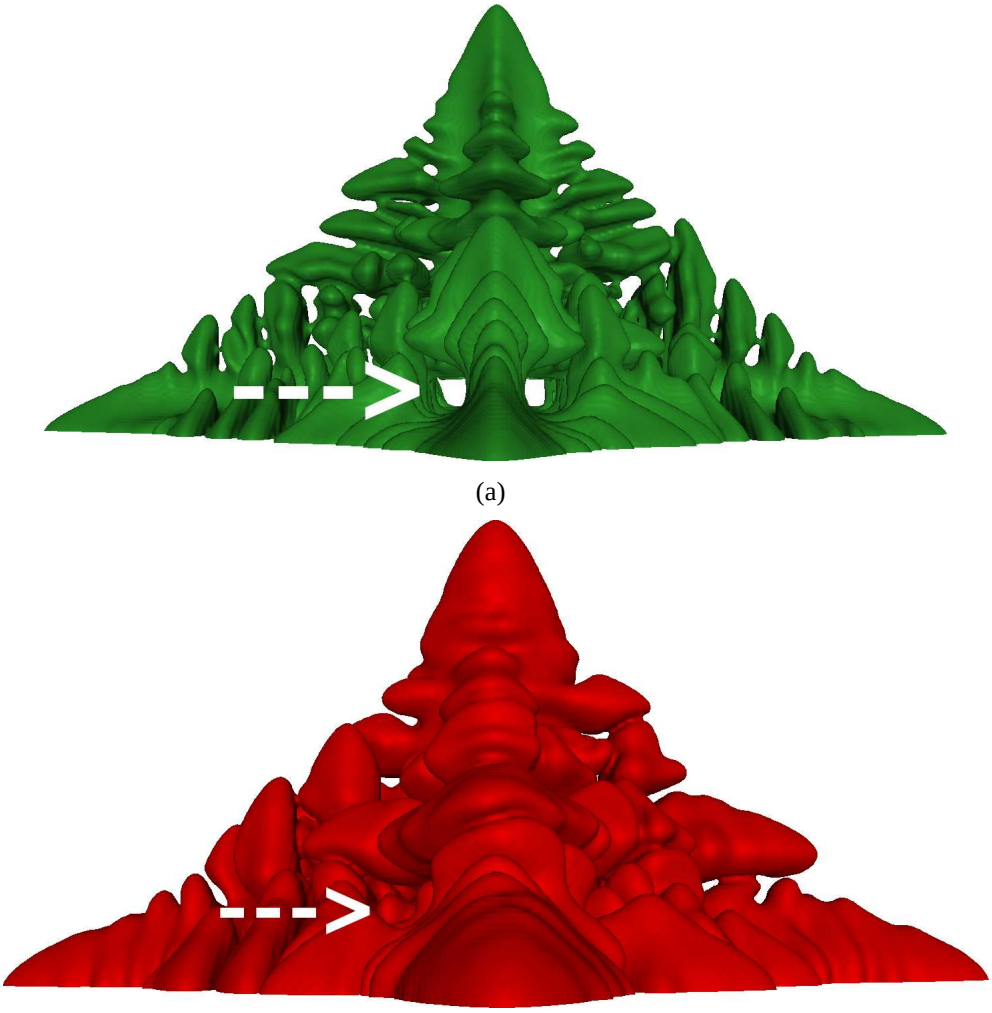
a tortuous path for the interdendritic liquid which might cause discontinuous precipitation or eutectic formation in these channels than compared with a more pronounced channel flow. The liquid flow would be continuous and this leads to a continuous and a larger channel for the flow as shown in Fig.7.4 and Fig.7.5.

There could be situations where prediction of such a channel formation in 3-D dendrite growth is important but the availability of the strength of anisotropy for an alloy system is limited. In such situations, a length scale bridging with an atomistic simulation technique such as Molecular Dynamics (MD) could be of importance. Such studies to obtain model parameters for phase-field from MD are available (Bragard *et al.* (2002); Danilov *et al.* (2009); Guerdane *et al.* (2010)). MD simulations on solidification are being attempted (Nestler *et al.* (2009); Sankarasubramanian and Kumar (2010)), as numerical computation involving a large number of atoms is now possible.

7.1.3 Thermal cycling

Thermal cycling shown in Fig.7.6, was done to study the morphological transition from dendritic to globular, where, apart from thermal and solutal gradients, the curvature also plays a significant role in modifying the morphology. An attempt was made to study the effect of thermal cycling on the grown dendrite and a qualitative study was made.

The amplitude and the wavelength used for the thermal cycling is shown in Table 7.2. A single cycle was done which resulted in the fragmentation of the secondary arms. Due to computational constraints, multiple cycling was not attempted. However, a fair correlation between the multiple cycling done on Al-Mg dendrites can be arrived. It was shown in Fig.6.22 that fragmentation of the secondary arms happen during initial cycles. These fragmented high curvature regions (secondary arms) are consumed by the low curvature (primary arms) regions. A similar scenario is observed in 3-D as shown in Fig.7.6, where secondary arm fragmentation is observed. It can be assumed that further cycling would cause the primary arm to coarsen at the expense of secondary arms. A qualitative assessment though made, it has to be pointed out



(a)

(b)

Figure 7.2: Interdendritic channel flow in the regions surrounding the dendrite. It is clearly shown that as the strength of capillary anisotropy is increased the formation of channels are more pronounced as evident from (a) for ϵ_4 =0.04 and (b) for ϵ_4=0.025. Though, branching is more pronounced at lower anisotropies, formation of channels are still evident suggesting that primary arms do play a major role in its formation

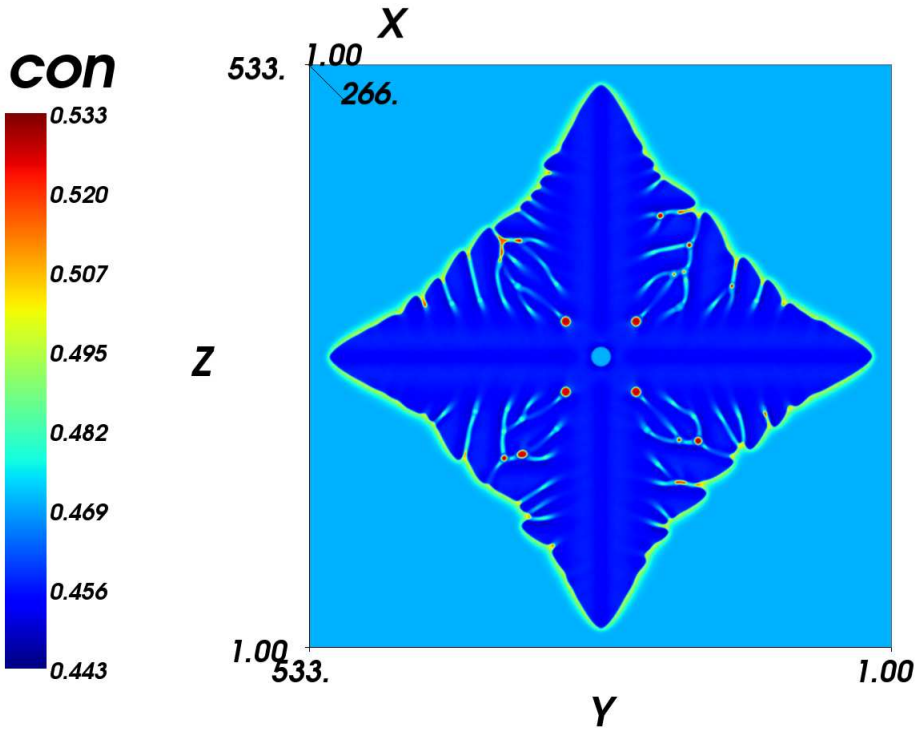


Figure 7.3: Solutal dendrite in 2-D showing solute entrapment. Channel formation evident in 3-D is suppressed in 2-D, which leads to morphological artefacts

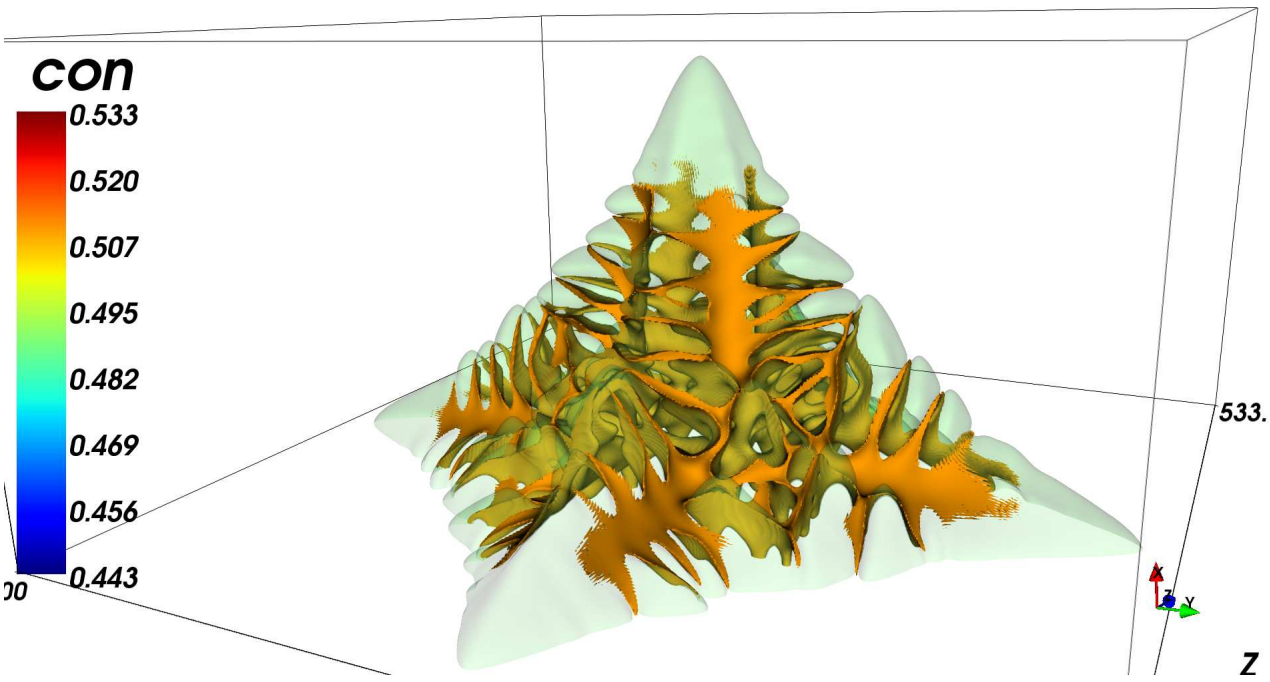


Figure 7.4: Solutal flow around a dendrite through the interdendritic channels

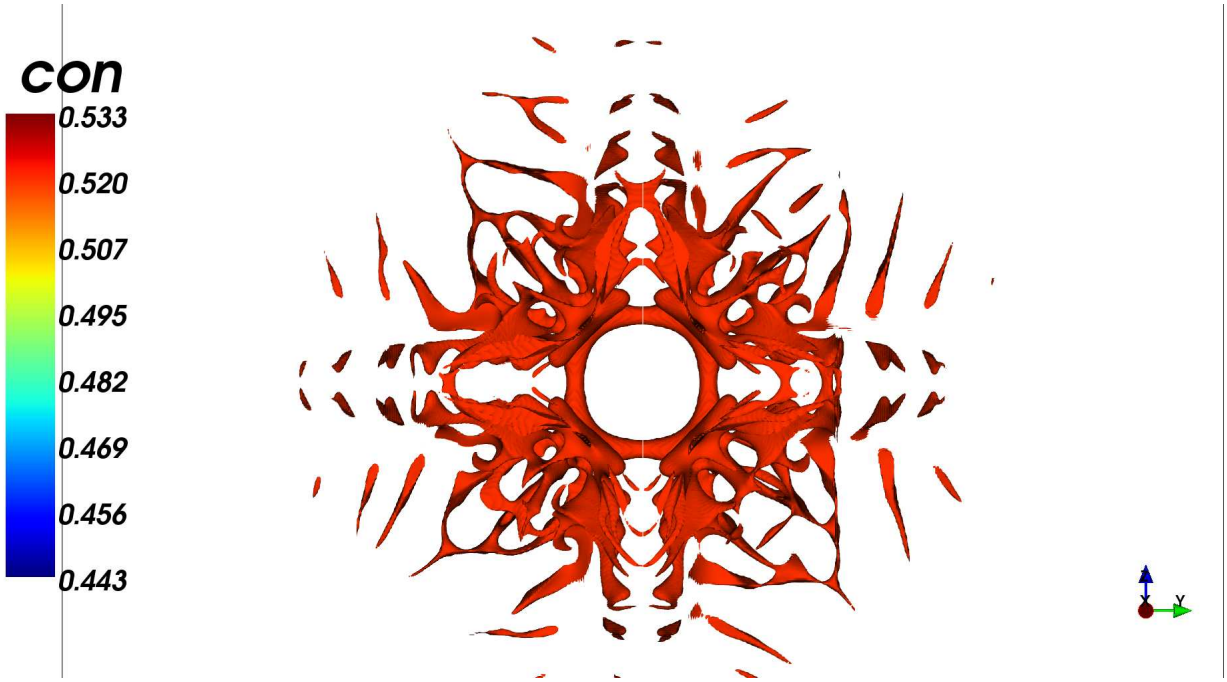


Figure 7.5: Interconnected solutal network through the channels showing that the channels are continuous along the primary arms

that further cycling operations are required to ascertain the solute redistribution analogy made in the case of 2-D.

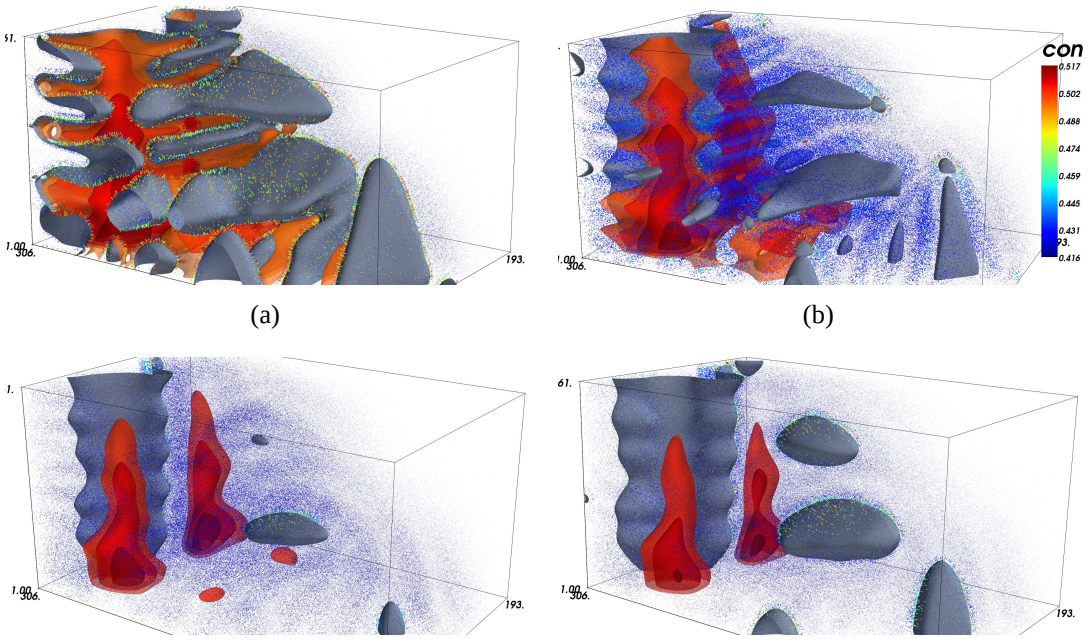
7.2 Al-Mg dendrite

Growth of an Al-Mg dendrite was simulated by incorporating the Gibbs energies for the solid and the liquid phases obtained through CALPHAD. Detailed explanations on the coupling aspects of the Gibbs energy function with the phase-field model are provided in Chapter 4 and Chapter 5. Kim *et al.* (1999) model was used for the simulations, lending the following governing equations for solidification of a binary alloy.

$$\frac{1}{M_\phi}\frac{\partial \phi}{\partial t} = \varepsilon^2 \nabla^2 \phi - W\frac{dg_w(\phi)}{d\phi} - \frac{dp(\phi)}{d\phi}\left(G^{\mathrm{S}} - G^{\mathrm{L}} + \mu_{\mathrm{B}}(x_{\mathrm{B}}^{\mathrm{S}} - x_{\mathrm{B}}^{\mathrm{L}})\right) \tag{7.6}$$

Based on the works of Tiaden *et al.* (1998), composition, $x_{\mathrm{B}}(\mathrm{x}, t)$, of a component in a phase of a binary alloy is defined by a simpler partition relation,

$$x_{\mathrm{B}}(\mathrm{x}, t) = \phi_{\mathrm{S}} x_{\mathrm{B}}^{\mathrm{S}} + \phi_{\mathrm{L}} x_{\mathrm{B}}^{\mathrm{L}} \tag{7.7}$$



(a) (b)

(c) (d)

Figure 7.6: Thermal cycling done on a 3-D Ni-Cu dendrite shown in Fig.7.1. The cycling proceeds as (a)-(b)-(c)-(d) for a single thermal cycle.

Modelling solute diffusion with phases of different diffusion coefficients was done by adopting volume averaging methods,

$$\frac{\partial x_{\mathrm{B}}}{\partial t} = \nabla \left[h(\phi) D_{\mathrm{S}} \nabla x_{\mathrm{B}}^{\mathrm{S}} \right] + \nabla \left[(1 - h(\phi)) D_{\mathrm{L}} \nabla x_{\mathrm{B}}^{\mathrm{L}} \right] \tag{7.8}$$

Table 7.3 details the parameters used for simulating the growth of a 3-D Al-Mg dendrite. A spherical seed is introduced at the origin of the cube with the initial composition of the melt x^0_{Mg}=0.165. The temperature of the melt was kept at 770 K. The domain size for the simulations was 0.38 μm and the grid dimensions. The simulations were executed on a NVIDIA Tesla M2070 GPU and the simulated dendrite in Fig.7.7 was taken at a time slice of 24000 (0.55 μs). Simulations were stopped once the diffusion boundary intercepted the domain walls. The full Al-Mg dendrite was obtained by mirroring the corner dendrite about the X=Y=Z=0 planes. This dendrite was chosen as the starting point for subsequent thermal cycling.

Table 7.3: Simulation parameters for the growth of Al-Mg dendrite in a supersaturated melt. Other material and simulation parameters are as provided in Table 6.5

Grid parameters		
Interface width	δ	$70\mathrm{d}_0$
Domain size	Nx=Ny=Nz	384
Spatial resolution	$\Delta\mathrm{x}$	δ/8
Time resolution	$\Delta\mathrm{t}$	0.05 $\Delta\mathrm{x}^2$

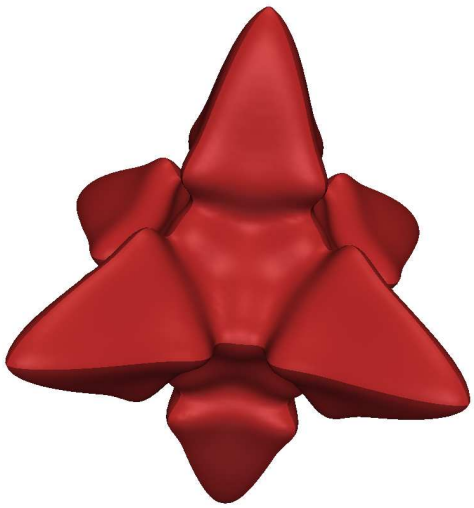


Figure 7.7: Al-Mg dendrite grown in the first octant (X,Y,Z>0) on a 384x384x384 grid with ϵ_4=0.01 with spherical seed nucleated at the origin and mirrored about the X=Y=Z=0 planes. The solid boundary surface is the ϕ=0.5 contour

7.2.1 Thermal cycling

Thermal cycling was attempted to understand the redistribution of the solute around the dendrite and also the effect of curvature. However, due to memory limitations on a single GPU, the domain size could not be scaled to domains larger than 512^3 for our simulations. Further, due to the computational time required, it was considered sufficient to stop execution within 100000 time steps. The parameters used for the thermal cycling are: wavelength(λ)=50000, amplitude(ΔT)=30 K. The total time taken to simulate two cycles was 2.3 μs. The simulations were done as detailed in Section 6.9 on stirring cycles.

7.2.2 Results and Discussion

Figure 7.8 shows the phase-field isosurface (in pink) for ϕ=0.5 and the concentration volume superimposed. The phase-field isosurface is the initial morphology and subsequent thermal cycles have rendered the final morphology, which is more clearly depicted by taking a section through the centre of the dendrite. The superimposed profiles through the midsection are shown in Fig.7.9. The pink contour indicates the phase-field profile (ϕ=0.5) of the initial dendrite. The image plane shows the evolution of the composition profiles for 100000 time steps. As each thermal cycle proceeded, subsequent remelting and coarsening had reshaped the dendrite tips to a more globular morphology.

It was observed that the growth rate was higher slightly away but towards the tip than at the root of the dendrite. This is a geometric confinement effect that comes up due to the accumulation of the solute at the root as the tip is growing. This solute accumulation prevents the growth of the roots at the same rate as the tip. The solute profiles taken along L1, L2 and L3 in Fig.7.10 are shown in Fig.7.11(a). L1 represents the body diagonal, L2, the surface diagonal and L3, the growth axis of the tip.

The profiles in Fig.7.11(a), shows that a higher concentration persists at the roots compared to the tip. This prevents the root to grow at the same rate as the tip. The phase-field profiles taken along L1, L2, L3 and shown in Fig.7.11(b), show the interface position in terms of radial distance from the center of the dendrite. A globular trend should indicate that the interface position across these three lines to be almost the same. This disparity in the interface position is due to more amount of solute accumulation at the roots than at the tip. Hence the variation in

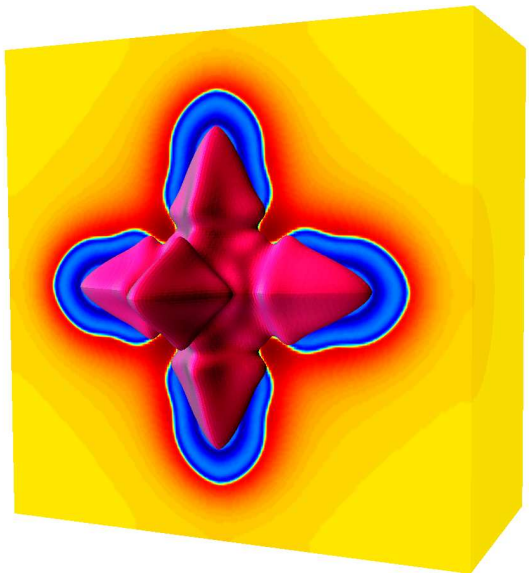


Figure 7.8: Phase-field isosurface (pink) superimposed on to the corresponding composition volume. The composition volume is sliced along X-Y depicting the concentration profiles along the mid-plane section.

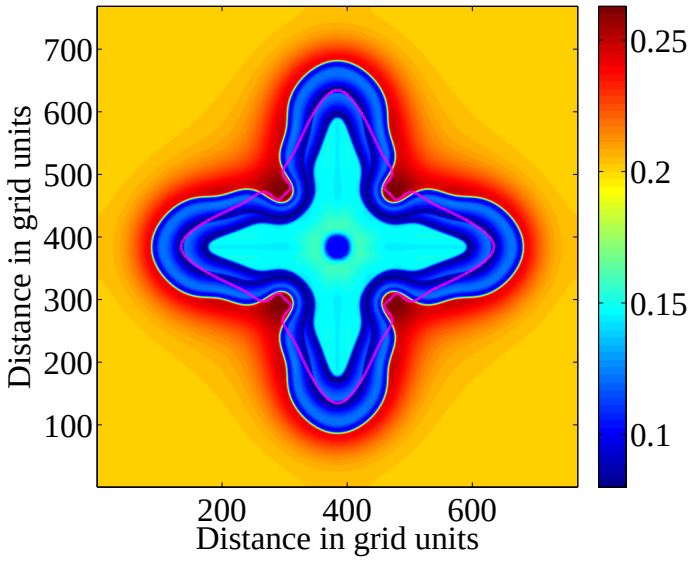


Figure 7.9: Evolution of composition profiles after two thermal cycles, with the phase-field contour in pink depicting the initial dendrite morphology chosen for thermal cycling. The composition profiles show the morphological change at the end of two thermal cycles, showing the gradual transition from dendritic to globular. Solute accumulation near the roots causes the root to grow slower than the tip as explained in Fig.7.11(a) and Fig.7.11(b)

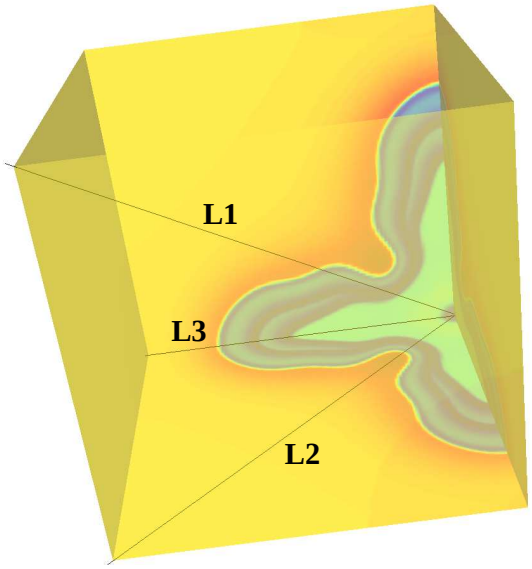
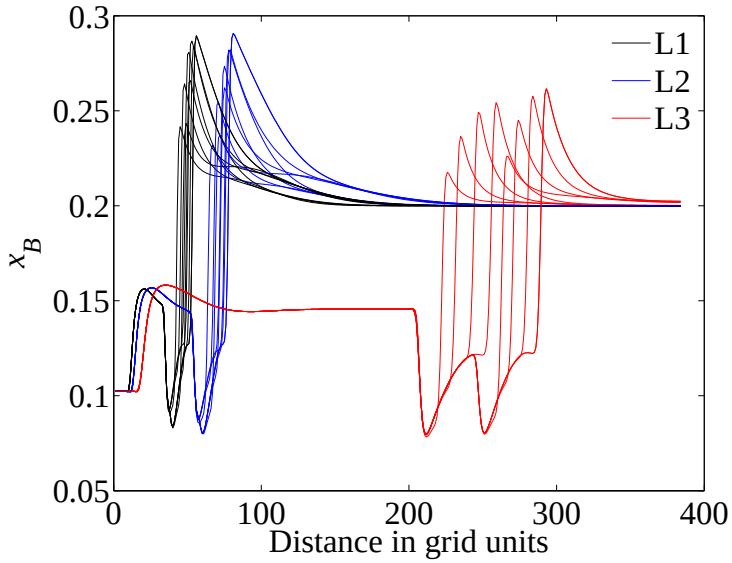


Figure 7.10: Lines L1, L2, L3 along which concentration profiles are taken. L1 is the body diagonal, L2 along the surface diagonal and L3 is along the axis

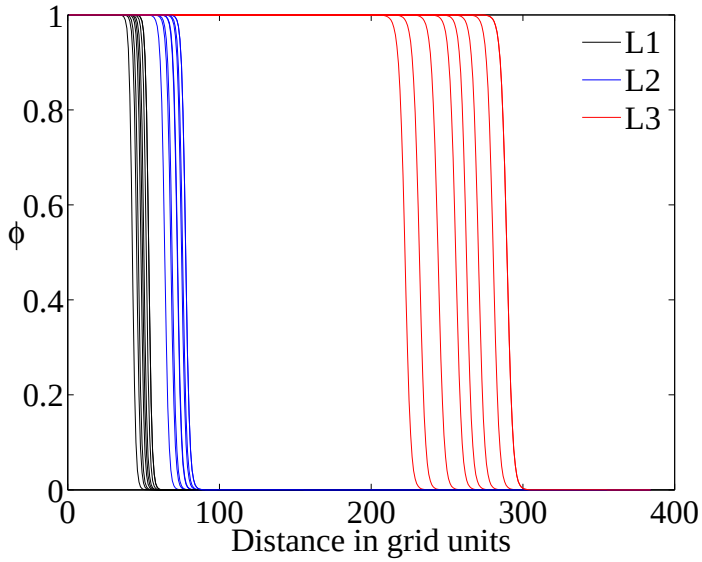
the interface position along the surface of the dendrite. In comparison with the 2-D simulations done on Al-Mg with multiple dendrites and multiple cycling, the following observation can be made. In the event of multiple dendrites being present, the growing tip, encounters other dendrites and its growth rate will be reduced. The presence of multiple dendrites creates a solute redistribution in the melt, as depicted by the morphological changes brought about by thermal cycling done in 2-D. This leads to a curvature driven diffusion along the surface of the dendrite. Further, cycling would bring about the morphological change from dendrite to globular as shown in Fig.6.22.

7.3 Dendrite growth with anti-trapping current

Dendritic growth in 3-D was simulated with the anti-trapping current for the Al-Mg alloy system. In this study, however, only the growth morphologies were studied. This was due to the memory restrictions faced when trying to simulate for larger domains on a single GPU. Grid dimensions greater than 512^3 were unable to be simulated on a single GPU. However, a very recent work on parallelization across multiple GPUs has provided with possibilities of extending the simulations across larger domains, in particular to thermal cycling. Two different sets of simulations were adopted, one was simulating the growth morphology in a single GPU. The other set of simulations were done across multiple GPUs. The simulation parameters for



(a)



(b)

Figure 7.11: Concentration profiles along L1, L2 and L3 from Fig.7.10, as coarsening proceeds shown in (a). Higher solute concentration near the roots hinder the growth rate at the root. This is caused by the solute accumulation at the roots by the moving tip. The interface position near the root and the tip are taken along the lines L1, L2 and L3. As coarsening proceeds, due to the solute accumulation near the roots, the interface motion is slower at the roots than the tip (b). In the case of a multiple dendrites configuration, the faster growing tip will be hindered by other growing dendrites (Fig.7.17) and solute redistribution in the melt will eventually cause the entire dendrite to undergo a globular transition

Table 7.4: Simulation parameters for the growth of Al-Mg dendrite in a supersaturated melt with anti-trapping current. Other material and simulation parameters are as provided in Table 6.5

Grid parameters		
Interface width	δ	$100\mathrm{d}_0$
Domain size	Nx=Ny=Nz	512
Spatial resolution	$\Delta\mathrm{x}$	$\delta/10$
Time resolution	$\Delta\mathrm{t}$	$0.1\Delta\mathrm{x}^2$

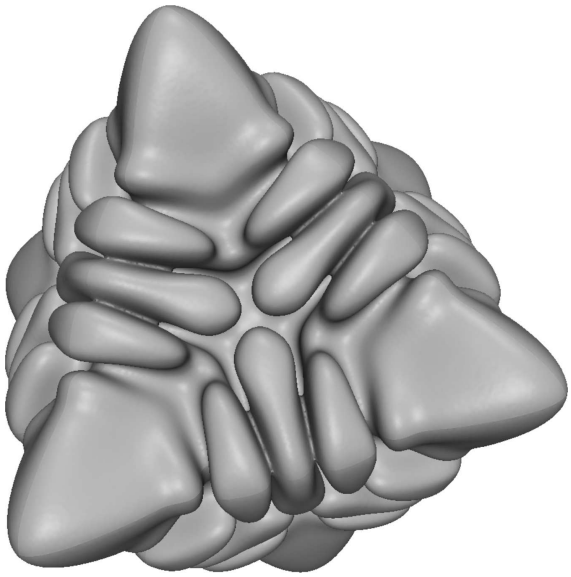


Figure 7.12: Al-Mg dendrite grown in the first octant (X,Y,Z>0) on a 512×512×512 grid with ϵ_4=0.01 with spherical seed nucleated at the origin and mirrored about the X=Y=Z=0 planes. The solid boundary surface is ϕ=0.5 isosurface

dendritic growth in a single GPU are provided in Table 7.4.

A dendrite is grown by nucleating a seed at the origin of the cube. The fully grown profile of the dendrite is obtained by a reflection about the X=Y=Z=0 planes. The phase-field profile shown in Fig.7.12, is taken at ϕ =0.5 depicting the interface. The grid limitation, due to memory requirement, resulted in a less developed growth morphology as shown in Fig.7.12. To simulate a full grown dendrite, it is necessary to address the memory issues. One method is to write algorithms for efficient memory management. The other method is a bit straightforward, where the entire domain is spread across different GPUs. This is achieved through distributed memory parallelization using MPI. Domain splitting and ghost cell exchange are done by the mpi routines and the solver computations on the GPU.

This is the first study where a dual parallelization strategy was adopted for phase-field simulations, that demonstrates the effectiveness of distributed parallel computing using MPI and

Table 7.5: Simulation parameters for the growth of Al-Mg dendrite in a supersaturated melt with anti-trapping current. Other material and simulation parameters are as provided in Table 6.5

Simulation parameters		
Interface width	δ	$90\mathrm{d}_0$
Spatial resolution	$\Delta\mathrm{x}$	$\delta/12$
Time resolution	$\Delta\mathrm{t}$	$0.15\Delta\mathrm{x}^2$
Initial melt Composition	x_{Mg}	0.15
Configuration-A: Corner seed		
Size of each sub-domain	Nx×Ny×Nz	$1022 \times 1022 \times 87$
Number of slices/ GPUs	ngpus	12
Domain length	$1022 \times 1022 \times 87$	$1.02\ \mu\mathrm{m} \times 1.02\ \mu\mathrm{m} \times 1.02\ \mu\mathrm{m}$
Configuration-B: Seeds nucleated on X-Z plane and grown along Y		
Size of each sub-domain	Nx×Ny×Nz	$576 \times 1046 \times 60$
Number of slices/ GPUs	ngpus	18
Domain length	$576 \times 1046 \times 1046$	$0.58\ \mu\mathrm{m} \times 1.06\ \mu\mathrm{m} \times 1.06\ \mu\mathrm{m}$
Configuration-C: Multiple dendrites		
Size of each sub-domain	Nx×Ny×Nz	$1010 \times 1010 \times 56$
Number of slices/ GPUs	ngpus	18
Domain length	$1010 \times 1010 \times 1010$	$1.02\ \mu\mathrm{m} \times 1.02\ \mu\mathrm{m} \times 1.02\ \mu\mathrm{m}$

OpenCL in simulating microstructure evolution for larger domains. A similar early attempt on dual parallelization using MPI and CUDA is also available (Shimokawabe *et al.* (2011); Yamanaka *et al.* (2011); Aoki *et al.* (2011)). In view of the applicability of OpenCL to multiple GPU platforms and for hybrid and heterogeneous computing platforms, it is felt that this study is one of a first kind. This approach being still in its early stages, only dendrite growth morphologies on large domains were realised. Further treatment on the aspect of thermal cycling needs to be delved into, which forms the basis for future work.

Al-Mg was chosen as the binary alloy system and the solid and the liquid phases were modelled using Gibbs energy functions, obtained through CALPHAD. Programming was done in C using a virtual topology for the MPI environment. A 1-D domain decomposition along Z was done by splitting the domain into many volume slices as given in Table 7.5. The MPI environment was initialised for the respective number of processors, so that each CPU handles a GPU. Each volume slice was solved by one GPU and the MPI routines take care of the boundary nodes exchange at every time step.

Three different growth morphologies across three different domains were simulated as shown in Table 7.5. Similar computation on a single GPU for 512^3 domain took 40 minutes for every 3000 time steps. The same set of computations, distributed across multiple GPUs took 27

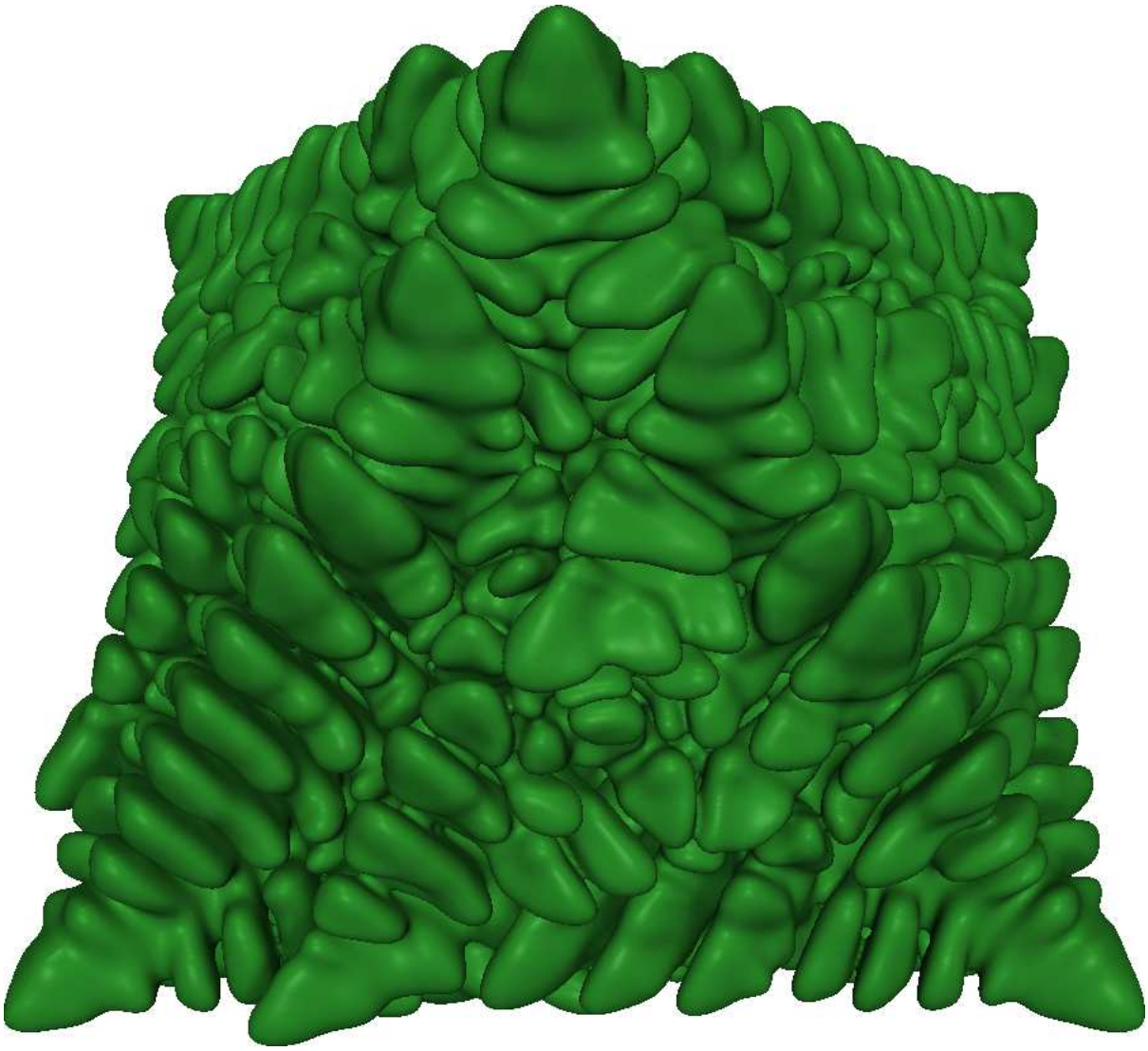


Figure 7.13: Configuration-A: Al-Mg dendrite grown in the first octant (X,Y,Z>0) on a 1022x1022x1022 grid with a spherical seed nucleated at the origin and mirrored about the X=Y=Z=0 planes. The isosurface shown is at ϕ=0.5

minutes across 18 GPUs and 31 minutes for 12 GPUs for the same 3000 time steps.

Configuration-A was a spherical seed nucleated at one corner of the cube. The growth morphology of the dendrite obtained after 45000 time steps (2.1μs) is shown in Fig.7.13. The domain length and other system parameters are given in Table 7.5. The dendrite shown in Fig.7.13 is obtained by mirroring the corner dendrite about the X=Y=Z=0 planes. The Al-Mg dendrite shows the existence of channel formation in primary arm for a minor deviation from the isotropic capillary strength of ϵ_4=0.01. Due to weak capillary anisotropy, a more rounded morphology is observed as is the case of Al-based alloys.

Configuration-B was to observe dendritic coalescence in the case of dendritic arrays. Three

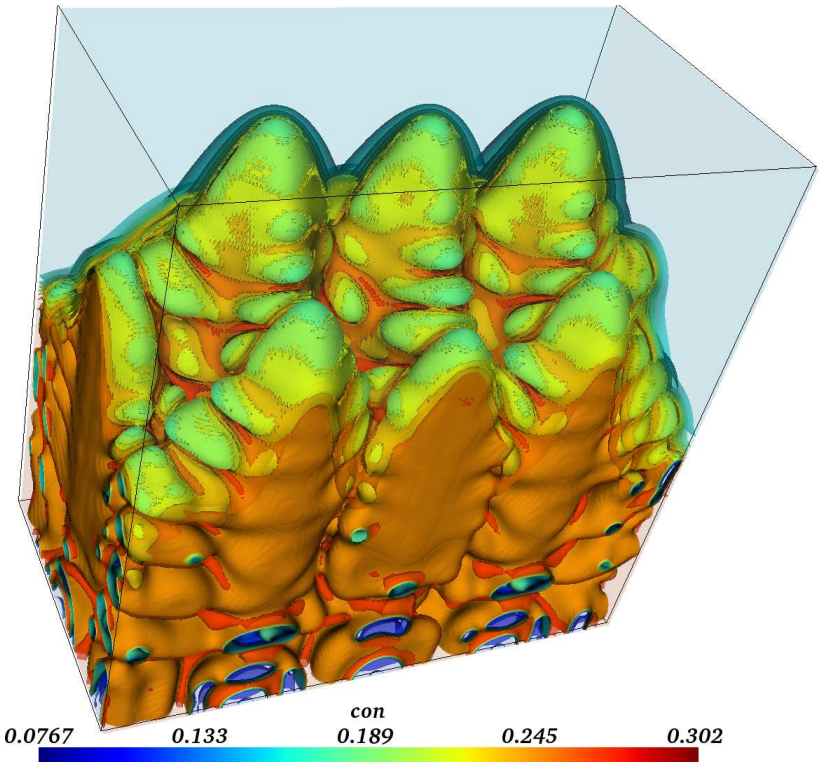


Figure 7.14: Configuration-B: Al-Mg dendrite seeded from the X-Z plane and grown along the Y-axis. Grid size is 576×1046×1046

seeds were placed on the X-Z plane and grown along Y axis as shown in Fig.7.14. No-flux boundary condition was applied on all boundaries. A cross-section through the centre of the 3 dendrites shows that certain regions were coalesced and certain regions were not. Closer observations of the section shown in Fig.7.15 and the superimposed composition profiles in Fig.7.16 reveal many features. It is observed that coalescence generally occurs when the thermal or solutal gradients of the coalescing regions are shallow. For alloys, in the absence of thermal gradient, the solutal gradient plays a significant role. In general, assuming no convection in the melt, a lower solutal gradient causes the tip of the moving dendrite to get coalesced with the surrounding solid. On the other hand, a higher solutal gradient does not allow the growing regions to coalesce as depicted by the regions marked with white arrows in Fig.7.15 and Fig.7.16.

Configuration-C was done for two reasons. One was to observe the effect of solute segregation on impingement and coalescence of multiple dendrites. The other was to extend the study effect of thermal cycling across multiple dendrites. In this section, only the effect of solute segregation on dendrite growth and impingement are discussed. Further studies on thermal cycling are not yet performed due to hardware requirements for performing the computations.

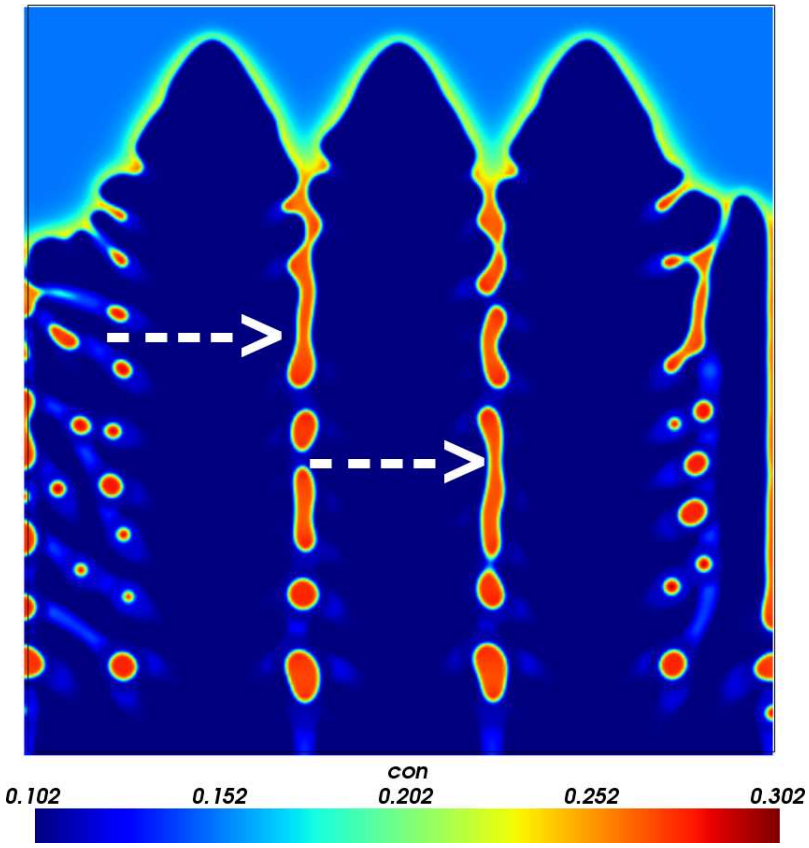


Figure 7.15: A section plane along the centre of the 3 growing dendrites with regions marked in white arrows to show solutal gradients, that does not allow coalescence to happen

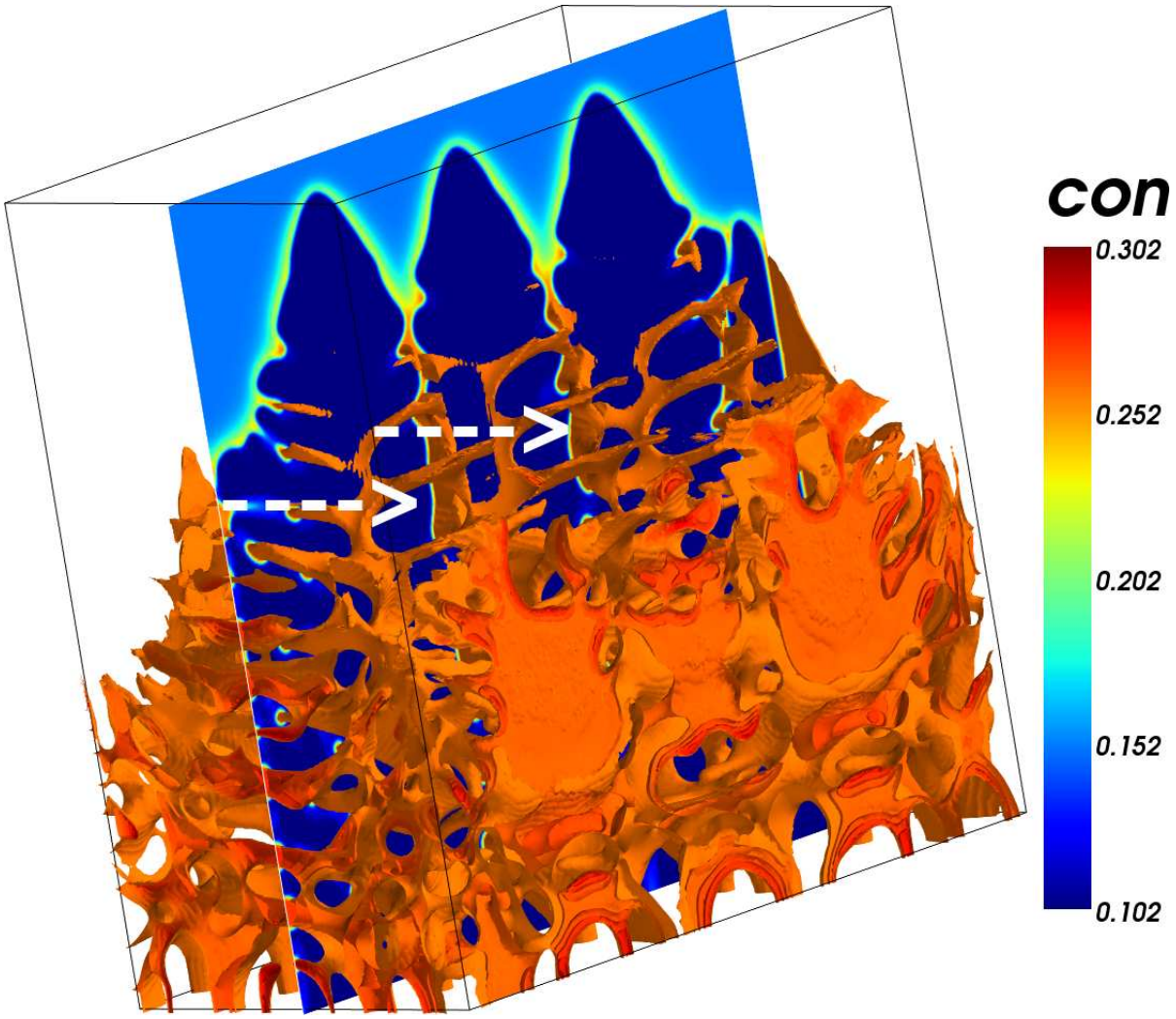


Figure 7.16: The section plane is superimposed with the solutal profiles. The magnitude of the solutal profiles shown are for higher segregation of Mg content (0.25x_{Mg} - 0.3x_{Mg}). This is done to show that the regions surrounding these profiles do not undergo coalescence as depicted by the white arrows in accordance with Fig.7.15

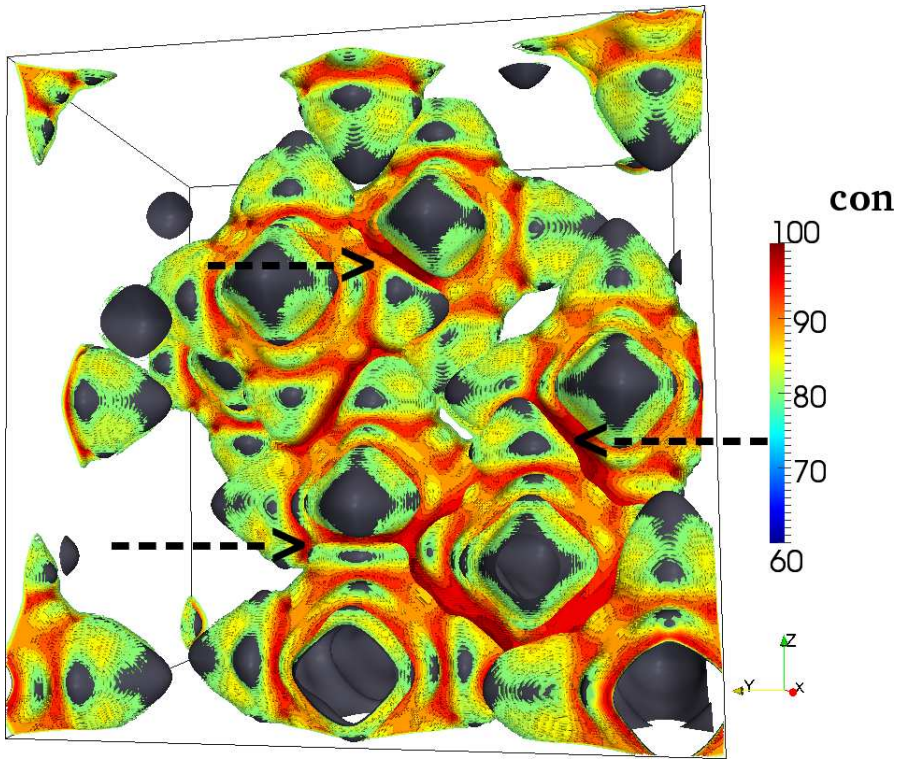


Figure 7.17: Configuration-C: Multiple Al-Mg dendrites grown at heuristically seeded locations in a domain size of 1010×1010×1010. Coalescence of the dendrites are not observed due to the excess amount of solute rejection as shown in the regions marked with arrows. Concentration is shown on a relative scale.

The treatment of thermal cycling in 3-D is important as one more effect apart from thermal and solute redistribution comes in *i.e.*, the effect of curvature. The study of this effect is not possible in 2-D, which implies that the thermal cycling in 3-D to be realistic.

Multiple dendrites were grown at arbitrarily seeded locations as shown in Fig.7.17. The domain size chosen was not large enough to see the development of secondary arms. However, it was sufficient to observe the impingement and coalescence of the primary arms. As shown by the regions marked with white arrows in Fig.7.17, the growing arms when encountered with high solutal gradients are unable to coalesce. However, coalescence is observed in those regions where the solutal gradients are lower.

7.4 Summary

Dynamics of solute flow at the inter-dendritic side is important and large scale simulations with multiple dendrites will pave way to benchmark homogenisation models that are at higher length

scales. These models can be calibrated against detailed numerical studies at microscales, so that an appropriate anisotropic porous medium formulation can be proposed for system scale simulations without losing sight of various physical phenomena occurring at the microscale. Anisotropic mushy zone formulation should take all these phenomena into account to be more realistic. Direct numerical simulation of 3-D dendrite growth could lead to the necessary correlations that aid in such a length-scale bridging in future and help in reaching predictive capability for solidification microstructures.

CHAPTER 8

CONCLUSIONS

1. CALPHAD formulation of Gibbs energy functions for the binary systems Al-Cu and Al-Mg was successfully incorporated in the phase-field model (Kim *et al.* (1999)) to perform faster simulations of dendrite solidification.

2. Thin-interface limit with antitrapping current allowed for increased interface widths, hence making possible simulation of large computational domains.

3. Using parallel programming approach by combining distributed computing (MPI) and GPU computing (OpenCL), the computational time could be reduced to a reasonable level for large scale 3-D simulations.

4. The stencil approach to the solver made the program massively parallel for computation on heterogeneous platforms.

5. Model parameters for numerically stable simulation were identified. It was found that improper choice of numerical parameters lead to artefacts.

6. The one-sided nature of the diffusion model used in the anti-trapping model led to artefacts at locations where local remelting takes place.

7. There is an offset between the imposed thermal cycle and the growth-remelting cycle. Equilibration of solute field led to solutal remelting during the beginning of the cooling portion of the thermal cycle.

8. Fractal dimension approach proved effective in capturing the branched nature of the dendrites and in quantifying the morphological transition where curvature driven diffusion plays a key role.

9. Geometric confinement of solute within interdendritic regions in 2-D is not present in 3-D simulations.

10. Channels observed in 3-D provide valuable insights into the formation of interdendritic regions and solute segregation within these regions. The strength of anisotropy has an important role on the growth morphology : stronger values of anisotropy lead to formation of clear channels for solute movement and weaker values forming a more tortuous path for the movement of solute.

11. During impingement and coalescence of multiple dendrites in 3-D, curvature effects and solute segregation play a crucial role. Solute segregation within the interdendritic regions alone cannot account for dendrite coalescence. The effect of the two principal curvatures also play an important role in the diffusion kinetics in shaping the final growth morphology.

CHAPTER 9

SCOPE FOR FUTURE WORK

The ever increasing computing power at an affordable cost can be now be exploited to gain more insights into the microstructure evolution. Realistic 3-D simulations provide a more accurate view of the morphological changes occurring during material processing techniques. The phase-field model presented in this work involved solidification studies on binary alloy systems. Technical alloys that are of industrial importance are usually multi-component in nature. The extension of the present work can be made on the following lines as follows:

1. Extension of the binary alloy model to multicomponent systems.
2. Non-isothermal solidification by introducing temperature dependency to the present isothermal solidification for applicable situations.
3. Length scale bridging of phase-field simulations with system scale could be attempted taking the morphological evolutions and channel formations into account for the solute diffusion in liquid.
4. Model parameters such as surface energies and interface kinetics coefficient required by phase-field could be obtained from MD simulations at an atomic scale. This is important so that the parameters that are experimentally difficult to measure but play an important role in morphological evolution have more realistic values.
5. Nuclei seeded in the melt can be made to grow in different orientations by introducing an order parameter for orientation (Warren *et al.* (2003))
6. Convection can be incorporated by the coupling of the Navier-Stokes equation with the phase-field governing equations along the lines of Beckermann *et al.* (1999)

Scope for improvements on numerical procedures:

1. Parallel implementation of phase-field method using an implicit scheme. This requires implementation of efficient CPU as wells as GPU based sparse matrix solvers for solving the equations.
2. Adaptive mesh refinement (AMR). Typically an AMR application is composed of four integral parts
   (a) Error Estimation Funcion to determine when to refine
   (b) Domain Decomposition to determine the size of the grids at each level

(c) Parallel data structure + data distribution over processors

(d) Data communication

The latter two (2(c) and 2(d)) describes the parallelization aspects of AMR. PARAMESH (MacNeice *et al.* (2000)) is one such open-source package which is designed to provide an application developer with an easy route to extend an existing serial code which uses a logically Cartesian structured mesh into a parallel code with AMR. Other tools which maybe of interest are CHOMBO (Applied Numerical Algorithms Group, Lawrence Berkeley National Laboratory (2005)), that provides a set of tools for implementing finite difference methods for the solution of partial differential equations on block-structured adaptively refined rectangular grids. BoxLib developed by (Rendleman *et al.* (2000); Center for Computational Sciences and Engineering, Lawrence Berkeley National Laboratory (2000–)), one more AMR tool, contains all the functionality needed to write a parallel, block-structured AMR application.

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LIST OF PAPERS BASED ON THESIS

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1. P. Gerald Tennyson, P. Kumar, H. Lakshmi, G. Phanikumar and P. Dutta. Experimental studies and phase field modeling of microstructure evolution during solidification with electromagnetic stirring. *Semi-Solid Processing for Alloys and Composites (S2P 2010)*, Beijing, September 16-18, 2010.

2. P. Gerald Tennyson and G. Phanikumar. Phase-field simulations of dendritic to globular transition using a non-isothermal binary alloy solidification model *National Metallurgists Day - Annual Technical Meeting (NMD-ATM 2010)*, IISc Bangalore, November 14-16, 2010.

3. P. Gerald Tennyson and G. Phanikumar. Computational modelling of dendritic to globular transition using an isothermal binary phase-field model. *International Symposium for Research Scholars (ISRS 2010)*, IIT Madras, December 20-22, 2010.

4. P. Gerald Tennyson and G. Phanikumar. Phase-field modelling of dendritic to globular transition coupled with CALPHAD description of Gibbs energies. *National Metallurgists Day - Annual Technical Meeting (NMD-ATM 2011)*, Hyderabad, November 14-16, 2011.

5. P. Gerald Tennyson, K.C. Hari Kumar and G. Phanikumar. Phase-field modelling of dendritic solidification incorporating CALPHAD description of Gibbs energies. *International Conference on Solidification Science and Processing (ICSSP5 2012)*, IIT Bhubaneswar, November 19-22, 2012.

6. P. Gerald Tennyson. Computational modelling of dendritic solidification using MPI and OpenCL. *HPC Symposium 2013*, IIT Madras, March 28, 2013.

Papers to be communicated

1. P. Gerald Tennyson, Karthik G.M. and G. Phanikumar. Phase-field simulation of alloy dendritic growth across heterogeneous platforms.

2. P. Gerald Tennyson, Karthik G.M. and G. Phanikumar. MPI+OpenCL parallelization of a phase-field model for microstructure evolution on GPU clusters.

3. P. Gerald Tennyson, S. Sandhya, K.C. Hari Kumar and G. Phanikumar. Computational studies on dendritic to globular transition using a phase-field model incorporating CALPHAD description of Gibbs energies.

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